

Intermediate Macroeconomics

A Self-Contained Course Companion

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How to Read This Book

This is a course companion for a one-semester **intermediate macroeconomics** course taught in the neoclassical tradition. It is written to be read straight through, like a short textbook, rather than skimmed like a formula sheet.

Macroeconomics asks two questions. The first is about the *long run*: why are some countries rich and others poor, and what makes an economy grow over decades? The second is about the *short run*: why does output fluctuate around its trend, and can policy do anything about it? The book follows that division. We begin by learning how the aggregate economy is *measured* — output, prices, and the labor market — because every later argument is ultimately a claim about these numbers. We then build the theory of long-run growth from the Solow model up through its microfounded descendants and the Malthusian world that preceded modern growth. Next we study money and banking and the quantity theory that links money to prices. Only then do we turn to the short run: the Keynesian cross, the IS–LM model, aggregate demand and supply, fiscal policy and the limits set by Ricardian equivalence, and finally the Phillips curve, expectations, and the rules-versus-discretion debate.

A warning about the subject. Macroeconomics is less tidy than microeconomics. The same fact can be read through more than one model, the data rarely settle an argument cleanly, and a good lecturer will digress into institutional detail — how China’s central bank actually operates, why a particular policy failed in a particular year — that does not fold neatly into any single theorem. We have kept those digressions where they illuminate the theory, set off in **remark** boxes, rather than forcing every observation into a model it does not quite fit. Read the boxes as the texture of a living subject, not as loose ends.

How Each Chapter Is Organized

Each chapter opens with a short *story* in plain English explaining what question we are answering and why we are about to introduce a particular tool. We then formalize the idea, work through the algebra, and read the economics off the diagrams. Definitions and key results are boxed so you can find them quickly when reviewing. Every figure in this book has been redrawn from scratch; none are scanned.

Box Color Code

- **Green: Definition**

— the objects we build on (GDP, the natural rate of unemployment, the money multiplier, ...).

- **Blue: Theorem / Result**

— the central results of each model.

- **Pink: Assumption**

— the maintained hypotheses of a model, gathered in one place.

- Worked **examples**, boxed **remarks**, and emphasis boxes appear throughout.

A Note on Notation

Throughout, Y is aggregate output (real GDP), C consumption, I investment, G government purchases, and $NX = X - M$ net exports. K denotes the capital stock and L (or N) labor; lower-case $k = K/L$ and $y = Y/L$ are their per-worker counterparts. The savings rate is s , the depreciation rate δ , population growth n , and labor-augmenting technical progress g ; a steady-state value carries a star, as in k^* . Prices and inflation: P is the price level and π its growth rate, with a superscript e (as in P^e , π^e) denoting an expectation. Interest rates: r is the real rate and $i = r + \pi^e$ the nominal rate. In intertemporal problems β is the discount factor and $\mathcal{L}$ the Lagrangian. Money aggregates are $M_0 \subset M_1 \subset M_2$.

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Chapter 1

Introduction: The Macroeconomic Way of Thinking

Intermediate microeconomics leaves a student remarkably well equipped to reason about a single household, a single firm, or a single market. A consumer takes prices and income as given and chooses the bundle she likes best; a firm takes the wage and the rental rate as given and chooses how much to produce. The hard analytical work goes into the *choice*, and the numbers that enter the choice — this price, that income — arrive from outside, already measured. Macroeconomics begins where that convenience ends. Its object of study is not a household or a firm but the economy *as a whole*: the total output of a country, the average level of all prices, the fraction of willing workers who cannot find a job, the rate at which the standard of living rises over a generation. Nobody hands us those numbers. Before we can build a single model of the aggregate economy, we have to learn how to *measure* it.

This is the first thing that makes macroeconomics different, and it is the reason the book opens with three chapters on measurement before any model appears. But there is a second, deeper difference, and it shapes the entire subject. The behavior of the whole is not simply the sum of the behaviors of its parts. One household's decision to save more raises that household's wealth; if every household tries to save more at once, total spending falls, firms sell less, incomes drop, and the economy may end up saving no more than before. Aggregation is not addition. To say anything reliable about the whole we need theory — and, as we will see, macroeconomists have built that theory in two very different styles, one prizing logical purity and the other prizing usefulness. This chapter sets up the questions the book will answer, the methods it will use, and the two intellectual traditions it draws on. It is the map for everything that follows.

1.1 What Macroeconomics Studies

Macroeconomics studies the economy as a whole rather than its individual pieces. The shift in object brings a shift in the first problem we must solve. In microeconomics, measurement is comparatively easy: a consumer treats income and prices as given exogenously, and a firm

needs to track its costs, but the quantities involved are concrete and bounded. When we move to the level of an entire country, even the most basic quantities have to be *constructed*. We do not observe “the price” the way a single shopper does; we observe millions of individual prices and must somehow combine them into a single *price level* (Chapter 3). We do not observe “output”; we observe the production of countless firms in incommensurable units and must aggregate them into a measure such as gross domestic product (Chapter 2).

Macroeconomics: the study of the economy as a whole

Microeconomics analyzes individual agents and markets, taking prices and incomes as *given*. Macroeconomics analyzes aggregate quantities — total output, the price level, the unemployment rate, the long-run growth rate — which are not given but must first be *measured and aggregated*. This is why the book begins with national income accounting (Chapter 2), price indices (Chapter 3), and the labor market (Chapter 4) before turning to any theory.

Because these aggregates are built rather than read off directly, they are also, to some degree, *constructed numbers* — and constructed numbers can be shaped by the choices made in constructing them. This is not the same as fabrication. A figure such as GDP is assembled through a definite statistical procedure from real underlying data, but the procedure involves judgment calls, and those judgment calls can be leaned on. A careful reader of macroeconomic data therefore cross-checks the headline aggregates against harder-to-manipulate physical proxies, and against the same series in other economies.

Remark (Cross-checking the official numbers).

Because headline aggregates rest on construction choices, analysts often triangulate them against physical indicators that are difficult to dress up. A well-known example is the so-called Keqiang index, which gauges Chinese activity from electricity consumption, rail freight volume, and bank lending rather than from the reported output figure alone; the premise is that it is harder to fake the electricity a factory burns than the value it claims to have produced. Comparing the same series across economies — China against the United States, say — is a further sanity check. None of this means the official numbers are wrong; it means a serious macroeconomist treats every aggregate as a measurement with a method behind it, and asks what that method might miss.

1.2 How Macroeconomists Work

Confronted with the aggregate economy, how does one actually do macroeconomics? The discipline proceeds in a characteristic way. It begins by collecting *stylized facts* — broad, robust regularities in the data, such as the observation that consumption moves closely with income, or that output and employment rise and fall together over the business cycle — and then tries to explain them. The explanation can be pursued through several distinct methods, and it is worth distinguishing them at the outset, because the rest of the book uses all of them.

- **Reduced-form analysis.** One runs regressions linking aggregate variables and reports the associations found in the data. This is evidence, and useful evidence, but it has two

well-known limits. A regression coefficient need not reveal the underlying *mechanism*; at best it documents a relationship. And the macroeconomy is a *dynamic, interlocking system* in which variables feed back on one another, so the clean one-directional causal story that a single regression seems to tell is rarely warranted.

- **Modeling.** One writes down an explicit model — assumptions about agents, technology, and markets — and derives its implications. The payoff is a *predicted outcome* that follows logically from stated premises, so the source of any prediction can be traced and the model interrogated.
- **Simulation.** One feeds historical data through a model to see how well its behavior matches what actually happened, testing the degree to which the theory fits the record.
- **Calibration.** One pins down a model’s parameters by drawing on accumulated empirical estimates — stacks of prior studies and data — rather than estimating every parameter afresh, and then asks what such a disciplined model implies.

These methods are complementary rather than competing. A stylized fact motivates a model; the model is calibrated from existing estimates; the calibrated model is simulated against history; reduced-form regressions supply both the original fact and a check on the model’s predictions. The reader will see this cycle repeated, in miniature, in nearly every chapter.

Remark (A note on sources and texts).

This course is taught in the neoclassical tradition, and the natural companion in English is Williamson’s *Macroeconomics*, which develops the subject from explicit microfoundations. For the institutional side — how macroeconomic policy is actually conducted in China — the lectures draw on material pairing macroeconomic theory with Chinese policy. For data, the standard wells are the National Bureau of Statistics of China, commercial macro databases such as CEIC and Wind, and, for international and U.S. series, the Federal Reserve Economic Data (FRED) service. Knowing where the numbers come from is part of knowing what they mean.

1.3 Two Traditions: Neoclassical and Keynesian

There is a fundamental methodological fork in macroeconomics, and understanding it explains much of why the field looks the way it does. The two branches differ on a single question: should one build up from the individual, or start directly from the aggregate?

1.3.1 The neoclassical tradition: microfoundations

The neoclassical view draws no sharp line between microeconomics and macroeconomics. In this picture there is, in principle, nothing distinctively “macro” going on at all: the economy is the aggregation of its optimizing consumers and firms, and so macroeconomic outcomes should be derivable from, and consistent with, the microeconomic behavior of individuals. Modern neoclassical macroeconomics accordingly insists on *microfoundations* — every aggregate relationship must be grounded in the choices of agents who maximize utility or profit subject to constraints.

The appeal of this discipline is real. A microfounded model is *logically tight*: its conclusions follow rigorously from explicit assumptions about preferences, technology, and constraints, so one can always say exactly why a result holds and trace it back to a premise that can be debated. The cost is equally real. Human society contains a great many moving parts, and a clean, internally consistent logical system often struggles to capture all of them. Insisting that every macroeconomic phenomenon be derived from first principles can leave the resulting models with surprisingly weak *explanatory power* for the messy economy we actually observe — a frustration familiar to anyone who has tried to fit a beautiful model to stubborn data.

1.3.2 The Keynesian tradition: let the data speak

The Keynesian response is to change the starting point. Rather than building the aggregate up from individual optimization, one studies the aggregate *directly*, reading the behavior of the whole off the aggregate data and letting the data speak — even at the cost of leaving the microeconomic foundations unspecified.

The canonical illustration is the aggregate consumption function. Instead of deriving how much a society consumes from each household's utility maximization, one looks at the aggregate consumption data and posits a simple relationship,

$$C = C_0 + \text{MPC} \cdot (Y - T),$$

where $Y - T$ is aggregate disposable income (output net of taxes), C_0 is autonomous consumption, and MPC is the *marginal propensity to consume* — the fraction of an extra dollar of disposable income that is spent. The point to notice is that MPC has no microeconomic pedigree. It is not derived from any preference structure; it is a parameter fitted to the data because the data behave roughly that way. The same spirit appears later in the Solow model (Chapter 5), which simply *assumes* a constant saving rate rather than deriving saving from an intertemporal optimization problem.

The Keynesian bargain

The Keynesian approach trades *scientific purity for usefulness*. By starting from the aggregate and fitting relationships such as $C = C_0 + \text{MPC} \cdot (Y - T)$ directly to the data, it forgoes a microeconomic derivation of those relationships. What it gains is a tractable, empirically grounded account aimed squarely at *understanding the data themselves* — a model that may explain the world well even though it cannot, by construction, say exactly why each agent behaves as the aggregate suggests.

Neither tradition is simply right. The neoclassical models are the more rigorous and will occupy most of our growth theory; the Keynesian models are the more immediately useful for short-run policy and will organize the back half of the book. A mature reading of macroeconomics holds both in mind at once.

Remark (The aggregate was studied long before Keynes).

It would be a mistake to think macroeconomic questions began with Keynes. Classical economists studied the wealth of nations as a whole well before him — Adam Smith’s *The Wealth of Nations* is, in a real sense, a work about the aggregate economy — using the classical methods of their day. What Keynes changed was not the subject but the method: he made it respectable to model the aggregate on its own terms, without first reducing it to the individual.

1.4 The Two Big Questions

For all its scattered detail, macroeconomics is organized around two questions, separated by their time horizon.

- **Long-run economic growth.** Why are some countries rich and others poor, and what determines the long-run *trend* of an economy’s output over decades? This is the domain of growth theory — the Solow model and its microfounded descendants (Chapters 5–7).
- **Short-run business cycles.** Why does output not grow smoothly along its trend but instead *fluctuate* around it, with booms and recessions, and can policy smooth those fluctuations? This is the domain of short-run macroeconomics — the Keynesian cross, IS–LM, and aggregate demand and supply (Chapters 10–12).

A compact way to hold the two questions together is to think of the path of output as a trend plus a deviation:

$$\underbrace{\text{observed output}}_{\text{what we see}} = \underbrace{\text{long-run growth trend}}_{\text{Chapters 5–7}} + \underbrace{\text{cyclical shocks around the trend}}_{\text{Chapters 10–12}}.$$

Growth theory studies the trend; business-cycle theory studies the deviations from it. The book follows exactly this division: measure the aggregates first, then explain the trend, then explain the fluctuations.

1.5 The Goals of Macroeconomic Policy

Why does any of this matter for policy? Macroeconomic policy is conventionally aimed at four objectives, and the rest of the book can be read as a study of what stands in the way of achieving them.

1. **Economic growth** — raising output and living standards over the long run.
2. **Full employment** — ensuring that everyone who actively wants work can find it.
3. **Controlling inflation** — keeping the price level stable enough that money remains a reliable yardstick.
4. **External balance** — keeping international payments in balance, a concern that arises once the economy is open to trade and capital flows.

Three clarifications, each of which a later chapter develops, are worth flagging now.

First, “full employment” does not mean a zero unemployment rate. It means that everyone actively searching for a job can find one, so that the *cyclical* component of unemployment is zero and the measured unemployment rate reflects only the *natural rate* — the frictional and structural unemployment that persists even in good times as people move between jobs and industries. We make this precise in the labor-market chapter (Chapter 4).

Second, the goals are not jointly attainable. When fluctuations are driven by demand, employment and inflation tend to move together, so that pushing toward fuller employment tends to raise inflation and vice versa: the two cannot be optimized at once. This tension is the subject of the Phillips curve (Chapter 14), and it is the central dilemma of stabilization policy.

Remark (External balance, Chinese style).

The external-balance objective looks different depending on a country’s position in the world economy. China has run large and persistent *trade surpluses* on the strength of its exports, and its central bank exercises near-complete control over the exchange rate. A standing question raised in the lectures is whether the resulting accumulation of foreign claims is worth as much as it appears: piling up external surpluses is, in a sense, lending the rest of the world your output, and the value of that “saving” depends on what it is eventually exchanged for. We return to open-economy questions only at the margins of this book, but it is worth noting early that “external balance” is a live and contested policy goal, not a settled accounting identity.

1.6 A Word on the Character of the Subject

It is worth being honest, at the outset, about what kind of subject this is. Microeconomics is tidy: a well-posed problem typically has a clean optimization at its center and a definite answer. Macroeconomics is deliberately less tidy. The same fact can often be read through more than one model; the data rarely settle a dispute cleanly; and the most illuminating remark in a lecture is sometimes a one-line institutional aside — how a central bank actually conducts an operation, why a particular policy failed in a particular year, how a particular country measures a particular series — that fits no chapter perfectly. We will not pretend otherwise, and we will not force every such observation into a theorem. Where there is a model, we will be rigorous about it; where the lecture digresses into the texture of real economies, we will keep the digression, set off in a remark box. Read those boxes as part of the subject, not as loose ends. With the questions, the methods, and the temperament of macroeconomics now in view, we turn in the next chapter to the first task that any of them requires: measuring the aggregate economy.

Chapter 2

National Income Accounting

Before we can explain why economies boom and slump, grow and stagnate, we need a way to measure what they produce. Macroeconomics is the study of aggregates — total output, total income, total spending — and an aggregate is only as useful as the number that stands for it. National income accounting is the system of definitions and identities that turns the sprawling activity of millions of households and firms into a handful of measured quantities. It is the macroeconomist's equivalent of a firm's financial statements: dull on its own, indispensable for everything that follows.

The central object is gross domestic product (GDP), the market value of everything an economy produces in a period. Most of this chapter is an extended unpacking of that one sentence, because almost every word in it is a deliberate choice with consequences. We then show that the same total can be reached three different ways — by counting production, by counting income, or by counting expenditure — and that the equality of these three routes is not a theorem to be proved but an identity built into the accounts. From the expenditure route we read off the famous decomposition of demand into consumption, investment, government purchases, and net exports; from the income route, the chain of refinements that runs down from GDP to disposable personal income; and from the financial market, the saving–investment identity that ties household thrift, the government deficit, and the trade balance together. We close with two practical matters: how output is sorted into industries, and how analysts guess at GDP between official releases.

A standing warning frames the whole enterprise. No single number summarizes an economy, and GDP in particular was designed to be *measurable*, not to be a complete measure of welfare. Keeping straight what it does and does not count is half the battle.

2.1 Gross Domestic Product

Definition 2.1: Gross Domestic Product

Gross domestic product (GDP) is the market value of all final goods and services produced within the geographic borders of an economy during a given period.

Read slowly, that definition contains five qualifiers, each of which excludes something. We take them one at a time.

2.1.1 The Five Qualifiers

(1) A given period — GDP is a flow. GDP is always reported for a span of time, normally a quarter or a year; every country in the world publishes annual and quarterly figures. Because it is measured *over an interval*, GDP is a *flow*, not a *stock*. A flow is a quantity measured per unit of time; a stock is a quantity measured at a point in time. The distinction is not pedantry — it is the single most common source of confusion in macroeconomics, and one should form the habit of asking of every macro variable whether it is a stock or a flow.

Wealth, for instance, is a stock: at this instant the economy holds a definite total of assets. GDP is a flow, and the two are not the same thing. When Chinese GDP overtook Japanese GDP around 2010, it meant only that from that year onward China produced more goods and services each year than Japan did; it did not mean that China's accumulated *wealth* had overtaken Japan's, which very plausibly had not.

Remark (Income versus wealth, and why finance nets out).

It helps to keep three related flows and stocks apart. *Income* (a flow) is labor income — wages, bonuses — plus capital income — returns *on* assets, such as dividends and interest, but *not* the capital stock itself. *Wealth* (a stock) is the sum of real assets and financial assets. Macroeconomics is principally about income; wealth is more the concern of the individual. Note too that financial assets are not net wealth for society as a whole: every financial asset is some agent's claim on another agent, and summed across the economy these claims cancel. This is why macro accounting can ignore them and still add up.

Remark (When monthly GDP appears).

There is normally no monthly GDP, because the data cannot be cleaned in time. But in special circumstances the authorities estimate it: during the 2022 pandemic disruptions in China, monthly figures were tracked closely to set growth targets and steady expectations — roughly -2.2% in April, -0.1% in May, and $+4\%$ in June 2022, summing to about $+0.5\%$ for the second quarter. In China, a preliminary income-method estimate is released around the 16th–20th of the month; monthly real-economy data are largely *nominal*, since there is no time to deflate them. The full expenditure-method annual accounts, which include the shares of household and government consumption, appear the following May or June.

(2) Market value. Because GDP adds up apples and haircuts and software, it must value everything in a common unit, and that unit is the market price. The reliance on market prices has an immediate corollary: things that are never traded in a market have no observed price and so are, in principle, hard to count — subsistence production, household chores, and illegal transactions among them. What matters is that the good or service *can and does* change hands at a market price, not that a transaction has literally occurred for each unit (inventories, discussed below, are produced output that is counted before it is sold).

(3) Geographic scope — “domestic.” GDP is defined by *territory*. China's GDP covers production within China's borders, whether the producer is a Chinese national or

a foreigner. This is the natural choice in a globalized economy and matches the way local policy is tied to local conditions; we contrast it with the national measure (GNP) below.

(4) Final, not intermediate, goods. Whether a good is *final* or *intermediate* depends on its use. A good or service used up as an input into further production is an intermediate good; a final good is one bought for consumption or investment. GDP counts only final goods. The reason is to avoid *double counting*: if we added the flour and the cake, the flour would be counted twice. Note that the very same physical good can be either, depending on use — flour bought by a household to bake a cake at home is a final good, while flour bought by a bakery to bake cakes for sale is an intermediate good.

(5) Marketed or imputed production. GDP counts what is produced and sold at market, plus a few things that are not literally sold but *should* be valued as if they were, by imputation. The leading example is the housing services that owner-occupiers “buy” from themselves: an imputed rent is added to GDP. (There is no contradiction between counting the purchase price of a house and counting the flow of housing services it later yields — they are different things, as we explain under investment.)

2.1.2 What Counts and What Does Not

The five qualifiers settle most cases, but the boundary is fiddly enough that it is worth collecting the standard rulings in one place. The organizing principle is simple: *GDP counts newly created value, once.*

Remark (Edge cases in what GDP counts).

- *Nothing newly created, nothing counted.* Government taxes and transfer payments are not in GDP — they merely redistribute existing funds. (The *wages* of government employees *are* in GDP, as payment for services produced.) Stock trades are not in GDP, being transfers of ownership with no new output; the stamp duty on a trade is not in GDP, being a transfer to the government; but a *commission* is in GDP, because brokerage is a service produced.
- *Only the newly created part.* The full sale price of a new house is not all GDP, because part of it is the value of the land. Only the value of the newly constructed building counts. The taxes paid to the government on purchase do not count.
- *Inventories count as investment.* Goods produced this year but not yet sold are this year’s output, whenever they eventually trade. In expenditure terms, when an inventory good is later sold, consumption (or investment) spending rises while the inventory falls by the same amount, and the two offset.
- *Imputed values.* Output not sold in a market but conceptually marketable is imputed and counted — owner-occupied housing services being the standard case.
- *R&D is now investment.* Research and development spending, once treated as an intermediate input to production, is now counted as investment.
- *Services are valued at price, not quality.* A service enters at its price regardless of how good it was.

- *Defensive expenditure inflates GDP.* Activity that merely offsets a negative externality is still counted, which overstates true output and welfare. If a firm pollutes while producing, its output adds to GDP, and the later cleanup adds to GDP again — GDP is inflated twice over.

The last point is the sharpest reminder that GDP is a production tally, not a welfare index. The definition was chosen partly for its meaning and partly for its measurability, and as a result it deliberately leaves a great deal of value uncounted.

2.1.3 From Domestic to National: GNP and GNI

The geographic qualifier has a natural cousin. Where GDP draws its boundary around a *territory*, gross national product draws it around a *people*.

Definition 2.2: Gross National Product

Gross national product (GNP) is the market value of all final goods and services produced by the *nationals* of a country during a given period, wherever in the world they produce them.

The two measures differ only in who is included, and they are linked by a clean adjustment:

$$\text{GNP} = \text{GDP} + (\text{production by nationals abroad}) - (\text{production by foreigners at home}).$$

For a large economy the gap between GDP and GNP is small. The World Bank uses per-capita GNP — now reported as gross national income (GNI) — to gauge how rich a country's people are, classifying economies as low-, middle-, or high-income; China is currently in the upper-middle-income band.

2.2 The Circular Flow and the Fundamental Identity

Why should the various ways of measuring output give the same answer? The reason is that one agent's expenditure is another agent's income, and the economy's payments chase each other around a closed loop. That loop is the *circular flow*, sketched in Figure 2.1.

The analysis centers on the *household*. Households are the hinge of the whole system because they are responsible both for income and for spending: broadly, the household is the supplier on the factor market and the demander on the goods market. Income flows to households and is then split among consumption C , saving S , and taxes T , ultimately reappearing as the demands C , I , G , and net exports.

Three markets organize the picture. The *goods market* links the main macroeconomic agents — firms, households, the government, and the rest of the world. Firms supply this market; the demand comes from households, the government, and abroad. The *factor market* closes the loop: the revenue firms earn on the goods market must flow back to pay the factors of production, and the suppliers of those factors — earning the corresponding income — are the households. The *financial market* ties saving to investment, lets the

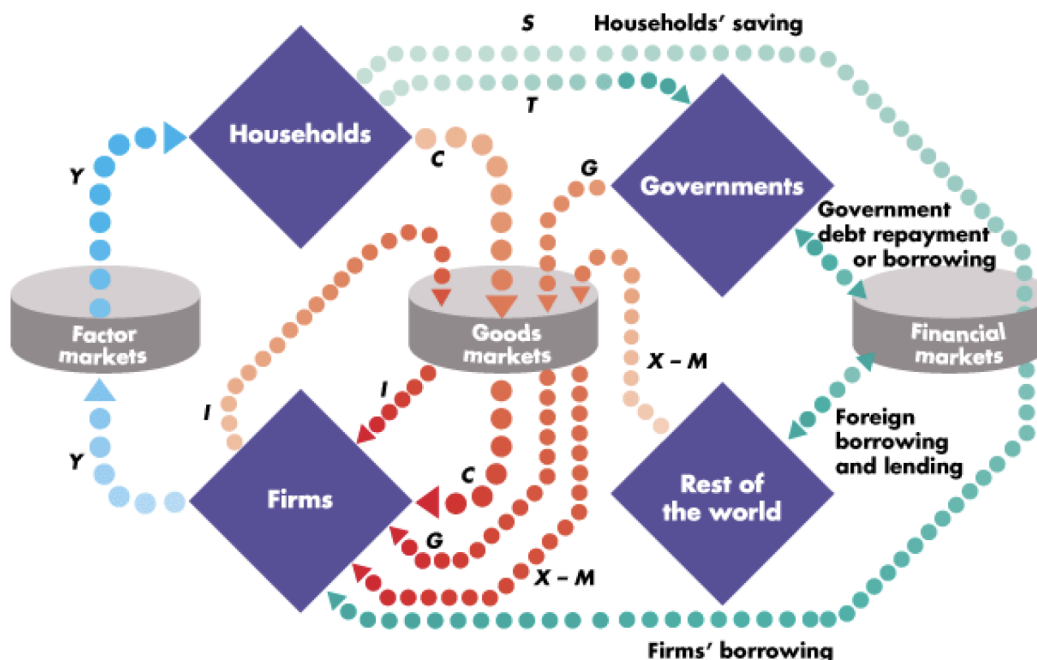


Figure 2.1: The circular flow of income: households, firms, the government, and the rest of the world transact across the goods market, the factor market, and the financial market, so that aggregate income equals aggregate expenditure.

economy lend to foreigners so that net exports can occur, and is closely bound up with the government's finances as well.

Because income and expenditure are the same payments seen from two sides, aggregate income (AI) equals aggregate expenditure (AE):

$$AI = Y = C + I + G + (X - M) = AE. \quad (2.1)$$

The identity (2.1) holds *ex post*: realized income equals realized expenditure by construction. Aggregate demand is the *ex ante* notion — the planned, intended spending built up from desired $C + I + G + NX$ — and short-run fluctuations arise precisely when planned demand differs from realized output. The decomposition $C + I + G + NX = AD$ is therefore a short-run object, constantly fluctuating, and its components are sorted by the *agent* doing the spending.

2.2.1 The Three Drivers of Growth

Sorting spending by agent runs into trouble in an economy like China's, where the state invests heavily and ownership is often blurred — many state-owned enterprises operate under mixed or even private ownership, so it is frequently impossible to say whether a given outlay is public or private. The fix is to sort government spending G not by *who* spends but by the *nature* of the spending, reassigning each piece of G to either consumption or investment. Government *consumption* folds into C , government *investment* into I . (In principle I then splits into private and public investment, but the statistics cannot separate

them.) Sorting by nature collapses (2.1) into

$$\text{GDP} = C + I + \text{NX}. \quad (2.2)$$

These three terms are the celebrated “three horses” pulling growth: final consumption expenditure C , gross capital formation I , and net exports of goods and services $\text{NX} = X - M$. The first two together, $C + I$, are *domestic demand*; net exports are *external demand*; and one can compute the contribution of each to a year’s growth.

2.3 Three Routes to the Same Total

Equation (2.1) already hints that output can be counted in more than one way. There are three standard methods, and by construction they all arrive at the same GDP.

2.3.1 The Production (Value-Added) Method

Definition 2.3: Value Added

A firm’s *value added* is its sales revenue minus its spending on intermediate inputs (raw materials). GDP by the production method is the sum of value added across all productive units in the economy.

This method is the most direct unpacking of the definition: value added is exactly the new value a firm creates, and summing it avoids double counting the intermediate goods that passed through. Sales, intermediate purchases, value added, and GDP are all flows.

A subtlety worth stressing: the amount subtracted is *only* intermediate-input cost, *not* financial costs, labor costs, or selling costs such as wages and interest. The production method looks at *production* — the value created by productive units. In effect it treats productive units as the source of GDP and does not separately count the value households create by supplying factors; it captures the social production process as a whole. By contrast, summing *all* sales across the economy yields gross industrial-and-agricultural output, in which intermediate sales are double-counted; a value-added tax, as levied in China, strips that double counting out.

2.3.2 The Income Method

Definition 2.4: The Income Method

GDP by the *income method* is the sum of incomes earned by all agents in the economy, since value added accrues as income to someone.

The incomes that sum to GDP are wage income, interest income (the return on working capital), tax income, and *accounting* profit — where accounting profit includes both economic profit and *rent*. The rent term captures the fact that machinery and equipment used in one’s own production could instead have been rented out on the market; using it oneself is, in effect, producing and consuming the rental service. Inventories enter here too, understood as value created this year that earns its return next year.

2.3.3 The Expenditure Method

Definition 2.5: The Expenditure Method

GDP by the *expenditure method* is the sum of spending on final goods and services: $C + I + G + (X - M)$.

The focus is the goods market. Set the government and trade aside for a moment: every final good on the goods market is bought either by a household or by a firm. Machinery and equipment bought by firms as capital goods are investment spending I ; almost everything else is household consumption spending C — with the standard exception that a house purchase counts as investment, not consumption. Reintroducing the government and trade changes nothing essential; the focus stays on the goods market. By convention, goods bought by households — housing aside — are counted entirely in the period of purchase.

Three methods, one number

Production, income, and expenditure are three vantage points on the same circular flow. The value added in production accrues as income, and that income is spent. The three methods therefore measure the same GDP, and their equality is an accounting identity, not an empirical finding.

2.4 Financial-Market Accounting and the Saving–Investment Identity

The financial market connects the same four agents — households, the government, firms, and foreigners. Household *saving* S is the chief source of funds; firms draw on those funds to finance investment I ; the government, when it runs a deficit $(G - T)$, must borrow from the market; and foreigners, when they buy the country’s net exports $(X - M)$, must borrow as well. Following the GDP-accounting logic, treat saving S as the total “income” supplied to the financial market and $I + (G - T) + (X - M)$ as the total “use” of those funds.

We can derive the relationship cleanly. Income is used in exactly three ways — consumption, saving, and taxes — which is the financial-flow identity

$$Y = C + S + T. \quad (2.3)$$

Setting the uses of income (2.3) equal to the expenditure identity $Y = C + I + G + (X - M)$ and cancelling C gives $S + T = I + G + (X - M)$, that is,

$$S = I + (G - T) + (X - M). \quad (2.4)$$

This matches the intuition above: saving funds private investment, plus the government’s deficit, plus foreigners’ net borrowing to buy our exports. Solving instead for investment,

$$I = S + (T - G) + (M - X). \quad (2.5)$$

Investment thus draws on three pools of funds:

- private saving, S ;
- the government's budget surplus, $T - G$;
- net borrowing from foreigners, $M - X$.

Here S alone is *private saving*, and $S + (T - G)$ is *national saving*.

Remark (Reading the signs).

Two of the terms in (2.5) flip sign relative to (2.4), and it is worth seeing why. Taxes T are redistribution — a second round of allocation following a fairness principle, as with a progressive income tax. (Note that Chinese taxation is essentially a tax on wages, not on property.) The term $T - G$ says that the government raises funds by taxation and spends on the goods market; a surplus is released *into* the financial market, while a deficit is borrowed *from* it. The term $M - X$ says that when foreigners buy our net exports they must pay in our currency, which they obtain by borrowing from us, draining funds from the domestic financial market; hence $M - X = -(X - M)$, the mirror image of the foreign borrowing that appears in (2.4).

Remark (Saving in the macro sense).

Three cautions on saving. First, *national saving* is total income minus total consumption, and China's national saving rate is far higher than most countries'; saving and investment are very strongly positively correlated. Second, saving in the macro sense is not the everyday notion of putting money aside — it is funds that enter and circulate through the financial market. Third, foreign direct investment brings, more than money, advanced technology and management know-how.

2.5 Investment and Capital Formation

Among the three drivers, investment is the one most easily misunderstood, so it deserves its own treatment. The measured counterpart of investment in the accounts is *gross capital formation*, and its job is to track changes in the economy's stock of real capital.

Definition 2.6: Capital Law of Motion

Capital is a stock that depreciates and is replenished by the flow of investment. Writing K_t for the capital stock, I_t for gross investment, and δ for the depreciation rate,

$$\Delta K = K_2 - K_1 = I_1 - \delta K_1, \quad \text{equivalently} \quad K_2 = (1 - \delta) K_1 + I_1.$$

This is investment in the true sense, and it has two components: gross fixed capital formation and the change in inventories (the latter a small share of the total). The distinction between investment and mere asset trading is essential. Apart from the change in inventories, only outlays that *form capital* — meaning physical, real capital — are investment in the

macroeconomic sense. Buying a share of stock, by contrast, transfers ownership of a financial asset; the asset's price may change in the process, but no new capital good is produced.

Remark (Fixed-asset investment versus capital formation).

Two related statistics should not be confused. *Fixed-asset investment*, an accounting category, includes spending on land, on used equipment, and on used buildings — but these are *not* in gross capital formation, since they do not create new capital. Below-scale projects, and intangible investments such as computer software, are also excluded. Land-purchase fees (from land auctions) are in fact a primary revenue source for local governments. As a result the gap between reported fixed-asset investment and gross capital formation is large. Cross-province fixed-asset investment, moreover, can be double-counted, so the provincial figures can sum to *more* than the national total. China's statistical bureau publishes fixed-asset investment as a cumulative year-to-date figure, since single-month values swing wildly; investment data are harder to collect than consumption data and involve much messier computation, and the January and February figures are reported jointly to absorb the timing of the Spring Festival. The three most important sub-indicators of fixed-asset investment are real estate, infrastructure, and manufacturing.

Remark (Housing is always investment).

Any house purchase, of any type, counts as investment, never as consumption — the holding period is long enough and the sum large enough that it has investment character. This matters for the macroeconomy: new-home sales in China have lately fallen by more than 20%, and the broader weakness of the economy is tightly linked to the real-estate downturn.

It is convenient at this point to fix some factor-price notation we will reuse. Let r denote the *capital rental price*, which is *not* the same concept as the interest rate, and let w denote the labor rental price (the wage). Output is produced from capital and labor through a production function $Y = F(K, L)$. The deeper contrast between raw materials and capital is one of durability: raw materials are consumed in one shot, whereas capital is a durable good whose “consumption” is precisely depreciation, and whose replenishment runs through the process of investment. Human capital shares this attribute — it too depreciates with the passage of time.

2.6 From GDP to Disposable Income

GDP is a *gross* measure: it includes the part of output that merely replaces capital worn out during the year. By the same token, gross investment I includes the part spent replacing depreciated capital. To move from the gross production total toward what people actually have to spend, we peel off depreciation and then make a series of further adjustments. The chain is mechanical but worth having in one place.

Definition 2.7: From GDP down to Disposable Personal Income

$$\begin{aligned}
 \text{NNP} &= \text{GDP} - \text{depreciation}, \\
 \text{National Income} &= \text{NNP} - \text{indirect business taxes}, \\
 \text{Personal Income} &= \text{National Income} - \text{corporate profits} - \text{net interest} \\
 &\quad + \text{dividends} + \text{government transfers} + \text{personal interest income}, \\
 \text{Disposable Personal Income} &= \text{Personal Income} - \text{personal taxes}.
 \end{aligned}$$

Net national product (NNP) strips depreciation out of the gross total; subtracting indirect business taxes gives national income, the total earned by factors of production; personal income adjusts national income for the income that is earned but not received (corporate retained earnings and net interest) and for the income received but not earned in production (transfers, dividends, personal interest); and removing personal taxes leaves disposable personal income, the amount households are free to consume or save.

2.7 The Government Budget and Fiscal Goals

The government's accounts enter the circular flow through taxes raised and purchases made, and the difference between the two is the budget balance.

Definition 2.8: Government Budget Balance

$$\text{Government Budget Balance} = T - G,$$

the excess of tax revenue T over government spending G ; it is a surplus when positive and a deficit when negative.

Fiscal policy can be pointed at two quite different goals. The first is a *balanced-budget* strategy — living within one's means, keeping $T - G$ at zero. The second takes *economic growth* as the central purpose of fiscal policy, accepting deficits when they serve growth.

Remark (State capacity and China's fiscal conservatism).

Which goal a state can pursue depends on its fiscal capacity, and history is instructive. England's Glorious Revolution and Bill of Rights sharply raised the state's ability to tax and borrow, helping clear the path for the Industrial Revolution. Late-imperial China, despite a more absolute concentration of power, taxed weakly and laid only a light per-capita burden, so its fiscal capacity — and therefore its military and defensive strength — was correspondingly weak. The modern Chinese state still carries something of that legacy: it keeps tight control over the deficit ratio and prefers debt as low as possible, a conservative fiscal tradition. Yet today's downward pressures make the ancient balanced-budget instinct ill-suited to circumstances; the appropriate strategy now is growth, not balance. China's reported ratio of fiscal activity to GDP is modest, but its *hidden* liabilities are large.

2.8 The Industrial Composition of GDP

Total output can be sorted not only by who spends it but by where it is produced. The standard partition is into three sectors.

Definition 2.9: The Three Sectors

- The *primary sector* comprises agriculture, forestry, animal husbandry, and fishing.
- The *secondary sector* comprises industry and construction.
- The *tertiary sector* is services.

Remark (Sector conventions and structural change).

Several practical notes. China classifies the *services* within agriculture, forestry, animal husbandry, and fishing as part of the primary sector, so the tertiary sector differs slightly from “services” as such, though the two are numerically close; agricultural goods cannot be used to balance trade. “Industry” within the secondary sector covers three broad categories: mining, manufacturing, and the production and supply of electricity, heat, gas, and water. Sector shares are usually read both as a share of GDP and as a share of total employment, and their movement is used to gauge a country’s structural transformation. China’s fastest-growing sector today is the tertiary, at roughly 60% of the economy.

A word of caution about reading these shares as policy targets. Structural change should be understood as a *result*, not a *cause*. One cannot copy another country’s admired sector mix onto one’s own economy and expect to copy its success. Manufacturing remains a powerful engine of growth; even if a larger service share is desirable, what is needed is *better* manufacturing, not *less* of it — just as the answer to environmental problems is not a romantic retreat to a pastoral economy and a single-minded cut in carbon, but rapid growth that lets the economy reduce carbon as it advances.

Remark (Consumption rate up, investment rate down).

China is now a consumption-driven economy: its consumption rate has risen and its investment rate has fallen. The temptation, when short-run growth disappoints, is to blame weak consumption and to stimulate it. But the lesson of reading structure as a result rather than a cause cuts the other way. Rapid growth has historically rested on rapid *investment*, not rapid consumption. China’s investment rate is in fact far higher than America’s (on the order of 40% versus 16%), and the claim that China over-invests is overstated: the problem is not too much investment but investment in the wrong places — investing *badly*, not investing too much. Equally, the weakness of consumption is not a problem of consumption *itself*: slowing consumption growth reflects problems with income and the social safety net, the main cause being a rapid decline in disposable income, and so cannot be cured by stimulating consumption alone.

2.9 Proxy Indicators for GDP

GDP is in a sense a theoretical construct, computed and released with a lag. Between releases, analysts lean on faster real-world series as side indicators. Each carries useful signal and characteristic noise.

1. *Electricity consumption and freight volume.* Both track physical activity, but both depend on the *structure* of the economy: it is entirely possible for electricity and freight growth to slow even as overall economic activity speeds up, if the activity shifts toward less energy- and transport-intensive sectors.
2. *New credit.* A monthly series, but tightly bound to the central bank's monetary policy, to macroprudential policy, and to the level of financial development; its movements need not reflect real activity at all.
3. *The purchasing managers' index (PMI).* A monthly, seasonally adjusted index, with 50% as the dividing line between expansion and contraction. The PMI is the first comprehensive macro reading available each month. Its *new-orders* component is a leading indicator that forecasts future production and correlates strongly with real-economy series; the central bank uses the PMI in its own forecasting.
4. *Retail sales of consumer goods.* Retail sales of physical goods plus food-and-beverage services. This is *not* the same as consumption C in the national accounts — services such as travel and education are excluded — but it is easier to collect, being gathered mainly from firms and merchants.

Remark (What retail sales hide, and durables versus perishables).

The single most important item within retail sales is the automobile, much as pork dominates the consumer price index. Consumption splits into *perishable goods* (one-off consumption) and *durable goods*; the longest-lived, highest-value purchase is typically the car, within which new-energy vehicles are the main driver — and which carry an element of investment. The pandemic sharply compressed the settings and means of consumption. Perishables and services cannot be deferred, but durable consumption can be postponed, sometimes producing compensatory spending later — weddings (jewelry) and automobiles being the leading cases. (Hence the “lipstick effect”: trading down to small luxuries in a downturn.)

Chapter 3

Prices, Real Output, and Inflation Indices

The national accounts of Chapter 2 deliver a single number for the value of everything an economy produces in a year. But that number is measured in money, and money is a slippery yardstick: a 10% rise in GDP could mean the economy produced 10% more goods, or that it produced exactly as much and merely charged 10% higher prices, or any mixture of the two. Before we can say whether a country is growing, or compare it with its own past or with another country, we must separate the part of a change in spending that reflects *more output* from the part that reflects *higher prices*. That separation is the business of this chapter.

We proceed in two steps. First we split nominal GDP into a real quantity and a price level, which gives us the *GDP deflator* and a clean decomposition of nominal growth into real growth plus inflation. With that in hand we can finally compare GDP across time and across countries, and we pause over the practical reporting conventions — year-on-year versus quarter-on-quarter, and the seasonally-adjusted annualized rate — that make published growth figures comparable. Second, we survey the family of price indices an economist actually consults: the consumer price index (CPI) and its “core” variant, the producer price index (PPI), and the personal consumption expenditures (PCE) index. The recurring theme is that there is no single “the” price level; each index answers a different question, and choosing the wrong one quietly distorts the answer. We close by comparing the CPI and the GDP deflator directly, because the differences between them — fixed versus changing weights, imported versus domestic goods — are exactly the kind of measurement subtlety that shows up in real policy debates.

A note on scope. This chapter is about *measuring* prices and inflation, not about *explaining* them. Why the overall price level moves, what inflation costs society, the “inflation tax” levied by a government that prints money, and the quantity theory that ties money growth to inflation are all deferred to Chapter 9. Here we only build the rulers; the theory of why the things being measured move comes later.

3.1 Nominal and Real GDP

A change in nominal GDP from one year to the next confounds two distinct events: quantities may have changed, and prices may have changed. To isolate quantities we recompute output holding prices fixed at the level of some chosen *base year*. This is the single idea behind the distinction between nominal and real GDP.

Definition 3.1: Nominal and Real GDP

Nominal GDP values each good at the prices prevailing in the year it was produced,

$$\text{Nominal GDP}_t = \sum_i p_{i,t} q_{i,t},$$

where $p_{i,t}$ and $q_{i,t}$ are the price and quantity of good i in year t . *Real GDP* values the same quantities at the fixed prices of a base year 0,

$$\text{Real GDP}_t = \sum_i p_{i,0} q_{i,t}.$$

Because real GDP holds prices constant, every change in it reflects a change in physical output rather than in prices.

Remark (Chinese data conventions).

China's National Bureau of Statistics reports the two series under names that translate as “current-price GDP” (*xianjia GDP*), which is nominal GDP, and “constant-price GDP” (*bubianjia GDP*), which is real GDP measured at a stated base year. The headline “GDP growth rate” and the published “GDP index” both refer to the growth of *real* GDP. When reading any such series the first question to ask is always: which base year, and is this a current-price or constant-price figure?

3.1.1 The GDP Deflator

If real GDP captures quantities, then whatever is left over when we compare nominal to real GDP must capture prices. That residual is the economy's broadest price index.

Definition 3.2: GDP Deflator

The *GDP deflator* is the ratio of nominal to real GDP, conventionally scaled by 100,

$$\text{GDP deflator}_t = \frac{\text{Nominal GDP}_t}{\text{Real GDP}_t} \times 100 = \frac{\sum_i p_{i,t} q_{i,t}}{\sum_i p_{i,0} q_{i,t}} \times 100.$$

It is a price index for *all* of the goods that enter GDP, weighted by the *current* year's quantities.

The deflator equals exactly 100 in the base year, where nominal and real GDP coincide. As long as prices have, on average, risen since the base year — the usual case in any economy with positive inflation — the numerator exceeds the denominator and the deflator sits above

100; before the base year it would lie below 100. A handy mnemonic is that the deflator is greater than 100 most of the time, precisely because the base year normally lies in the past.

3.1.2 Decomposing Nominal Growth

We can now make precise the loose statement that “nominal growth is real growth plus inflation.” Write g^N for the growth rate of nominal GDP, g^R for the growth rate of real GDP, and π for the growth rate of the GDP deflator (the deflator inflation rate). Since nominal GDP is the product of real GDP and the deflator, its gross growth factor is the product of the other two:

Theorem 3.3: Growth Decomposition

The gross growth factors satisfy

$$1 + g^N = (1 + g^R)(1 + \pi).$$

Expanding and dropping the second-order cross term $g^R\pi$, which is small when both rates are small, gives the convenient approximation

$$g^N \approx g^R + \pi.$$

Proof. By definition Nominal GDP = Real GDP $\times$ (deflator/100), so taking the ratio across two adjacent years,

$$1 + g^N = \frac{\text{Nominal GDP}_{t+1}}{\text{Nominal GDP}_t} = \frac{\text{Real GDP}_{t+1}}{\text{Real GDP}_t} \cdot \frac{\text{deflator}_{t+1}}{\text{deflator}_t} = (1 + g^R)(1 + \pi).$$

Multiplying out, $1 + g^N = 1 + g^R + \pi + g^R\pi$, so $g^N = g^R + \pi + g^R\pi$. For rates on the order of a few percent the product $g^R\pi$ is on the order of hundredths of a percent and is dropped. $\square$

The decomposition is exact in its multiplicative form and only approximate when written additively; the gap is the cross term, which matters only when growth or inflation is large. This is the same compounding logic that governs interest: once a base year is fixed, in normal times nominal GDP drifts ever further above real GDP, and the deflator drifts ever further above 100, because each year’s price increase compounds on the last.

3.2 Comparing GDP Across Time and Space

The reason we went to the trouble of stripping prices out of GDP is that we want to *compare*: this year against last, one country against another. Real GDP is what makes those comparisons meaningful.

3.2.1 Cross-Time and Cross-Country Comparison

To compare an economy with its own past we must remove the price factor, which is exactly what real GDP does: a rise in real GDP is a genuine increase in the volume of goods and

services, not an artifact of inflation. To compare across countries we additionally convert to a common currency (and, for welfare comparisons, often adjust for differences in price levels via purchasing power parity, a refinement we set aside here). GDP thus furnishes a common basis both for tracking growth over time and for ranking economies at a point in time.

It is worth keeping the limits of the measure in view even as we use it. GDP records the market value of production; it is silent on the distribution of that production, on wealth as opposed to income, on environmental degradation, on corruption, and on whether people are any happier for the activity it counts. None of this makes GDP useless — across countries and across decades it is robustly and positively associated with the things we do care about, from life expectancy to literacy. The right conclusion is not that GDP can be discarded or replaced by a single better number, but that human welfare is multidimensional and is best read off several indicators at once, with GDP among the most informative of them.

3.2.2 Year-on-Year, Quarter-on-Quarter, and the Annualized Rate

High-frequency data raise a reporting problem that annual figures hide. A given quarter's output can be compared either with the *same* quarter one year earlier (year-on-year growth) or with the *immediately preceding* quarter (quarter-on-quarter growth), and the two tell different stories.

- **Year-on-year** growth compares like quarter with like quarter, so it automatically nets out the regular seasonal pattern — the fact that, say, retail sales always spike around a holiday. Its drawback is that it is slow: because it averages over a full year of intervening change, it can be late to reveal a turn in the economy.
- **Quarter-on-quarter** growth is timely, registering a change as soon as it happens, but for the same reason it is contaminated by seasonality. To be usable it must be *seasonally adjusted*, with the regular within-year pattern statistically removed before the growth rate is computed.

Remark (Two series that should agree, but need not).

The two conventions are not independent — a consistent set of quarterly levels should reconcile the year-on-year and quarter-on-quarter growth rates — yet in practice published Chinese year-on-year and quarter-on-quarter GDP figures have at times failed to confirm one another, a reminder that the seasonal-adjustment step is itself an estimate and that high-frequency macro data should be read with some tolerance.

Several economies — the United States, Japan, and the euro area among them — report a *seasonally adjusted annualized rate* (SAAR). The idea is to take the current quarter's seasonally adjusted quarter-on-quarter growth and ask: *if* the economy kept growing at this quarterly pace for four consecutive quarters, what annual growth rate would result? Formally, if g is the quarter-on-quarter growth rate, the annualized rate is

$$\text{SAAR} = (1 + g)^4 - 1.$$

The annualized rate is timely and intuitive — it states a quarterly reading in the familiar units of an annual rate — but because it compounds a single quarter’s growth four times, it also magnifies that quarter’s noise, so the SAAR is considerably more volatile than year-on-year growth.

Remark (Smoothed data and the scar effect).

A statistical agency may deliberately “shave the peaks and fill the valleys” to make a reported series look smoother than the underlying economy. Smoothing the *output* series is one thing — GDP can absorb it — but the underlying shock to *income* is not so easily papered over: a genuine downturn cuts incomes permanently for those affected, and spending falls by even more, both because incomes are lower and because households respond to the shock with precautionary saving, durably shifting their consumption and saving habits. The lasting damage a sharp contraction leaves on the level of activity, long after the contraction itself has passed, is called a *scar effect*.

3.3 Price Indices in Practice

The GDP deflator is comprehensive, but it is published infrequently and with a lag, and it is not the index most closely watched for the cost of living or for the pressures most relevant to monetary policy. In practice economists consult a small family of price indices, each constructed for a different purpose. We take them in turn.

3.3.1 The Consumer Price Index

Definition 3.4: Consumer Price Index (CPI)

The *consumer price index* measures the cost of a *fixed basket* of goods and services representative of a typical household’s consumption. Each category of consumption is assigned a weight reflecting its share in the representative household’s budget, and the index tracks the cost of that fixed, weighted basket over time. It is, in short, a measure of the cost of living.

The CPI is built to answer one question: how much more (or less) does it cost this month to buy the same bundle of consumer goods that a typical family bought before? Because the basket and its weights are held fixed — in China they are revised only about once every five years, and the detailed category weights are not published, though they can be backed out indirectly from other released figures — the index isolates price changes from changes in what people buy.

Remark (What is in the basket — a China snapshot).

Food carries a large weight in the Chinese CPI; pork alone has historically been the single most heavily weighted commodity, and because pig farming follows a production cycle that is hard for policy to tame, pork prices swing widely and drag the index with them. The broader food-and-tobacco category sits around a quarter of the index. There is a general regularity here: the *smaller* the food share of consumption, the *higher* the standard of living — this is Engel’s law, that richer households spend a smaller fraction

of income on food and more on everything that raises the quality of life. Housing enters the CPI chiefly through *rents*: because owner-occupancy is high in China and buying a home is treated as investment rather than consumption, swings in house *prices* do not enter the CPI directly, so a rapid run-up or collapse in property prices need not show up in the index at all. For the same reason, stock and bond prices — assets, not consumption goods — are excluded.

A standard convention treats CPI inflation above 2% as inflation proper and inflation below 2% as disinflation shading into deflation. Its great practical virtue is frequency: the CPI is published monthly, which makes it far more useful for real-time analysis than the quarterly, lagged deflator.

Remark (The CPI is not the whole picture).

Relying on the CPI alone gives an incomplete reading of the economy. Through the mid-2000s the U.S. CPI was quiet even as house prices and subprime mortgage markets moved dramatically; the index simply did not see the build-up of financial risk that culminated in the 2008 crisis. The lesson drawn afterward was that consumer prices must be watched alongside asset prices — housing and financial assets — and that policymakers must attend to a wider set of macro indicators and to systemic risk, the stance now called *macroprudential* policy.

3.3.2 Cost-Push and Demand-Pull Inflation

Why prices rise matters as much as how fast, because the source of the pressure determines whether policy can do anything about it. It is useful to separate two channels.

Definition 3.5: Cost-Push and Demand-Pull Inflation

Cost-push inflation originates on the *supply* side: a rise in the cost of inputs — energy, food, raw materials — pushes prices up even when demand is unchanged. *Demand-pull inflation* originates on the *demand* side: stronger spending pulls prices up against a supply that adjusts only slowly.

The two have characteristic signatures in the data. Food and energy prices are driven largely by supply conditions — harvests, weather, oil markets — so a sharp rise in energy prices is the textbook case of cost-push inflation. Manufacturing output, by contrast, is produced under relatively stable supply conditions, so movements in manufactured-goods prices are typically traced to changes in demand, the textbook case of demand-pull inflation. The distinction is not merely descriptive: as we will see, it tells us which inflations monetary policy can hope to address.

3.3.3 Core CPI and the Logic of Monetary Policy

Definition 3.6: Core CPI

Core CPI is the consumer price index recomputed with the *food and energy* components removed from the basket. By stripping out the most volatile, supply-driven prices, it tracks the underlying trend in prices more smoothly than the headline CPI.

The motivation for core CPI is precisely the cost-push/demand-pull distinction. Food and energy are the prices most subject to supply shocks — the cost-push component — and they are also the most volatile. Removing them leaves an index that fluctuates less and that, by construction, reflects mostly demand-side pressure rather than transient supply shocks.

This is exactly what a central bank needs to see, and the reason is a structural one.

Why central banks watch core inflation

Monetary policy works through aggregate *demand*: easing or tightening shifts how much the economy wishes to spend. It does *not* act on the supply side — it cannot make oil cheaper or harvests larger. A central bank should therefore respond to the part of inflation it can actually influence, namely demand-pull inflation. Because core CPI nets out the supply-driven, cost-push component, it is the more reliable guide for monetary policy than the headline CPI.

The practical force of this principle is best seen when it is ignored, and the next two remarks record episodes in which the source of inflation was misdiagnosed.

Remark (China 2008: tightening into a supply shock).

An unusually cold winter in early 2008 killed large numbers of pigs in southern China; pork prices spiked and a single month's CPI reading reached around 8%. At that spring's session of the National People's Congress the leadership proposed tightening the supply of base money. But the inflation was overwhelmingly cost-push — a supply-side shock to pork, with underlying demand little changed — so monetary tightening was the wrong instrument and accomplished little. Worse, the U.S. financial crisis was just beginning to reach China, and with exports so important to the economy the prudent move would have been to prepare to *ease*. Growth slowed markedly in the second half of 2008, downward pressure mounted, and the autumn party conference duly called for loosening. A complete reversal of stance within a single year revealed how little forward-looking the policy framework was at the time.

Remark (China 2018 and the pork problem).

A familiar quip captured the 2018 surge in pork prices: “add pork and it's all inflation; take pork out and it's all deflation.” Monetary policy cannot touch the supply side, yet faced with a high *headline* number — and the public and market reactions it provokes — a central bank can feel forced to tighten anyway. The danger is that tightening into a supply-driven inflation produces *stagflation*: the inflation, being cost-push, is not cured, while the economy is pushed further down.

Remark (The U.S. supply-side inflation).

A contemporaneous U.S. episode made the same point. The inflation of the period was driven by energy shortages, a tight labor supply, and strained logistics — all supply-side problems. The tight labor supply, for instance, owed partly to a generous welfare system depressing labor-force participation, not to overheated demand. Demand-side monetary policy was poorly matched to a supply-side problem and risked aggravating inflation rather than resolving it.

3.3.4 The PCE Index**Definition 3.7: Personal Consumption Expenditures (PCE) Index**

The *personal consumption expenditures* price index is the U.S. Federal Reserve's preferred measure of consumer inflation. It is constructed much like the CPI and moves broadly with it, but is computed separately by the Fed, which weights inflation heavily in its policy deliberations.

That the Fed maintains its own consumer-price measure, rather than simply taking the published CPI, signals how seriously a central bank takes the choice of price index. The PCE and CPI track each other closely but differ in coverage and weighting; the gap between them is small in normal times but real, and a central bank that targets inflation cares about which ruler it is holding.

3.3.5 The Producer Price Index**Definition 3.8: Producer Price Index (PPI)**

The *producer price index* measures prices at the *factory gate* — the prices producers receive for their output, rather than the retail prices consumers pay. The factory-gate price is, roughly, cost plus profit plus tax, and it is also the wholesale price a merchant pays to acquire goods for resale.

The CPI and the GDP deflator both look at retail prices, the prices buyers face at the point of final sale; the PPI looks one step upstream, at the prices producers charge. This makes it a different and complementary instrument. The PPI excludes agricultural products, is tied closely to demand and directly to firms' profits, and is correspondingly sensitive to the business cycle, which makes it informative about firms' investment decisions.

Its key threshold is zero. When the PPI is positive, output prices are rising and producers' margins are expanding; when it turns negative, producers' profits are squeezed, and the appetite to produce and invest contracts. This can become *self-reinforcing*: falling producer prices depress profits and investment, which weakens activity and pushes prices down further, drawing the economy into a vicious circle in the short run — or, with the signs reversed, a virtuous one.

Remark (A deflationary reading).

In one illustrative month — October 2022 — China’s CPI ran a little above 1% while the PPI had already turned negative. With consumer-price inflation weak and producer prices outright falling, both consumption and investment demand were soft, and the economy was sliding toward deflation. Read together, the two indices diagnosed the slowdown more clearly than either could alone.

3.4 CPI versus the GDP Deflator

We now have two broad price indices — the CPI and the GDP deflator — and it pays to compare them carefully, because they differ in three ways that matter, and because the comparison reveals a systematic bias in the CPI.

Fixed weights and substitution bias. The CPI is built on a *fixed* basket: it asks what it costs to buy the same quantities, year after year, regardless of how relative prices move. The GDP deflator, by contrast, weights goods by the *current* year’s quantities, which adjust as relative prices change. This difference has a direction. When a good becomes relatively more expensive, households substitute away from it toward cheaper alternatives, so the true cost of maintaining a given standard of living rises by less than the cost of the old, fixed basket. By holding quantities fixed, the CPI ignores this substitution and therefore *overstates* the rise in the cost of living — it suffers from *substitution bias*. The deflator, embedding the substitution through current-year weights, does not. To the same effect, the CPI is slow to credit the arrival of new products and the quality improvement of existing ones, both of which raise welfare without raising the cost of the fixed basket.

Imports. The two indices also differ in scope of origin. The GDP deflator covers *domestically produced* goods only — GDP, by definition, excludes imports. The CPI covers what consumers actually buy, which includes imported goods. A jump in the price of imported oil, for instance, raises the CPI directly but enters the GDP deflator only insofar as it affects the prices of domestic output.

Domestic scope of goods. Finally, the two indices cover different slices of the economy. The CPI tracks consumer goods alone. The GDP deflator tracks everything in GDP — not just consumption but also investment, government purchases, and net exports. Consumer prices are an important component of the deflator, but only a component; the deflator’s coverage is far broader. The two indices answer different questions, and in any given year they can diverge.

Reading the two indices

The CPI is a fixed-basket cost-of-living index, published monthly, that includes imports and tends to *overstate* inflation through substitution bias and the slow incorporation of new and improved goods. The GDP deflator is a current-weighted, domestic-only, all-of-GDP price index that incorporates substitution but is published less often and with a lag. Each is right for a different question: use the CPI for the cost of living and for timely readings, the deflator for the economy-wide price level

behind real GDP.

This chapter has equipped us with the rulers — nominal versus real magnitudes, the deflator, and the consumer- and producer-price indices — and with a working sense of when each is the right one. What these instruments measure is the overall price level and its rate of change, inflation. *Why* that level moves, what inflation does to borrowers and lenders, the revenue a government extracts by printing money, and the quantity-theory link between money and prices are the subject of Chapter 9; we now turn first, however, to the other great aggregate the accounts only hinted at — the labor market and unemployment.

Chapter 4

The Labor Market and Unemployment

Of all the numbers a macroeconomist watches, the unemployment rate is the one that travels furthest from the seminar room. A point of GDP growth is an abstraction; a person without work is not. In many democracies the unemployment rate is read as a verdict on the government, and a long list of policies — from interest-rate decisions to extensions of jobless benefits — is tied to it more or less explicitly. This chapter builds the apparatus needed to read that number honestly: how the working-age population is sorted into those who count as unemployed and those who do not, why the unemployment rate is almost meaningless without the participation rate beside it, and why a healthy economy has positive unemployment rather than none at all.

The plan is straightforward. We first partition the population, since the unemployment rate is a ratio and we must be clear about what goes into the numerator and the denominator. We then define the two headline ratios — the unemployment rate and the labor-force participation rate — and explain why they must always be read together. With the accounting in place, we classify unemployment into three types according to its cause: *frictional*, *structural*, and *cyclical*. That classification delivers the chapter’s central concept, the *natural rate of unemployment*, which in turn defines what economists mean by “full employment” and by an economy’s *potential output*. Throughout, we keep an eye on the institutional and data-quality issues that make these numbers harder to interpret in practice than in theory, especially in the Chinese context.

4.1 Partitioning the Population

To turn “unemployment” into a measurable quantity we must first decide who is even a candidate for being unemployed. The official statistics build the partition in two stages, asking two questions in turn: *can* this person work, and, if so, *do they want to*?

The first cut is by **ability to work**. Some members of the population are outside the labor market not by choice but by circumstance or rule: children, the institutionalized, full-time students, active-duty military personnel, and the incarcerated. These groups are excluded at the first stage regardless of their preferences, because they are not free to take

a market job.

Among those who are able to work, the second cut is by **willingness to work**. A person who is able to work and wants to work is counted in the *labor force*; a person who is able to work but does not want to work — a homemaker who has chosen to stay out of paid employment, an early retiree living off savings — is counted as *not in the labor force*. The labor force is thus the intersection of the two conditions: able *and* willing.

Definition 4.1: Labor Force and Its Components

Among the working-age, non-institutionalized population:

- The *labor force* is everyone who is both *able* to work and *willing* to work. It splits into the *employed* and the *unemployed*.
- The *unemployed* are those who are able and willing to work — actively seeking a job — but are not currently working.
- Those *not in the labor force* are able to work but unwilling to work at prevailing conditions. This group includes homemakers and others who produce value without producing measured market output, as well as *discouraged workers*.

Two features of this partition deserve emphasis because they are exactly where the official count can mislead.

First, the unemployed are not simply “people without a job.” A retiree and a toddler also lack jobs, but neither is unemployed: the retiree is unwilling, the toddler unable. To be counted as unemployed one must clear both bars — able and willing — and still be without work. Unemployment is the gap between the desire to work and the chance to do so.

Second, *discouraged workers* sit on the boundary and reveal a subtle weakness in the count. A discouraged worker is someone who would take a job but has given up actively searching, convinced that none is available. By the willingness criterion as the surveys apply it — which keys on active search — discouraged workers are classified as out of the labor force, not as unemployed. They therefore vanish from the numerator and the denominator of the unemployment rate alike. When a downturn pushes many job-seekers into discouragement, the measured unemployment rate can fall even though the labor market has worsened, precisely because the discouraged have been reclassified out of the labor force. This is one reason the unemployment rate cannot be read in isolation, a point we return to below.

Remark (Two ways to count the jobless, and why China switched).

For a long time China relied on the *registered* urban unemployment rate, which counted only those who walked into a local labor bureau and registered as unemployed to claim benefits. Because registration was incomplete and incentive-driven, this measure was a poor gauge of actual labor-market slack. China later began publishing a monthly *surveyed* urban unemployment rate, built from household surveys in the manner of international practice, which improved both the reliability and the cross-country comparability of the data. The broader caveat is worth stating plainly: Chinese labor-market data have historically been weaker than the corresponding output and price data, and employment

targets in official plans tend to be set conservatively (low), so that they are reliably met. Within the employment data, two groups — factory workers and new university graduates — are the most politically sensitive and the most closely watched.

Remark (Unemployment and the political cycle).

The weight placed on unemployment differs sharply across economies. In many Western democracies the unemployment rate is a central electoral and policy variable, and a great deal of macroeconomic policy is pegged to it. China's policy framework reacts far less directly to the unemployment number itself, leaning instead on other large real-economy indicators; but this does not make employment unimportant there, since labor-market conditions are tightly bound up with every other macroeconomic question. The difference is one of which dial the policymaker watches, not of whether employment matters.

4.2 The Two Headline Ratios

With the partition fixed, the two rates that summarize it are simple ratios. The first measures slack *within* the labor force; the second measures how large the labor force is relative to the population that could in principle supply it.

Definition 4.2: Unemployment Rate

The *unemployment rate* is the share of the labor force that is unemployed,

$$u = \frac{\text{unemployed}}{\text{labor force}} = \frac{\text{unemployed}}{\text{unemployed} + \text{employed}}.$$

Its denominator is the labor force, *not* the whole population: people who are unwilling or unable to work are excluded from both top and bottom.

Definition 4.3: Labor-Force Participation Rate

The *labor-force participation rate* is the share of the able-to-work population that is actually in the labor force,

$$\text{LFPR} = \frac{\text{labor force}}{\text{working-age population}}.$$

It measures, among those capable of working, the fraction who are *willing* to do so.

The two rates answer different questions, and each is blind to what the other sees. The unemployment rate looks only inside the labor force; it says nothing about how many able people have chosen to stay outside it. The participation rate counts who is inside the labor force at all; it says nothing about how those inside are faring. Reported alone, either can be read in opposite ways.

Always read the unemployment rate beside the participation rate

A falling unemployment rate is good news only if participation is holding up. If people leave the labor force — retiring early, returning to school, or simply giving up the search and becoming discouraged workers — both the numerator and the denominator of u shrink, and the unemployment rate can fall even as the labor market deteriorates. *A low unemployment rate does not by itself imply an ample supply of labor.*

The clearest illustration is an economy with generous transfers and benefits. When staying out of work is comfortable, many able workers choose not to participate; the participation rate falls, and because the discouraged and the voluntarily idle are not in the labor force, the measured unemployment rate falls too. Yet the result is not a booming labor market but a shrinking one — the United States has at times shown exactly this pattern, a declining participation rate alongside a low unemployment rate and complaints of labor shortage. The unemployment rate looked strong; the participation rate revealed that the pool of available labor had thinned.

4.2.1 Nonfarm Payrolls

Survey-based rates are not the only labor-market data that move markets. In the United States the most closely watched single release is *nonfarm payrolls* (NFP), the monthly change in the number of paid jobs outside the agricultural sector.

Definition 4.4: Nonfarm Payrolls (NFP)

Nonfarm payrolls count the number of paid employees in the economy excluding the agricultural sector. The headline figure is the month-over-month change, interpreted as net job creation (or destruction).

Agriculture is stripped out deliberately. Farm employment is highly seasonal and volatile, swinging with the planting and harvest calendar in ways that say little about the underlying health of the labor market. Removing it leaves a cleaner read on how many jobs the economy is genuinely creating or shedding. Because NFP is a timely and direct measure of job creation, a release that deviates sharply from expectations can deliver a large shock to financial markets.

4.3 Why a Market Can Fail to Clear for People

We now turn from measurement to mechanism: *why* is there unemployment at all? In an idealized competitive market, a surplus drives the price down until quantity supplied equals quantity demanded, and nothing is left unsold. The persistent existence of unemployment tells us that the labor market does not clear so cleanly. As an organizing principle: *the presence of unemployment is a sign that there is friction in the market.* The three types of unemployment are three different kinds of friction.

4.3.1 Frictional Unemployment

The first kind of friction is simply that matching a worker to a job takes time. *Frictional unemployment* is the unemployment that arises while people search for and sort into positions, even in an otherwise healthy economy.

Definition 4.5: Frictional Unemployment

Frictional unemployment is unemployment arising from the time it takes to match workers with jobs. Workers lose or leave jobs for reasons unrelated to any deficiency in their ability, and finding the right new match — the right firm, occupation, or location — takes time to search and negotiate.

Some frictional unemployment is individual: a worker between jobs is in a matching spell, sampling offers until a suitable one appears. Some of it is sectoral: when an industry restructures, technology shifts, or economic activity migrates from one region to another, workers must move across firms, occupations, or locations, and the reallocation cannot happen instantly. Frictional unemployment is, in this sense, the unavoidable byproduct of a dynamic economy in which jobs are continually created and destroyed. It is not a symptom of weak demand — workers lose jobs for reasons that have nothing to do with their competence — but a symptom of the irreducible fact that *search and matching take time*.

4.3.2 Structural Unemployment

The labor market is a peculiar market because its commodity is attached to a person, and people are not bid down to a market-clearing wage the way bushels of wheat are. *Structural unemployment* is the unemployment that persists because the wage is held above the level that would clear the market, so that the quantity of labor supplied exceeds the quantity demanded and some willing workers cannot find jobs. There are three classic sources.

Definition 4.6: Structural Unemployment

Structural unemployment is unemployment that persists because the wage is held above its market-clearing level, creating a chronic excess supply of labor. Its three standard sources are labor unions, minimum-wage legislation, and efficiency wages.

Unions and collective bargaining. A labor union bargains collectively on behalf of its members, and the chief item on the agenda is higher wages. Starting from a market that would otherwise clear at the equilibrium wage, a negotiated (or administratively imposed) wage above that level raises the pay of those who keep their jobs but reduces the number of jobs firms are willing to offer, leaving some workers unemployed. In effect the union protects its *insiders* — the members who remain employed at the higher wage — at the expense of the outsiders who are priced out of work.

The minimum wage. A legislated minimum wage is the government's version of the same mechanism. If it is set above the market-clearing wage for low-skill labor, it raises

the incomes of those who keep their jobs while pricing the least productive workers out of employment — a well-intentioned policy that can do harm. This is the populist trap: a measure that sounds like pure generosity to workers can, by ignoring how firms respond, leave the most vulnerable workers worse off. It raises a genuinely hard question — what level of wage floor or welfare support is reasonable? — to which market forces give one answer and political pressure another.

Efficiency wages. The third source is subtler, because here firms raise wages *voluntarily*. Under *efficiency-wage* theory, a firm facing imperfect information about its workers' effort may find it profitable to pay above the market-clearing wage in order to raise productivity: a wage premium reduces turnover, attracts better applicants, and gives existing workers something to lose, so they work harder and shirk less. The firm trades a higher wage bill for higher effort per worker. But the same premium that buys effort also means that the wage sits above the clearing level, so that, in aggregate, fewer workers are hired than would be at the competitive wage — and the surplus shows up as unemployment.

Remark (Structural unemployment is about people, not commodities).

All three sources — unions, the minimum wage, and efficiency wages — share a root cause: the labor market traffics in human beings, not anonymous goods. Wages are sticky downward, contracts and laws constrain them, fairness and morale shape effort, and information about who is working hard is imperfect. These “human” features are exactly what a frictionless commodity market lacks, and they are why the labor market does not simply clear.

4.3.3 Cyclical Unemployment

The first two types are properties of the economy's normal functioning. The third is a property of the business cycle. *Cyclical unemployment* is the unemployment that rises and falls with aggregate economic conditions: a recession contracts the demand for labor and throws workers out of jobs, while a boom does the reverse.

Definition 4.7: Cyclical Unemployment

Cyclical unemployment is the component of unemployment caused by fluctuations in aggregate demand over the business cycle. It is positive in recessions, when the demand for labor falls, and can be negative in booms, when the economy is overheating.

In a boom, cyclical unemployment can in effect turn negative: firms cannot find enough workers and employees may be pushed into overtime, working more than they would freely choose. This overheating, however, is generally regarded as temporary rather than a durable state of the economy.

4.4 The Natural Rate, Full Employment, and Potential Output

We can now assemble the three types into the chapter’s central organizing idea. Frictional and structural unemployment are *long-run* features: they reflect the permanent architecture of the labor market — search frictions, institutions, information — and they would persist even in an economy with no demand shocks at all. Cyclical unemployment, by contrast, is the *short-run* component, the part that comes and goes with booms and recessions. The natural rate isolates the long-run part.

Definition 4.8: Natural Rate of Unemployment

The *natural rate of unemployment* is the sum of frictional and structural unemployment:

$$u^* = u_{\text{frictional}} + u_{\text{structural}}.$$

It is the unemployment rate that prevails when cyclical unemployment is zero. Because it is driven by long-run, structural factors, the natural rate cannot be moved by short-run macroeconomic policy.

The natural rate is what is left when the business cycle is switched off. A useful way to see this: when *actual* GDP equals *potential* GDP, exogenous demand shocks are essentially zero, cyclical unemployment is zero, and the observed unemployment rate is exactly the natural rate — the sum of frictional and structural unemployment. Since the natural rate is set by long-run forces, short-run stabilization policy cannot lower it; the proper target of short-run policy is the *cyclical* component alone.

This sharpens what economists mean by “full employment,” a term that is easy to misread.

Full employment means zero *cyclical* unemployment, not zero unemployment

Full employment is the state in which cyclical unemployment is zero and the measured unemployment rate equals the natural rate u^* . It does *not* mean that the unemployment rate is zero. Even at full employment there is frictional unemployment (people are still between jobs) and structural unemployment (wages still sit above market-clearing in parts of the market). An economy with zero measured unemployment would not be healthy; it would be one with no room for workers to move between jobs at all.

The output counterpart of full employment is *potential GDP*.

Definition 4.9: Potential Output and the Potential Growth Rate

Potential GDP is the level of output the economy produces at full employment — that is, when actual output has been stripped of short-run shocks and unemployment sits at its natural rate. The *potential growth rate* is the economy's long-run growth rate once short-run fluctuations are removed, driven by capital accumulation and technological progress.

Remark (Potential output is a belief as much as a measurement).

Potential output and the potential growth rate are theoretical constructs: we never observe the economy with its shocks cleanly removed. In practice, an estimate of potential growth is closer to a *belief*, usually formed by summarizing past experience and projecting it forward. When a government announces, say, a target to double the size of the economy by a given year — which works out to roughly five percent average annual growth — it is asserting a view about potential growth, not reporting a measured fact. The natural rate of unemployment has the same status: it is real and consequential, but it is inferred, not read off a dial.

To summarize the timing, frictional and structural unemployment are the long-run components, fixed by the structure of the labor market and untouchable by short-run policy; cyclical unemployment is the short-run component, the only part that demand-management policy can hope to move. Full employment is the point where that short-run component has been driven to zero and only the natural rate remains.

4.5 Demographics and Participation: Two Asides

The participation rate raises questions that lead naturally into demography, and two observations from the Chinese experience are worth recording, even though neither fits cleanly into the unemployment accounting above.

Remark (Why Chinese female participation is so high).

Despite a long cultural tradition of women keeping house, China has among the highest female labor-force participation rates in the world. Two forces push in the same direction. The first is the level of *household income*: where a single earner cannot support a family, both partners must work simply to get by, and participation is not really a choice. The second is *technological change*: as production has shifted away from physical labor toward mental labor, the range of jobs and the productivity at which women can perform them have converged with men's, eroding the old occupational divide. As physical labor matters less and mental labor more, the historical premium on male strength fades.

The same shift toward mental labor reshapes fertility, and here the economics is worth writing down. Consider a household choosing consumption C and the number of children

N , with utility $U(C, N)$ and the budget constraint

$$C + NP = w(1 - tN),$$

where w is the wage, P is the goods cost of raising one child, and t is the time required to raise each child. The term $w(1 - tN)$ is the household's earnings net of the time withdrawn from work to raise N children: each child consumes a fraction t of the parent's working time, so each child carries an *opportunity cost* of wt in foregone wages on top of its direct cost P .

Remark (The quantity–quality tradeoff and the demographic transition).

Reading the budget constraint as the wage w rises traces out the demographic transition. In the early stages of industrialization, w grows rapidly while the time cost of children is still modest, so families respond to higher income by having *more* children. As w rises further, the time cost wt of each child — the wages a parent gives up — comes to dominate, and children become expensive in the one resource that scales with income: the parent's time. Households then shift toward fewer children, investing more heavily in the *human capital* of each; this is the classic *quantity–quality tradeoff*. The fall in fertility that follows is the demographic transition, and it implies that population aging is not an accident of policy but an essentially unavoidable stage of development for a society built on mental labor.

Chapter 5

The Solow Growth Model

Why is the average American roughly fifty times richer than the average person two centuries ago, and why are some countries today twenty times richer than others? Questions about the *long run* — about the slow drift of living standards over decades and generations — are the province of growth theory, and they matter enormously: it is sustained growth in output per person that lifts consumption, raises life expectancy, and, by enlarging the total economic pie, makes redistribution and social stability possible even where inequality is high. Business cycles, the short-run fluctuations we study later, are best understood as deviations around this rising trend; in a useful slogan, the cycle is growth plus shocks. To make sense of the cycle we first need a theory of the trend.

This chapter develops the first and simplest such theory, the *Solow growth model*. Before introducing it, we pause to set up the general framework that all of macroeconomics's dynamic models share: a timeline, a state variable that summarizes where the economy is, control variables that the economy chooses, and a law of motion that carries the state forward in time. The Solow model is the leanest possible instance of this framework. Its defining shortcut — and the single assumption that separates it from the neoclassical model of Chapter 6 — is that the saving rate is fixed exogenously rather than chosen by optimizing households. We will build the baseline model, find its steady state, ask which steady state is best (the golden rule), trace how the economy adjusts when policy changes the saving rate, and then extend the model to accommodate population growth and technical progress. The model will not explain everything, but it will explain a great deal, and it organizes the questions that the rest of growth theory tries to answer.

5.1 The Dynamic Framework of Macroeconomics

Microeconomics is largely static: a consumer chooses a bundle, a firm chooses a quantity, and we compare equilibria. Macroeconomic growth is irreducibly *dynamic*. The central object is not a single choice but a sequence of choices linked through time, and the device that links them is the accumulation of *capital*. Every dynamic macro model therefore begins by laying down a timeline.

Always set up a timeline

To analyze any macroeconomic growth problem, write down a timeline indexed by periods $t = 1, 2, \dots$. In each period the economy inherits a stock of capital, makes choices that produce output and determine how much capital it carries into the next period, and the cycle repeats.

The capital stock plays a special role. At the start of period t the stock K_t is fixed — it is whatever past investment has bequeathed — so from the vantage point of period t it is *given*, a number the economy cannot change today. A variable of this kind is called a *state variable*: it summarizes everything the economy needs to know about its past in order to decide what to do now. Against the given state, the economy chooses how to split current output between consumption and saving. These choices are the *control variables*: consumption C_t and saving S_t are free to be set, and once they are set, next period's capital stock is determined. The rule that determines next period's stock is the *law of motion of capital*, which tracks how the stock accumulates over time.

Definition 5.1: State and Control Variables

In a dynamic model, a *state variable* is a quantity that is predetermined at the start of a period and summarizes the relevant history of the economy — here the capital stock K_t . A *control variable* is a quantity the economy chooses freely within the period — here consumption C_t and saving S_t . The state constrains the controls; the controls determine the next state.

To make these ideas concrete, consider a two-period example stripped of households and firms, displaying only the accounting identities that govern any economy. Output is produced from capital, $Y_i = F(K_i)$; in the first period output is split between consumption and saving, $C_1 + S_1 = Y_1$; the saving is invested, $I_1 = S_1$; capital depreciates at rate δ and is augmented by investment, so $K_2 = (1 - \delta)K_1 + I_1$; and in the second period the household consumes its output plus the gross return on its saving, $C_2 = Y_2 + (1 + r)S_1$. Collecting these,

$$\begin{aligned} C_1 + S_1 &= Y_1, \\ C_2 &= Y_2 + (1 + r)S_1, \\ K_2 &= (1 - \delta)K_1 + I_1, \quad I_1 = S_1, \\ Y_i &= F(K_i). \end{aligned}$$

These are the laws by which the economy mechanically operates; they involve no optimization. This is precisely the level at which the Solow model works — it imposes a fixed rule for how output is split and then lets the accounting carry the economy forward. The neoclassical model of Chapter 6 will instead add microfoundations, deriving the consumption–saving split from a household that maximizes lifetime utility and from firms and markets that clear in general equilibrium. Concretely, the optimizing household solves

$$\max \sum_{t=1}^{\infty} \beta^{t-1} U(C_t),$$

subject to the same laws of motion. But whether or not we add optimization, the structure is the same: in each period K_t is the state, and C_t and S_t are the controls.

This structure has a deep consequence. Standing at period t with the state K_t in hand, each choice of C_t pins down S_t , which pins down K_{t+1} . The map from today's state to tomorrow's state can therefore be written as a single function,

$$K_{t+1} = g(K_t),$$

called the *policy function*. Following the policy function period after period traces out the entire optimal path of the economy. Two features of this map are worth naming. First, next period's state depends only on this period's state, not on the whole history of how the economy arrived there; the problem has the *Markov* property. Second, the decision problem faced at every date has exactly the same structure — only the inherited value of the state differs — so the problem is *recursive*. The infinite-horizon sequence of choices, daunting at first sight, collapses into one problem solved over and over with a different starting point. This recursivity is what makes dynamic macro tractable, and it underlies every growth model in this book.

5.2 The Solow Model: Assumptions

We now specialize the framework to Solow's economy. The model makes six maintained assumptions, which we gather in one place.

Assumption 5.2: The Solow Model

1. **Aggregate production function.** Output is produced from capital and labor, $Y = F(K, L)$. The labor force L is held fixed; the only quantity that moves is capital K .
2. **Constant returns to scale (CRS).** F is homogeneous of degree one: scaling both inputs by any factor λ scales output by λ .
3. **Diminishing marginal products and the Inada conditions.** Each input is productive but with diminishing returns,

$$F_K > 0, \quad F_L > 0, \quad F_{KK} < 0, \quad F_{LL} < 0,$$

and the marginal product becomes unbounded as an input vanishes,

$$\lim_{K \rightarrow 0} F_K(K, L) = \infty, \quad \lim_{L \rightarrow 0} F_L(K, L) = \infty.$$

4. **Closed economy with no government.** Net exports and government spending are zero, $NX = G = 0$.
5. **Constant, exogenous saving rate.** A fixed fraction $s \in (0, 1)$ of output is saved, $I = sY$, with s given from outside the model.
6. **Capital accumulation.** Capital depreciates at rate δ and is augmented by investment,

$$K' = (1 - \delta)K + I,$$

where a prime denotes the next period.

Two of these assumptions deserve comment before we use them.

Constant returns to scale (Assumption 2) is what lets us pass freely between the aggregate economy and a representative “per-worker” description. Under CRS a firm can be split or merged without changing its productivity, so we may equally regard society as one giant firm hiring L workers and K machines, or as a single representative worker operating $k := K/L$ units of capital. Setting $\lambda = 1/L$ in the homogeneity property,

$$\frac{Y}{L} = F\left(\frac{K}{L}, 1\right) =: f(k),$$

which defines the *intensive* (per-worker) production function $y = f(k)$. The Inada conditions (Assumption 3) guarantee an interior solution: because the marginal product of capital is huge when capital is scarce and tiny when capital is abundant, the economy is always pulled away from the corners $k = 0$ and $k = \infty$ toward a sensible interior steady state.

The closed-economy assumption (Assumption 4) lets us identify saving with investment. With $NX = G = 0$ the output identity reads $Y = C + I$, while income is split into consump-

tion and saving, $Y = C + S$; subtracting gives the fundamental closed-economy condition

$$S = I.$$

In per-worker terms, $y = c + i$, with $i = sy$ and $c = (1 - s)y$.

What makes Solow “Solow”

The model’s simplicity — and the one assumption that sets it apart from the neo-classical model — is that the *saving rate s is exogenous and constant*. Households do not choose how much to save by trading off present against future utility; a fixed share of income is mechanically saved. The neoclassical model of Chapter 6 replaces this with an optimal, utility-maximizing consumption–saving choice. In this precise sense Solow describes a planned economy with a hard-wired saving rule.

5.3 The Baseline Model: No Population Growth

We start with the simplest case, in which the labor force is constant. Because L does not change, the per-worker law of motion has the same form as the aggregate one. Dividing $K' = (1 - \delta)K + I$ by the constant L ,

$$k' = (1 - \delta)k + i.$$

The change in the per-worker capital stock from one period to the next is therefore

$$\Delta k = k' - k = i - \delta k = sy - \delta k = sf(k) - \delta k. \quad (5.1)$$

Equation (5.1) is the heart of the Solow model. It is the law of motion of capital per worker, and reading it is reading the whole model. Capital per worker rises when gross investment per worker $sf(k)$ exceeds the capital lost to depreciation δk , and falls when it falls short. The dynamics are entirely intertemporal: today’s split between consumption and investment determines tomorrow’s capital stock, which determines tomorrow’s output, and so on. This is the contrast with the static comparisons of microeconomics — in macro growth, the action is in the law of motion.

5.3.1 The Steady State

The economy settles down when capital per worker stops changing. Setting $\Delta k = 0$ in (5.1) gives the *steady-state* condition

$$sf(k^*) = \delta k^*. \quad (5.2)$$

Definition 5.3: Steady State

A *steady state* of the Solow economy is a level of capital per worker k^* at which gross investment exactly replaces depreciated capital, $sf(k^*) = \delta k^*$, so that $\Delta k = 0$. At the steady state every per-worker variable — k^* , $y^* = f(k^*)$, $c^* = (1 - s)f(k^*)$, $i^* = sf(k^*)$ — is constant over time.

The steady state is easiest to see graphically. Figure 5.1 plots three curves against capital per worker k : the production function $f(k)$, rising but concave by diminishing returns; the investment curve $sf(k)$, the same shape scaled down by s ; and the depreciation line δk , a ray through the origin. Because $sf(k)$ starts above δk near the origin (the Inada condition makes investment's slope infinite there) and is eventually overtaken by the straight line (diminishing returns flatten f), the two cross exactly once at k^* .

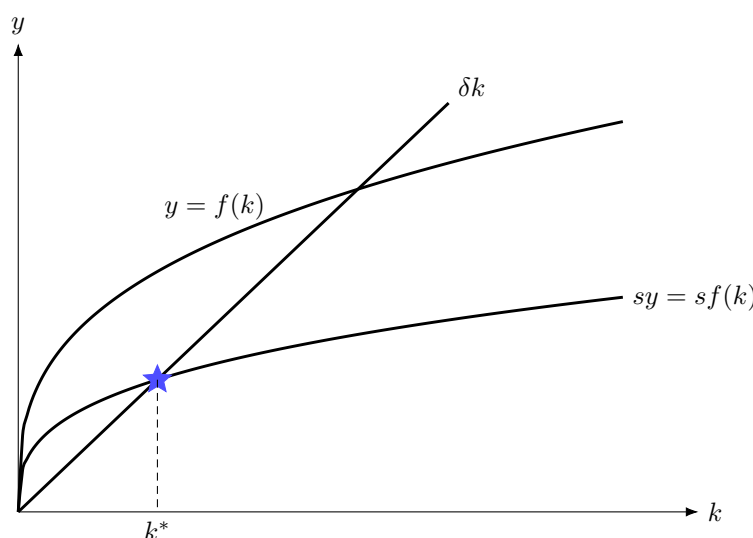


Figure 5.1: The baseline Solow diagram. The economy converges to the steady state k^* where the investment curve $sf(k)$ crosses the depreciation line δk ; to the left of k^* investment exceeds depreciation and k rises, to the right it falls.

The steady state is *stable*: whenever the economy is displaced from k^* , it returns. To the left of k^* , investment $sf(k)$ lies above depreciation δk , so $\Delta k > 0$ and capital grows toward k^* ; to the right, depreciation dominates, $\Delta k < 0$, and capital shrinks back toward k^* . The economics behind this convergence is the same diminishing returns that bent the production function: when capital is scarce, each additional machine is very productive and easily pays for its own depreciation, so the stock builds up; as capital accumulates, the marginal product falls until a new machine's output just covers the depreciation of the whole stock, at which point growth stops.

Growth from capital alone must stop

In the baseline Solow model, accumulation of capital per worker eventually ceases. Because the marginal product of capital diminishes while depreciation grows linearly,

there is a finite k^* at which the two balance, and the economy stalls there. Sustained capital deepening cannot, by itself, sustain growth.

This conclusion carries a famous corollary about convergence. Two economies with the same technology, saving rate, and depreciation rate share the same k^* and so converge to the same level of output per worker, regardless of where they started. A poor latecomer that begins with little capital grows fast at first — its scarce capital earns a high return — but slows as it approaches the common steady state. Early development is largely a story of capital accumulation, but precisely because that engine runs out, any *lasting* differences in income across countries must come from somewhere other than capital. The baseline model is thus too simple to explain the persistent gaps we observe in the world; identifying what it is missing will drive the extensions below.

5.4 The Golden Rule

A higher saving rate produces a higher steady-state capital stock k^* — but is a richer steady state a happier one? Welfare in this economy is measured by consumption, not capital, and there the answer is not obvious. Steady-state consumption per worker is

$$c^* = (1 - s)f(k^*) = f(k^*) - \delta k^*,$$

where the second equality uses the steady-state condition $sf(k^*) = \delta k^*$ to substitute investment for saving. Written this way, c^* is the vertical gap between the production function $f(k^*)$ and the depreciation line δk^* : of everything produced, what is not eaten by depreciation can be consumed.

Raising s has two opposing effects on this gap. On one hand it pushes the economy to a larger k^* , where the economy produces more, $f(k^*)$ is higher, and there is more to go around. On the other hand, sustaining a larger capital stock requires devoting more output to merely replacing depreciation, δk^* , which eats into consumption. Somewhere between the extremes — between saving nothing ($k^* = 0$, no output) and saving everything ($c^* = 0$, all output reinvested) — consumption is maximized. That best steady state is the *golden rule*.

Theorem 5.4: Golden-Rule Condition

Among all steady states, consumption per worker is maximized at the capital stock k_g^* satisfying

$$f'(k_g^*) = \delta.$$

The golden rule is the saving rate that delivers this steady state; combining the optimality condition with the steady-state condition $sf(k_g^*) = \delta k_g^*$,

$$s_g = \frac{\delta k_g^*}{f(k_g^*)}.$$

Proof. Steady-state consumption $c^*(k^*) = f(k^*) - \delta k^*$ is a function of the steady-state

capital stock that the saving rate selects. Maximizing over k^* ,

$$\frac{dc^*}{dk^*} = f'(k^*) - \delta = 0 \implies f'(k^*) = \delta.$$

Geometrically this is a tangency: consumption is the gap between $f(k^*)$ and δk^* , and the gap is widest where the slope of the production function equals the slope δ of the depreciation line. Concavity of f ensures this is a maximum. The expression for s_g then follows by reading the steady-state condition $sf(k_g^*) = \delta k_g^*$ at k_g^* . $\square$

Figure 5.2 makes the geometry explicit. Steady-state consumption is the vertical distance between $f(k^*)$ and δk^* , and this distance is largest at the point k_g^* where the production function runs parallel to the depreciation line — that is, where $f'(k_g^*) = \delta$, the marginal product of capital equals the depreciation rate.

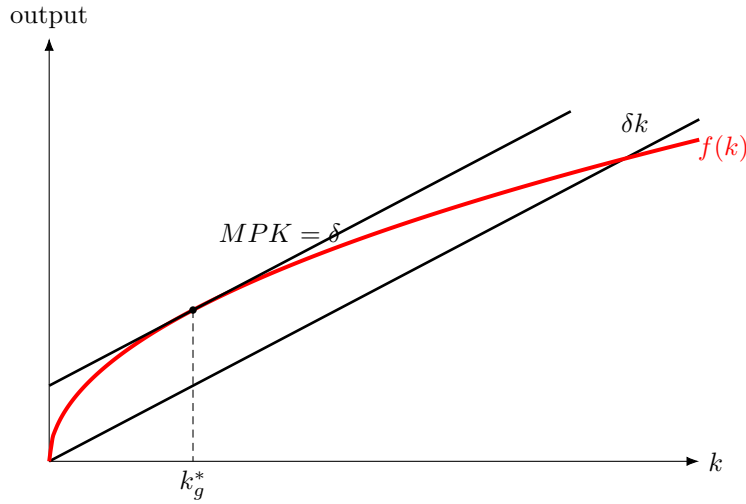


Figure 5.2: The golden rule. Steady-state consumption is the gap between output $f(k^*)$ and depreciation δk^* ; it is maximized at k_g^* , where the marginal product of capital equals the depreciation rate, $f'(k_g^*) = \delta$.

Remark ([]).

“Effective” depreciation] The δ in the golden-rule condition $f'(k_g^*) = \delta$ should be read as the *effective* rate at which capital per worker is worn down. In the baseline model this is just physical depreciation δ . Once we add population growth it becomes $\delta + n$, and once we add labor-augmenting technical progress it becomes $\delta + n + g$. The same condition — marginal product of capital equals effective depreciation — holds throughout; only what we plug in for “effective depreciation” changes.

5.4.1 The Golden Rule under Cobb–Douglas

The golden-rule saving rate takes a strikingly clean form when production is Cobb–Douglas. Let

$$Y = AK^\theta L^{1-\theta}, \quad 0 < \theta < 1,$$

so the intensive production function is $y = f(k) = Ak^\theta$.

Example (Golden-rule saving rate under Cobb–Douglas).

Show that with $Y = AK^\theta L^{1-\theta}$ the golden-rule saving rate is $s_g = \theta$.

Solution.

The intensive function is $f(k) = Ak^\theta$, with marginal product $f'(k) = \theta Ak^{\theta-1}$. The golden-rule condition $f'(k_g^*) = \delta$ gives

$$\theta A(k_g^*)^{\theta-1} = \delta \implies \delta = \theta \frac{A(k_g^*)^\theta}{k_g^*} = \theta \frac{f(k_g^*)}{k_g^*}.$$

Substituting into the saving-rate formula,

$$s_g = \frac{\delta k_g^*}{f(k_g^*)} = \frac{\theta f(k_g^*)/k_g^* \cdot k_g^*}{f(k_g^*)} = \theta.$$

The result $s_g = \theta$ is memorable: the golden-rule saving rate equals capital's share of output. In a competitive economy capital is paid its marginal product, so θ is exactly the fraction of national income that accrues to capital. The rule says, save and reinvest precisely the income that capital earns, no more and no less. Saving more than capital's share over-accumulates — the extra machines do not pay for their own upkeep — and saving less leaves the economy short of the consumption-maximizing stock. In this sense the golden rule asks that investment in capital be exactly cost-effective relative to capital's contribution to output.

Two comparative statics round out the picture. A higher depreciation rate δ lowers golden-rule consumption: the economy must set aside more output simply to maintain its capital, leaving less to enjoy. A more productive technology — a higher $f(k)$ at every k — raises golden-rule consumption, since the economy produces more efficiently; whether the accompanying golden-rule saving rate rises or falls, however, is ambiguous and depends on the shape of f .

5.5 Dynamic Adjustment after a Change in the Saving Rate

The steady state tells us where the economy ends up; it does not tell us how it gets there. Suppose a benevolent social planner judges the current saving rate to be too high — the economy has over-accumulated past the golden rule — and lowers it to s_g . How do consumption, investment, output, and capital respond over time? The answer turns entirely on a single distinction: which variables are *flows* and which are *stocks*.

Stocks adjust slowly, flows jump

A *flow* (a process measured per unit time, such as consumption c , investment i , or output y) can change discontinuously the instant a parameter changes. A *stock* (a quantity accumulated over time, such as capital k) cannot jump; it can only be built up or run down gradually through the flow of investment. Tracing any macroeco-

conomic adjustment comes down to keeping this distinction straight and knowing how each parameter acts.

At the moment the saving rate falls to s_g , capital per worker k is still at its old, higher value — a stock cannot change instantaneously. With k momentarily fixed, output $y = f(k)$ is also unchanged on impact. But the way that fixed output is divided changes at once: investment $i = s_g y$ *drops* immediately because s_g is smaller, and consumption $c = (1 - s_g)y$ correspondingly *jumps up* immediately. Both are flows, so both move on impact; and because output is momentarily unchanged, the fall in investment is exactly the rise in consumption.

After the jump, the slow adjustment begins. With the lower saving rate, investment now falls short of depreciation, $s_g f(k) < \delta k$, so $\Delta k < 0$ and capital per worker drifts down toward the new, lower steady state. As k falls, output $y = f(k)$ falls with it, following k closely because y is tied to k through the production function. As output falls, both consumption and investment — now fixed fractions of a shrinking output — gradually decline as well. Consumption, having leapt up on impact, eases back down toward its new steady-state level, which is nonetheless higher than before (that was the point of moving to the golden rule). Figure 5.3 shows the four time paths: the sharp discontinuous jumps in the flows i and c at the date of the policy change, the smooth slide of the stock k , and output y tracking k .

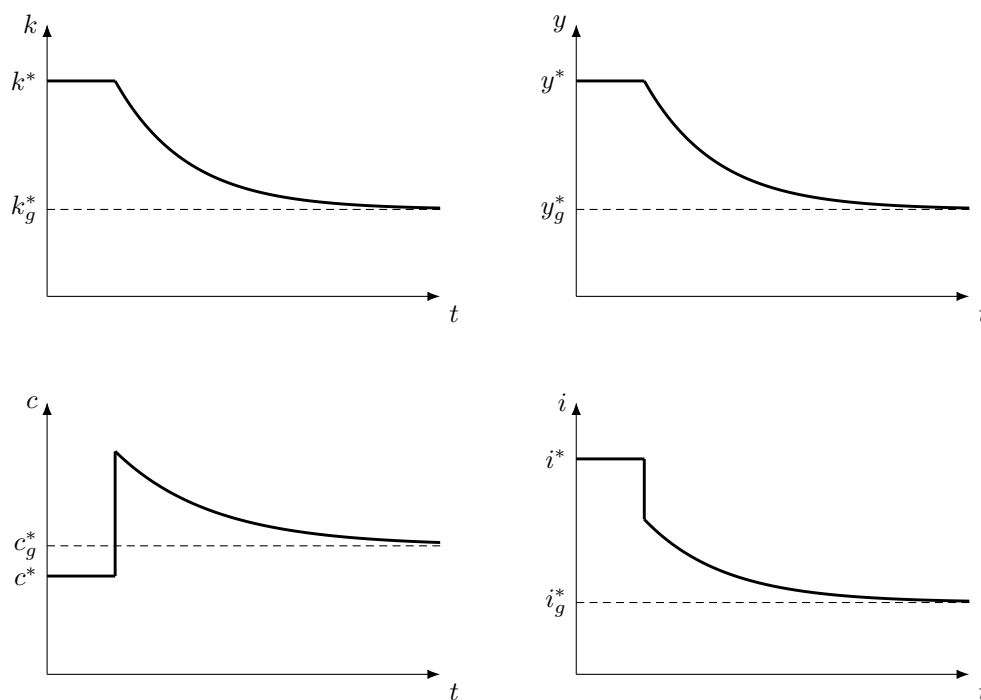


Figure 5.3: Time paths of capital k , output y , consumption c , and investment i after the saving rate is cut to s_g . The flows i and c jump discontinuously on impact (investment down, consumption up); the stock k then declines slowly, dragging y , c , and i down toward the new steady state.

One further feature of the path is worth naming: convergence to the new steady state is slow and decelerating. As k nears its new resting point the gap between investment and depreciation shrinks, so Δk shrinks, and the economy creeps toward the steady state ever

more gently. Reaching the new equilibrium is the work of many periods, not a single jump.

5.6 The Extended Solow Model

The baseline model holds the labor force fixed and assumes no technical change. Both are plainly counterfactual, and relaxing them is what lets the model speak to sustained growth. The guiding principle is the one stressed in the dynamic framework: a new factor enters only through the *law of motion*; the static relationships are unchanged. Think through the intuition before reaching for the algebra.

5.6.1 Population Growth

Let the labor force grow at a constant rate n ,

$$\frac{\Delta L}{L} = n.$$

The aggregate law of motion is untouched, $K' = (1 - \delta)K + I$. What changes is the translation into per-worker terms, because the worker count itself is now rising. Writing $k = K/L$ and dividing by next period's larger labor force,

$$k'(1 + n) = (1 - \delta)k + i.$$

Subtracting k and simplifying,

$$\begin{aligned} \Delta k &= k' - k = i - \delta k - nk' \\ &= i - \delta k - nk + n\left(\frac{K}{L} - \frac{K'}{L'}\right) \\ &\approx i - (\delta + n)k, \end{aligned}$$

where the term $n(K/L - K'/L')$ is a product of two small changes and is negligible to first order. We arrive at the central result for population growth,

$$\Delta k \approx sf(k) - (\delta + n)k. \quad (5.3)$$

The interpretation is immediate, and it is exactly what the intuition predicted. Compared with the baseline, the effective rate at which capital per worker is depleted rises from δ to $\delta + n$. Beyond replacing worn-out machines (the δk term), the economy must now also *equip the new workers* who join the labor force each period (the nk term), simply to keep capital per worker from falling. In effect, population growth acts like extra depreciation. The steady-state condition becomes $sf(k^*) = (\delta + n)k^*$, and the golden-rule condition becomes $f'(k_g^*) = \delta + n$.

Remark ([].)

Population growth, the Malthusian trap, and the limits of CRS] Because the return to capital is governed by diminishing marginal products while population growth keeps diluting the capital each worker has, faster population growth lowers steady-state capital

per worker and hence output per worker — everyone is poorer. Pushed to its extreme this is the logic of the *Malthusian trap* (Chapter 7), in which any gain in income is dissipated by a larger population. The mechanism rests squarely on the constant-returns assumption. In reality the clustering of people can generate positive externalities — ideas, specialization, dense markets — so that society may exhibit *increasing* returns. Moreover, fertility is a deliberate choice, and treating population growth as an endogenous decision rather than an exogenous parameter is the more satisfying modeling stance, one the Solow framework does not take.

5.6.2 Labor-Augmenting Technical Progress

To generate sustained growth in living standards we must add technical progress. Solow does so through a deliberately specific device: technology enters the production function as a multiplier on labor,

$$Y = F(K, E \cdot L),$$

where E measures the *efficiency of labor* and grows at a constant rate g . The product $E \cdot L$ is the supply of *effective workers*. We ordinarily picture technical progress as a rising total-factor productivity that improves the use of all inputs; modeling it as *labor-augmenting* is a tractable choice that lets the economy settle onto a clean balanced path. (If technology also augmented capital, it would amount to improving total-factor productivity directly.)

The law of motion is again unchanged, $K' = (1 - \delta)K + I$. Now we measure everything per *effective* worker, defining $k = K/(EL)$ and $i = I/(EL)$. Because effective labor grows at rate $n + g$ (population at n , efficiency at g), dividing the law of motion by next period's effective labor gives

$$k'(1 + n)(1 + g) = (1 - \delta)k + i.$$

Expanding and dropping the second-order term ng ,

$$k'(1 + n + g) = (1 - \delta)k + i,$$

which rearranges, by the same steps as before, to

$$\Delta k \approx sf(k) - (\delta + n + g)k. \quad (5.4)$$

The effective depreciation rate is now $\delta + n + g$: capital per effective worker is worn down by physical depreciation, diluted by population growth, and diluted again by the growth in labor efficiency. In steady state $\Delta k = 0$ gives

$$(\delta + n + g)k^* = sf(k^*) = i^*.$$

The crucial question is what this implies for actual living standards, which are measured per *person*, not per effective worker. In steady state $k^* = K/(EL)$ is constant, so capital and output *per person* are

$$\frac{K}{L} = k^*E = k^*E_0(1 + g)^t, \quad \frac{Y}{L} = y^*E = y^*E_0(1 + g)^t.$$

Although adding g to effective depreciation lowers the steady-state k^* (and hence y^*) mea-

sured in effective-worker units, the multiplicative factor $E_0(1+g)^t$ grows without bound and dominates completely. Output and capital per person therefore grow at the rate g forever. This is the model's central achievement: *labor-augmenting technical progress is the only source of permanent growth in income per person*. Meanwhile, aggregate output Y , being EL times its effective-worker value, grows at the rate $n+g$.

Remark ([].

Technology as a public good] Why does labor-augmenting technology raise everyone's income rather than commanding a factor payment of its own? The key, in the model, is that efficiency E is *non-rivalrous*: my using an idea does not stop you from using it. Capital and labor are rivalrous factors, traded in factor markets and earning a return; technology is not a factor in this sense but a public good whose benefits accrue to the whole workforce. One can picture E as a multiplier that lets society achieve, with L people, what would otherwise require $E \cdot L$ people; when we tally human welfare we collapse those E effective workers back into the L real people, and the gains show up as ever-rising income per person. To the extent that patents, secrecy, or other frictions make technology partly excludable in reality, outcomes will diverge from the frictionless model.

5.6.3 Steady-State Growth Rates: Summary

The three versions of the model differ only in what grows in the long run. The intensive variables (per effective worker) are constant in every steady state by construction; the difference is in the per-capita and aggregate growth rates, summarized in Table 5.1.

Table 5.1: Long-run growth rates of variables in the steady state across the three versions of the Solow model.

In steady state	Baseline	Population growth	Population & technical growth
Per effective worker	constant	constant	constant
Per capita	constant	constant	g
Aggregate	constant	n	$n+g$

The table reads off the discussion above. With no growth in either labor or technology, everything is flat. With population growth alone, output per person is still flat in the long run — the economy is bigger but no richer per head — while aggregate output grows at n . Only when labor-augmenting technical progress is present does income per person grow, at rate g , with the aggregate growing at $n+g$.

5.7 More on the Solow Model

The Solow framework, for all its simplicity, supports a rich set of further observations. We collect the most useful here.

5.7.1 Three Stages of Growth

Empirically, countries appear to pass through three broad stages, each governed by a different growth mechanism.

1. **The poverty trap** (GDP per capita below roughly \$3,000). Capital per worker is very low, so the marginal product of capital — and hence the return to investment — is very high. Economies in this stage can grow rapidly simply by accumulating capital, a process called *capital deepening*. Rapid catch-up growth of this kind, such as China's, is exactly what the baseline Solow model predicts for a capital-scarce economy, not an anomaly.
2. **The middle-income transition.** As capital deepens, diminishing returns set in and the easy growth from accumulation fades. To keep growing, the economy must shift from factor accumulation to innovation-driven growth. This transition is difficult — the failure to make it is the so-called *middle-income trap* — and navigating it typically requires sustained institutional reform.
3. **The high-income stage.** Here growth comes overwhelmingly from technological innovation, since further capital deepening yields little. This is the regime in which the g term, not the saving rate, drives living standards.

5.7.2 Growth Accounting and the Solow Residual

To measure where growth actually comes from, suppose output follows a Cobb–Douglas path with total-factor productivity z ,

$$Y = z K^\alpha L^{1-\alpha}.$$

Taking logs and differentiating yields the *growth-accounting* decomposition,

$$\frac{dY}{Y} = \frac{dz}{z} + \alpha \frac{dK}{K} + (1 - \alpha) \frac{dL}{L}. \quad (5.5)$$

Output growth is the sum of productivity growth and the contributions of capital and labor growth, each weighted by its income share. Historically $\alpha \approx 1/3$. In per-worker terms, $y = zk^\alpha$, and the decomposition collapses to

$$\frac{dy}{y} = \frac{dz}{z} + \alpha \frac{dk}{k}.$$

The term z is the *Solow residual*: the part of output growth that is left over after the measured contributions of capital and labor have been subtracted out. It is residual precisely because it is not observed directly but inferred from (5.5) as $dz/z = dY/Y - \alpha dK/K - (1 - \alpha) dL/L$.

Definition 5.5: Solow Residual

The *Solow residual* z (often called total-factor productivity) is the component of output not accounted for by the measured quantities of capital and labor. It captures everything that raises output given the inputs, including

- human capital (the education and skill embodied in workers);
- technological progress;
- externalities (public-good spillovers, the opposite of excludable, fee-charging goods);
- institutional quality (the organization of firms and managerial know-how, patent protection, control of corruption).

Note the contrast with the labor-augmenting E of the previous section: E improved only labor, whereas the residual z multiplies the whole production function and so raises the productivity of both capital and labor. A useful way to think about z is that it is itself an *outcome* — the equilibrium product of investments in education, research, and institutions — so one can analyze its level and growth with the same supply-and-demand logic used for any other equilibrium object. The policy levers that promote growth follow directly from the residual's contents: raise the saving rate, invest in human capital, encourage technological progress, and build the right institutions.

5.7.3 The Balanced Growth Path and Convergence

The steady state of the extended model is a *balanced growth path* (BGP): a trajectory along which output, capital, and consumption all grow at constant (and mutually consistent) rates. As with the baseline steady state, the BGP is an attractor — if the economy is knocked off it, the variables converge back to it over time.

Two productivity measures help locate an economy along the path. *Labor productivity* and *capital productivity* are

$$\frac{Y}{L} = z \left(\frac{K}{L} \right)^\alpha, \quad \frac{Y}{K} = z \left(\frac{K}{L} \right)^{\alpha-1}.$$

Remark ([]).

Labor productivity is not the wage] Labor productivity Y/L should not be confused with the average wage, because L here is the number of *employed* workers, not the whole population; unemployment, labor-force participation, and demographics drive a wedge between output per worker and income per person.

The Solow logic now delivers its sharpest lesson. Growth in the capital-labor ratio K/L is *not* sustainable — diminishing returns guarantee it peters out — whereas growth in the residual z *is* sustainable, and it raises both labor and capital productivity at once. Mature economies grow chiefly through rising total-factor productivity; the United States today, for instance, owes most of its growth to the residual rather than to capital deepening.

Finally, the model speaks to convergence with an important qualification. The unconditional convergence of the baseline model — everyone ends up equally rich — holds only when economies share the same fundamentals. Because institutions and the path of technology differ across countries, economies converge not to a single common steady state but to their *own* balanced growth paths. Countries that share similar institutions and technologies do converge to a common path, a phenomenon known as *conditional* or *club convergence*. The endogeneity of technical progress — the fact that g itself depends on policy, institutions, and research — is precisely what the Solow model leaves unexplained, and it is where later growth theory begins.

What Solow does and does not explain

The Solow model explains why capital-poor economies grow fast and then slow, why capital accumulation cannot sustain growth, and why long-run growth in living standards must come from technical progress. It does not explain where that technical progress comes from, nor why some countries' technology and institutions improve faster than others. It does not explain everything — but it explains a great deal, and it frames the questions the neoclassical model (Chapter 6) takes up by replacing Solow's mechanical saving rule with optimizing households.

Chapter 6

Microfoundations of Growth: The Neoclassical Model

The Solow model of Chapter 5 took us a long way: with a single accumulation equation it explained why poor countries grow faster than rich ones, why an economy converges to a steady state, and why long-run growth must rest on technological progress rather than on thrift alone. But it bought this reach at a price. The saving rate s was simply handed down from outside the model, a constant fixed by assumption. An economy in which households save a constant fraction of their income no matter what is, in effect, a *planned* economy: nobody is asked whether saving that much is in their own interest, and the model is silent on whether the resulting capital stock is too high, too low, or just right.

The neoclassical growth model repairs this gap by supplying the *microfoundations* Solow lacked. We populate the economy with optimizing agents—households that choose consumption and saving to maximize utility, and firms that choose inputs to maximize profit—and we require that markets clear. The saving rate then ceases to be a parameter we impose; it becomes the *outcome* of households trading off consumption today against consumption tomorrow, given the returns the market offers. Remarkably, when we work the optimization through, the long-run predictions look very much like Solow's: a stable steady state, convergence, and growth driven by technology. The difference is that now those predictions rest on individual choice, so we can ask welfare questions Solow could not even pose—is the competitive outcome efficient? could a planner do better?—and answer them.

We build the apparatus in stages. We start with a static general-equilibrium economy to fix the logic of household optimization, firm optimization, and market clearing in their simplest setting. We then add time: a two-period endowment economy introduces the Euler equation and consumption smoothing; an overlapping-generations economy turns that two-period logic into a genuine theory of capital accumulation with a law of motion and a steady state; and an infinite-horizon cake-eating problem and its productive cousin, the social planner's problem, complete the picture and let us state the two welfare theorems that tie competition and efficiency together.

6.1 The Static General-Equilibrium Model

We begin with no capital accumulation at all. In a single period the capital stock cannot change—investment takes time to become productive—so we hold K fixed and study how a representative household and a representative firm interact in two markets, for goods and for labor, within that period. This strips the problem down to its general-equilibrium core: optimizing agents on each side, prices that adjust until supply equals demand.

Assumption 6.1: The static neoclassical environment

- Agents are identical, so we may study a single *representative* household and a single *representative* firm.
- The household maximizes utility taking prices as given; the firm maximizes profit taking prices as given.
- Markets clear: in each market, demand equals supply.
- The capital stock K is exogenous within the period (a state variable, not a choice).

It is worth flagging at the outset where macroeconomics will differ in emphasis from the microeconomics the reader has seen. Two household choices organize everything that follows. The choice between *labor and leisure*—how much to work—is fundamentally *static*: it is resolved within a single period. The choice between *consumption and saving*—how much to set aside for the future—is fundamentally *dynamic*, and it is exactly this choice that, period after period, builds up the capital stock K . The static model isolates the first; the rest of the chapter develops the second.

6.1.1 The Household: Consumption and Leisure

The representative household derives utility from consumption C and leisure l ,

$$U = U(C, l),$$

and we assume the usual regularity: both goods are desirable, $U_C > 0$ and $U_l > 0$; the marginal rate of substitution is diminishing (indifference curves are convex); and both consumption and leisure are normal goods.

The household is endowed with a fixed amount of time H , which it splits between leisure l and work $H - l$. We normalize the price of the consumption good to one and let w denote the real wage, so that w is also the price of leisure—each hour not worked costs the household w units of forgone consumption. Besides wage income, the household receives the firm's distributed profits π and pays a lump-sum tax T . Its budget constraint is

$$C = w(H - l) + \pi - T.$$

Moving the wage value of leisure to the left-hand side puts the constraint in its most revealing form,

$$C + wl = wH + \pi - T,$$

which reads as a *full-income* budget: the household's total resources, the right-hand side, equal the value of its entire time endowment plus profits net of taxes, and it spends those resources on consumption and on leisure priced at w . The right-hand side is a number the household takes as given.

Maximizing $U(C, l)$ subject to this constraint, the standard tangency condition is that the marginal rate of substitution between leisure and consumption equals the relative price of leisure,

$$\text{MRS} = w \implies C^*, l^*.$$

The household's optimal labor supply is then $N^s = H - l^*$.

6.1.2 The Firm: Production and Labor Demand

The representative firm produces output with capital and labor through

$$Y = z F(K, N^d),$$

where z is total factor productivity and N^d is labor hired. We assume F exhibits constant returns to scale, which is why we may treat the whole production side as a single price-taking firm: with CRS the scale of the individual firm is indeterminate, and the aggregate behaves like one representative producer. The technology has positive but diminishing marginal products,

$$F_K > 0, \quad F_N > 0, \quad F_{KK} < 0, \quad F_{NN} < 0.$$

Remark (A correction to the classroom statement).

The lecture notes wrote the second-order conditions as $F_{KK} > 0$ and $F_{NN} > 0$. That is a slip: diminishing marginal products—the very feature that makes the model well behaved and gives convergence in the growth versions below—require the *second* derivatives to be *negative*, $F_{KK} < 0$ and $F_{NN} < 0$. The first derivatives are positive; the second derivatives are negative.

Within the period, K is a fixed state variable, so the firm chooses only how much labor to hire. Its problem is

$$\max_{N^d} \{z F(K, N^d) - w N^d\}.$$

The first-order condition equates the marginal product of labor to the real wage,

$$\text{MPN} = z F_N(K, N^d) = w \implies N^{d*}.$$

Note the division of labor between the two conditions: $\text{MRS} = w$ is the household's optimality condition, $\text{MPN} = w$ is the firm's. The classroom shorthand “ $\text{MPN} = \text{MRS} = w$ ” compresses both into one line, but they belong to different agents; they hold simultaneously only in equilibrium, when the single wage w has adjusted to make both sides content.

6.1.3 Equilibrium and Walras' Law

An equilibrium is a wage w at which both markets clear. In the labor market, the household's supply equals the firm's demand,

$$H - l^* = N^{d*}.$$

In the goods market, the household consumes exactly what the firm produces,

$$C^* = Y^*.$$

These two clearing conditions are not independent. By the household's budget constraint, the value of what it demands always equals the value of what it supplies; summed across markets, the excess demands must cancel. This is *Walras' law*: in an economy with two markets, if one clears the other must clear too.

Walras' law

In a general equilibrium with n markets, the value of aggregate excess demand is identically zero. Consequently, if $n-1$ markets clear, the last one clears automatically. In our two-market economy, **once the labor market clears, the goods market clears as well**—we need only find the single wage w that equates labor supply and labor demand.

It can be shown that there exists a *unique* wage w for which $H - l^* = N^{d*}$. At that wage the firm's profits π are determined, the household's full income is determined, and the entire allocation (C^*, l^*, N^{d*}, Y^*) falls out.

6.1.4 Income and Substitution Effects of a Wage Change

How does the equilibrium respond to a change in the wage? Because the wage is both the household's income per hour worked and the price of leisure, a change in w sets off two competing forces.

Consider a rise in w . The *income effect*: the household is now richer, and since both consumption and leisure are normal goods, it wants more of each—consumption rises, and leisure also rises (work falls). The *substitution effect*: leisure has become more expensive relative to consumption, so the household substitutes away from leisure toward consumption—consumption rises, and leisure falls (work rises).

The two effects push consumption in the same direction, so a higher wage *unambiguously raises consumption*. But they push leisure—and therefore labor supply—in opposite directions, so the net effect on hours worked is ambiguous in general. Whether labor supply rises or falls with the wage depends on which effect dominates.

A clean special case makes the cancellation exact. Normalize the time endowment to one ($H = 1$, so $C = w(1 - l)$ when $\pi = T = 0$) and take logarithmic preferences,

$$\max_{C, l} \log C + \beta \log l \quad \text{s.t.} \quad C + wl = w.$$

Example (Log utility: income and substitution effects cancel).

Solve the household's problem and show that labor supply does not depend on the wage.

Solution.

Substitute the budget constraint $C = w(1 - l)$ into the objective to make it a function of l alone:

$$U(l) = \log(w(1 - l)) + \beta \log l = \log w + \log(1 - l) + \beta \log l.$$

The first-order condition is

$$\frac{dU}{dl} = \frac{-1}{1-l} + \frac{\beta}{l} = 0 \implies \beta(1-l) = l \implies l^* = \frac{\beta}{1+\beta}.$$

Hence labor supply is

$$1 - l^* = \frac{1}{1 + \beta},$$

which is a constant: the wage w has dropped out entirely. Under log utility the income and substitution effects on leisure exactly offset, so hours worked are independent of the wage. The entire effect of a higher wage flows into consumption: from the budget, $C^* = w(1 - l^*) = w/(1 + \beta)$ rises one-for-one with w .

This knife-edge result is not a coincidence of algebra; logarithmic utility is precisely the case in which the elasticity of substitution and the income elasticity are tuned so that the two effects cancel. It is the reason log utility recurs throughout this chapter: it gives the simplest closed-form solutions and isolates the dynamic mechanism we care about from distracting wealth effects on labor.

6.2 The Two-Period Endowment Economy

We now introduce time in its simplest form. Strip out production and firms for a moment and consider a household that lives for exactly two periods and receives an exogenous income in each—a windfall endowment, like manna. There is nothing to produce and no labor to supply; the only decision is *when* to consume. This bare setting is enough to deliver the central object of dynamic macroeconomics, the Euler equation.

The household maximizes lifetime utility

$$\max_{C_1, C_2} U(C_1, C_2) = \log C_1 + \beta \log C_2,$$

where $\beta \in (0, 1)$ is the *discount factor*—a number near 0.95 that captures impatience, the fact that a unit of utility tomorrow is worth less than a unit today.

Definition 6.2: Discount factor

The *discount factor* $\beta \in (0, 1)$ is the weight a household places on next period's utility relative to this period's. A larger β means more patience. The associated *discount rate* ρ satisfies $\beta = 1/(1 + \rho)$.

Income arrives as Y_1 in period 1 and Y_2 in period 2, both exogenous. The household can

move resources between periods only by saving an amount S at the market interest rate r . Its period-by-period budget constraints are

$$\begin{aligned} C_1 + S &= Y_1, \\ C_2 &= Y_2 + (1 + r)S. \end{aligned}$$

Saving in the first period earns gross return $1 + r$ in the second. As in the static model, we treat r , Y_1 , and Y_2 as given from the household's point of view: the macroeconomic forces that determine the interest rate lie outside this single household's control.

Saving S is the bridge linking the two periods. Eliminating it—solve the second constraint for S and substitute into the first—collapses the two one-period budgets into a single *intertemporal budget constraint*,

$$C_1 + \frac{C_2}{1 + r} = Y_1 + \frac{Y_2}{1 + r}.$$

The interpretation is exactly what one would hope: the present value of lifetime consumption equals the present value of lifetime income.

6.2.1 The Euler Equation

At the optimum, the household cannot raise utility by shifting a marginal unit of consumption between periods. The tangency condition is that the intertemporal marginal rate of substitution equals the gross interest rate—the relative price of consumption today in terms of consumption tomorrow:

$$\text{MRS} = - \frac{dC_2}{dC_1} \Big|_U = \frac{\text{MU}_1}{\text{MU}_2} = 1 + r.$$

With $U = \log C_1 + \beta \log C_2$ we have $\text{MU}_1 = 1/C_1$ and $\text{MU}_2 = \beta/C_2$, so

$$\frac{1/C_1}{\beta/C_2} = \frac{C_2}{\beta C_1} = 1 + r,$$

which rearranges to the *Euler equation*.

Theorem 6.3: Consumption Euler equation (two periods, log utility)

The optimal consumption path satisfies

$$\frac{C_2^*}{C_1^*} = \beta(1 + r), \quad \text{equivalently} \quad C_2^* = \beta(1 + r) C_1^*.$$

Proof. From the intertemporal budget $C_1 + C_2/(1 + r) = W$ where $W := Y_1 + Y_2/(1 + r)$ is lifetime wealth, the Lagrangian is

$$\mathcal{L} = \log C_1 + \beta \log C_2 + \lambda \left(W - C_1 - \frac{C_2}{1 + r} \right).$$

The first-order conditions are

$$\frac{1}{C_1} = \lambda, \quad \frac{\beta}{C_2} = \frac{\lambda}{1+r}.$$

Dividing the first by the second eliminates λ and gives $\frac{C_2}{\beta C_1} = 1+r$, i.e. $C_2^* = \beta(1+r) C_1^*$. $\square$

One can reach the same condition without forming the intertemporal budget by substituting the constraints directly and maximizing over S :

$$U(S) = \log(Y_1 - S) + \beta \log(Y_2 + (1+r)S), \quad \frac{-1}{Y_1 - S} + \frac{\beta(1+r)}{Y_2 + (1+r)S} = 0,$$

which again yields $C_2 = \beta(1+r)C_1$ once we substitute back $C_1 = Y_1 - S$ and $C_2 = Y_2 + (1+r)S$.

6.2.2 Consumption Smoothing

The Euler equation is the key to how households spread consumption over time. Suppose for a moment there were no impatience and no interest, $\beta = 1$ and $r = 0$. Then the Euler equation reads $C_2^* = C_1^*$: the household chooses to consume exactly the same amount in both periods, regardless of the time profile of its income. This is *consumption smoothing*.

Why concavity implies smoothing

Whenever utility is strictly concave (marginal utility positive and diminishing, $U' > 0$, $U'' < 0$), the household prefers a smooth consumption path to a volatile one with the same present value. A unit of consumption taken from a high-consumption period and moved to a low-consumption period raises utility, because the marginal utility lost is smaller than the marginal utility gained. Smoothing is the behavioral content of diminishing marginal utility.

Once we restore impatience and interest, the path tilts but does not break. The Euler equation $C_2^* = \beta(1+r)C_1^*$ says the household smooths *around* the factor $\beta(1+r)$. The intuition is exactly as one would guess: saving a unit into the next period earns interest, so resources grow by $(1+r)$, which tilts consumption toward the future; but next period's utility is discounted by β , which tilts it back toward the present. When $\beta(1+r) = 1$ these forces balance and consumption is flat; when $\beta(1+r) > 1$ the return outweighs impatience and consumption grows over time; when $\beta(1+r) < 1$ impatience wins and consumption declines.

6.2.3 The Value Function and Shadow Prices

Once the parameters (Y_1, Y_2, r) are fixed, the optimization pins down a unique solution (C_1^*, C_2^*, S^*) and therefore a unique maximized utility. Viewed as a function of those parameters, the maximized utility is the *value function*,

$$U^* = V(Y_1, Y_2, r).$$

The value function records the best the household can do given its circumstances. Its derivatives carry economic meaning. Differentiating with respect to first-period income, $\partial V/\partial Y_1$, measures how much an extra unit of windfall income in period 1 raises lifetime welfare; this is exactly the *shadow price* of period-1 resources—the Lagrange multiplier λ from the optimization. At the optimum the multiplier equals the ratio of each good’s marginal utility to its price,

$$\frac{\partial U/\partial x_i}{p_i} = \lambda \quad \text{for every choice } x_i,$$

which is the familiar statement that, in equilibrium, the household equalizes the marginal utility per dollar across all margins. We will meet the value function and its shadow prices again in the infinite-horizon problems below, where this recursive way of thinking becomes indispensable.

6.3 The Overlapping-Generations Model

The two-period economy taught us how a single household allocates consumption over time, but it had no production and no capital, so it could not be a theory of *growth*. The overlapping-generations (OLG) model repairs this by stacking two-period lives on top of one another so that, at any date, a young generation and an old generation coexist. The young work and save; their saving becomes the capital the next period inherits. In this way the dynamic saving decision of the two-period model becomes the engine of capital accumulation.

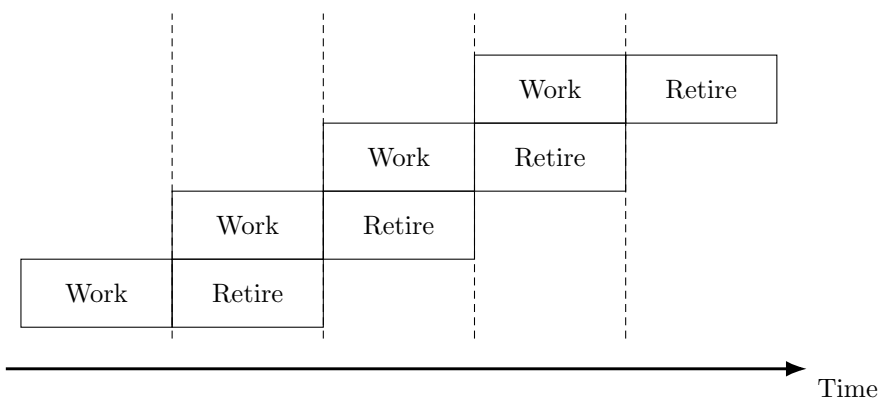


Figure 6.1: The overlapping-generations timeline: each cohort works and saves when young, then lives off its accumulated saving when old, so that at every date a working young generation overlaps with a retired old generation.

The structure is shown in Figure 6.1. Each individual lives two periods. When young she supplies one unit of labor, earns the wage, consumes part of it and saves the rest; when old she no longer works and lives off her saving (now in the form of capital she rents to firms). Although any one person lives only two periods, the chain of overlapping generations extends without end, so the model is effectively an *infinite-horizon* economy viewed one cohort at a time.

6.3.1 The Household

An individual born at t supplies one unit of labor when young, so her wage income is w_t . Her budget constraints are

$$\begin{aligned} C_t + S_t &= w_t, \\ C_{t+1} &= (1 + r_{t+1}) S_t, \end{aligned}$$

where S_t is saving and r_{t+1} is the net return earned on it. The crucial accounting link is that one cohort's saving is held in the form of capital, so a young person's saving becomes the next period's capital stock,

$$S_t = K_{t+1}.$$

This single equation is what turns a sequence of two-period problems into a theory of accumulation: the capital a society works with tomorrow is exactly what the young choose to set aside today.

With log preferences,

$$U(C_t, C_{t+1}) = \ln C_t + \beta \ln C_{t+1},$$

the saving decision is the same problem we just solved. The Euler equation $C_{t+1}/C_t = \beta(1 + r_{t+1})$ together with the budget constraints gives an explicit saving rule.

Example (Saving in the OLG model).

With log utility and the budget constraints above, show that the young save a constant fraction of their wage, and hence that

$$K_{t+1} = \frac{\beta}{1 + \beta} w_t.$$

Solution.

Substitute the constraints into the objective. With $C_t = w_t - S_t$ and $C_{t+1} = (1 + r_{t+1})S_t$,

$$U(S_t) = \ln(w_t - S_t) + \beta \ln((1 + r_{t+1})S_t).$$

The first-order condition in S_t is

$$\frac{-1}{w_t - S_t} + \frac{\beta}{S_t} = 0 \implies \beta(w_t - S_t) = S_t \implies S_t = \frac{\beta}{1 + \beta} w_t.$$

The gross return $1 + r_{t+1}$ cancels out—another consequence of log utility—so saving is a fixed fraction $\beta/(1 + \beta)$ of the wage, independent of the interest rate. Since $S_t = K_{t+1}$,

$$K_{t+1} = \frac{\beta}{1 + \beta} w_t.$$

We keep K explicit rather than substituting it away because it is the variable that links the two sides of the market—households supply it through saving, firms demand it for production—and it is also the variable that links one period to the next.

6.3.2 The Firm

The firm rents capital and hires labor each period to maximize profit,

$$\max_{K, L} F(K, L) - wL - rK,$$

with the Cobb–Douglas, labor-augmenting technology

$$F(K, L) = K^\alpha (zL)^{1-\alpha}, \quad 0 < \alpha < 1.$$

The first-order conditions set each factor price equal to its marginal product,

$$\begin{aligned} w_t &= (1 - \alpha) z_t^{1-\alpha} K_t^\alpha L_t^{-\alpha}, \\ r_t &= \alpha z_t^{1-\alpha} K_t^{\alpha-1} L_t^{1-\alpha}. \end{aligned}$$

6.3.3 Market Equilibrium and the Law of Motion

In equilibrium the price of each factor is the same on the supply and demand sides, so we may substitute the firm's wage into the household's saving rule. The result is a difference equation for capital,

$$K_{t+1} = \frac{\beta}{1 + \beta} w_t = \frac{\beta}{1 + \beta} (1 - \alpha) z_t^{1-\alpha} L_t^{-\alpha} K_t^\alpha.$$

Because there is no population variable anywhere in the problem and we normalized labor supply to one unit per young person, we set $L_t = 1$ and read every quantity as a per-capita magnitude. Provisionally fixing productivity at a constant level $z_t = z$, the law of motion becomes

$$K_{t+1} = \frac{\beta(1 - \alpha)}{1 + \beta} z^{1-\alpha} K_t^\alpha.$$

Because the exponent α is less than one, the right-hand side is a concave, increasing function of K_t , and it crosses the 45° line exactly once (Figure 6.2). The crossing is the steady state, where capital reproduces itself, $K_{t+1} = K_t = K_{ss}$.

Example (Steady-state capital).

Solve for K_{ss} from the law of motion.

Solution.

At a steady state $K_{ss} = \frac{\beta(1 - \alpha)}{1 + \beta} z^{1-\alpha} K_{ss}^\alpha$. Dividing both sides by K_{ss}^α ,

$$K_{ss}^{1-\alpha} = \frac{\beta(1 - \alpha)}{1 + \beta} z^{1-\alpha} \implies K_{ss} = z \left(\frac{\beta(1 - \alpha)}{1 + \beta} \right)^{\frac{1}{1-\alpha}}.$$

By applying the law of motion period after period we have, in effect, extended the two-period OLG structure to infinitely many periods. And since no population term appears, every variable carries a per-capita reading. The law of motion also reproduces the Solow lessons: a country that starts with little capital lies far to the left in Figure 6.2, where the curve is steep, so its capital—and output—grows quickly; similar economies converge to the

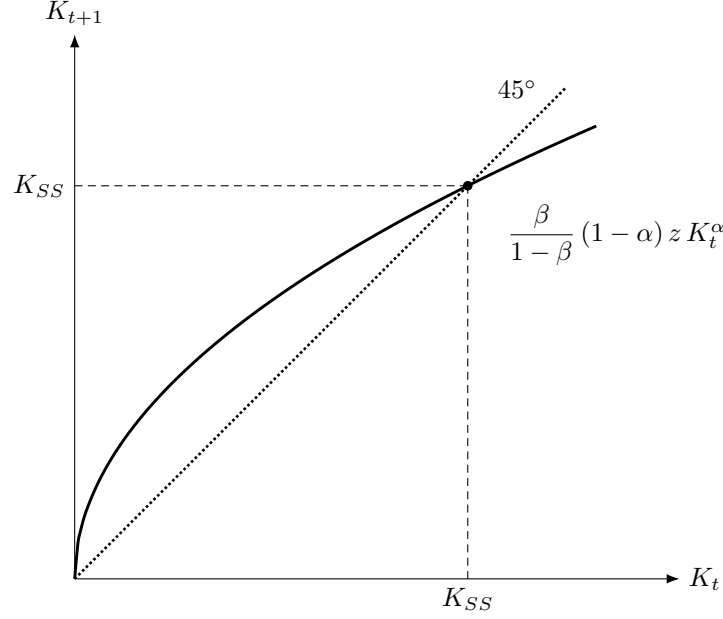


Figure 6.2: The law of motion $K_{t+1} = f(K_t)$ for capital. Because $\alpha < 1$ the curve is concave and crosses the 45° line once at the steady state K^* , to which the economy converges from any positive starting point.

same steady state. The deep reason, once again, is the diminishing marginal product of capital encoded in $\alpha < 1$.

6.3.4 Stable Versus Unstable Steady States

It is worth pausing on a general point about difference equations, because not every fixed point is a destination. Consider any first-order difference equation

$$x_{t+1} = f(x_t), \quad t = 1, 2, \dots$$

This is the canonical *law of motion*, or Markov-chain structure: tomorrow's value depends only on today's value, not on the full history. A steady state is a fixed point,

$$x_{ss} = f(x_{ss}),$$

a point where the graph of f crosses the 45° line. But solving $x = f(x)$ tells us only that a steady state *exists*; it does not tell us whether the economy will actually move toward it.

Stability is governed by the slope of f at the crossing. If the curve crosses the 45° line from above, with slope less than one in absolute value (Figure 6.3), trajectories are pulled back toward the intersection and the steady state is *stable*—it is a genuine attractor. If the curve crosses with slope greater than one (Figure 6.4), the intersection *repels* nearby paths, and the steady state is *unstable*: it solves $x = f(x)$ but the economy never settles there. Our concave capital law of motion has slope $\alpha K_{ss}^{\alpha-1} \cdot (\text{const}) < 1$ at the crossing, so K_{ss} is stable—the good case in Figure 6.3.

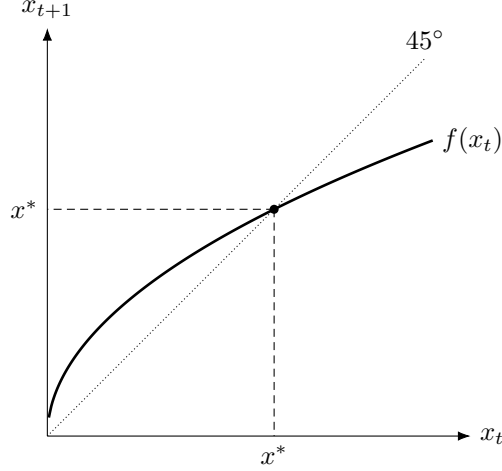


Figure 6.3: A *stable* steady state: the law of motion crosses the 45° line with slope less than one in absolute value, so trajectories started nearby are drawn back to the intersection.

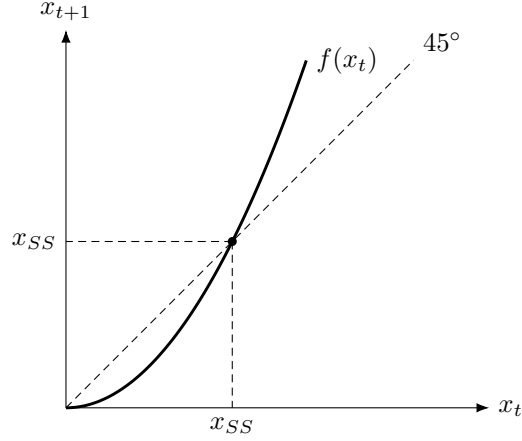


Figure 6.4: An *unstable* steady state: the law of motion crosses the 45° line with slope greater than one, so the intersection repels nearby trajectories rather than attracting them.

6.3.5 Productivity Growth

So far we froze productivity at $z_t = z$. But z measures the efficiency of labor—technology, organization, know-how—and it does not stand still. Suppose instead that productivity grows at a constant rate g ,

$$z_{t+1} = (1 + g) z_t.$$

The law of motion still reads

$$K_{t+1} = \frac{\beta}{1 + \beta} w_t = \frac{\beta(1 - \alpha)}{1 + \beta} z_t^{1-\alpha} L_t^{-\alpha} K_t^\alpha,$$

and if we set $L_t = 1$ and temporarily treat z_t as fixed, the same algebra as before gives a target level

$$K_{ss} = z_t \left(\frac{\beta(1 - \alpha)}{1 + \beta} \right)^{\frac{1}{1-\alpha}}.$$

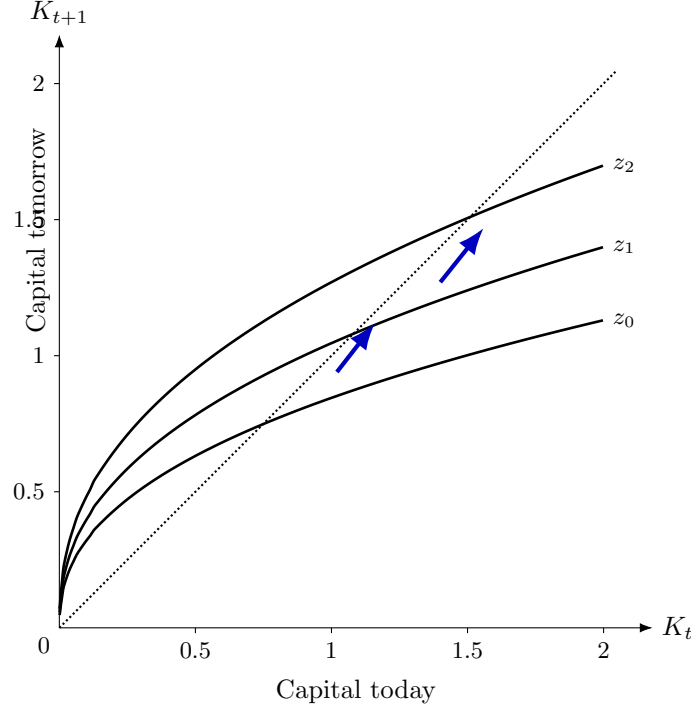


Figure 6.5: Productivity growth shifts the law of motion upward each period: as z_t rises, the curve $K_{t+1} = f(K_t)$ moves out, dragging the moving target $K_{ss}(z_t)$ steadily to the right and pulling the capital stock along a growth path.

But now z_t is itself growing, so strictly speaking $K_{t+1} \neq f(K_t)$ with a time-invariant f , and the “ K_{ss} ” above is not a true steady state. Instead, it is a *moving target*: a growth path that drifts upward as z_t rises (Figure 6.5). If the capital stock strays off this path, the diminishing-returns force still pulls it back toward the path. In the long run, then, capital grows at the same rate as productivity,

$$K_{t+1} = (1 + g) K_t,$$

since K_{ss} is proportional to z_t .

Feeding this balanced growth into the production function, with $L_t = 1$,

$$Y_{t+1} = K_{t+1}^\alpha (z_{t+1} L_{t+1})^{1-\alpha} = [(1 + g) K_t]^\alpha [(1 + g) z_t L_{t+1}]^{1-\alpha} = (1 + g) Y_t,$$

so output grows at the productivity rate g as well. Long-run growth is technology-driven, exactly as in Solow.

The same growth rate of Y and K has sharp implications for ratios and returns. The ratio Y/K approaches a constant, since both grow at g . The return to capital is then also constant,

$$r_t = \alpha z_t^{1-\alpha} K_t^{\alpha-1} L_t^{1-\alpha} = \frac{\alpha Y_t}{K_t},$$

a fixed multiple of the (constant) output–capital ratio. The wage, in contrast, grows with

productivity,

$$w_t = \frac{(1 - \alpha) Y_t}{L_t},$$

rising at rate g as output per worker rises.

The Kaldor stylized facts of postwar growth

The neoclassical model reproduces the broad regularities of advanced economies since World War II:

1. the output–capital ratio Y/K is roughly constant;
2. the real return to capital r is roughly constant;
3. the capital share of income α is roughly constant;
4. output per worker Y/L and capital per worker K/L grow at a common rate g .

Both the Solow model and the neoclassical model are consistent with these facts; the neoclassical model earns them from optimization rather than assumption.

6.4 The Cake-Eating Problem

We turn now from two-period lives to a single decision-maker with an infinite horizon. The *cake-eating problem* is the cleanest infinite-horizon optimization: a consumer owns a stock of a resource—a cake—and must decide how fast to eat it. It has no production and no prices, only a stock that shrinks as it is consumed, and it teaches the recursive structure that underlies all the dynamic models in this chapter.

The consumer maximizes discounted lifetime utility

$$\max_{\{C_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t U(C_t)$$

subject to the resource transition

$$C_t + H_{t+1} = (1 - \delta) H_t, \quad \forall t,$$

where H_t is the stock of cake at the start of period t and δ is a depreciation rate—the cake spoils a little each period. The vocabulary here is fundamental and recurs everywhere in dynamic macroeconomics.

Definition 6.4: State and control variables

A *state* variable is what the decision-maker inherits at the start of a period and cannot change within it; here it is the stock H_t . A *control* variable is what she chooses during the period; here it is C_t (equivalently H_{t+1} , since the two are linked by the resource constraint). Each variable plays one role then the other: today's control H_{t+1} becomes tomorrow's state.

6.4.1 Solving the Problem

Attach a multiplier λ_t to each period's constraint and form the Lagrangian

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t U(C_t) + \sum_{t=0}^{\infty} \lambda_t ((1 - \delta)H_t - C_t - H_{t+1}).$$

The first-order conditions in C_t , C_{t+1} , and the bridging stock H_{t+1} are

$$\begin{aligned} C_t : \quad & \beta^t U'(C_t) = \lambda_t, \\ C_{t+1} : \quad & \beta^{t+1} U'(C_{t+1}) = \lambda_{t+1}, \\ H_{t+1} : \quad & \lambda_{t+1}(1 - \delta) = \lambda_t. \end{aligned}$$

The stock H_{t+1} appears in the period- t constraint (with a minus sign) and in the period- $(t + 1)$ constraint (multiplied by $1 - \delta$), which is why its first-order condition ties the two multipliers together. Eliminating the multipliers gives the Euler equation,

$$\frac{U'(C_t)}{\beta U'(C_{t+1})} = 1 - \delta.$$

Specializing to log utility $U(C_t) = \log C_t$, so that $U'(C) = 1/C$,

$$\frac{C_{t+1}}{C_t} = \beta(1 - \delta).$$

Definition 6.5: Policy function

A *policy function* maps the current state into the optimal control—it traces out the path along which the optimized decision actually runs. Here it expresses consumption as a function of the inherited stock.

The Euler equation $C_{t+1} = \beta(1 - \delta) C_t$ is the law of motion for consumption; combined with the resource constraint it delivers the policy function.

Example (The cake-eating policy function).

With log utility, find consumption as a function of the stock, and verify that it solves the problem. Let the total initial stock be $H_1 = H$.

Solution.

Guess that consumption is proportional to the stock, $C_t = \gamma H_t$ for some constant γ to be determined. The resource constraint then gives the stock's transition,

$$H_{t+1} = (1 - \delta)H_t - C_t = (1 - \delta - \gamma) H_t,$$

and therefore $C_{t+1} = \gamma H_{t+1} = \gamma(1 - \delta - \gamma) H_t$. Imposing the Euler equation $C_{t+1}/C_t = \beta(1 - \delta)$,

$$\frac{\gamma(1 - \delta - \gamma)H_t}{\gamma H_t} = 1 - \delta - \gamma = \beta(1 - \delta) \implies \gamma = (1 - \delta) - \beta(1 - \delta) = (1 - \beta)(1 - \delta).$$

Hence the policy function and the resulting paths are

$$C_t = (1 - \beta)(1 - \delta) H_t, \quad H_{t+1} = \beta(1 - \delta) H_t.$$

Starting from $H_1 = H$, the stock decays geometrically, $H_t = (\beta(1 - \delta))^{t-1} H$, so consumption is

$$C_1 = (1 - \beta)(1 - \delta) H, \quad C_t = (1 - \beta)(1 - \delta) (\beta(1 - \delta))^{t-1} H.$$

If $\delta = 0$ this collapses to the classic cake-eating answer $C_t = (1 - \beta) \beta^{t-1} H$: eat a constant fraction $1 - \beta$ of what remains each period.

Remark (A correction to the classroom statement).

The lecture notes wrote the consumption fraction as $1 - \beta(1 - \delta)$, giving $C_1 = (1 - \beta(1 - \delta))H$. The correct constant is $(1 - \beta)(1 - \delta)$. The two agree only in the special case $\delta = 0$, where both reduce to $1 - \beta$; with positive depreciation the verified answer above is the right one, as one can check directly against the Euler equation.

6.4.2 What the Cake-Eating Problem Teaches

The value of this exercise is conceptual rather than computational. Two lessons carry forward.

First, identify the state and the control. Here the stock H plays exactly the role that *saving* played in the earlier models: it is the resource carried from the past into the present. Recognizing this correspondence is what lets us see all these models as variations on a single theme.

Second, notice what the constraint represents. In the budget-constrained problems of the earlier sections, relative prices appeared and governed the trade-offs. In the cake-eating problem there is no price at all—the constraint is a pure *resource constraint*, stating only how much of the good is physically available. This is precisely the form a constraint takes when a single decision-maker, rather than a market, allocates resources. It is the bridge to the planner’s problem, to which we now turn.

Concepts over algebra

The recurring discipline in dynamic macro is to be clear about what is a *state* and what is a *control*, and whether a constraint reflects *prices* (a market allocation) or pure *resources* (a planner’s allocation). The mathematical form is secondary; the economic reading of state, control, and constraint is what makes the models comparable.

6.5 The Social Planner's Problem and the Welfare Theorems

In a competitive general equilibrium, households maximize utility and firms maximize profit, and the whole allocation is driven by the interaction of supply and demand through prices. Imagine instead a benevolent *social planner* who commands all the information in the economy and allocates resources directly, with no prices at all. When are the two allocations the same? The answer is one of the deepest results in economics.

The planner solves

$$\max_{\{C_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t U(C_t)$$

subject to the economy's resource constraint

$$c_t + i_t = y_t,$$

which says output is split between consumption and investment—and notice, crucially, that no price appears. Output is produced from capital and labor by the same technology the firms used,

$$y_t = F(k_t, n_t),$$

and capital accumulates through

$$k_{t+1} = (1 - \delta) k_t + i_t,$$

with k_0 given. The cake-eating problem is the special case $F \equiv 0$, in which the only “resource” is the stock left over from last period; the planner's problem generalizes it by letting the leftover stock be *productive*—as if the uneaten cake were lent out to earn an extra output $F(k_t)$.

Setting $n_t = 1$ and combining the resource constraint with the accumulation equation eliminates i_t ,

$$c_t + (k_{t+1} - (1 - \delta)k_t) = F(k_t),$$

so the planner's problem is

$$\max_{\{c_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t U(c_t) \quad \text{s.t.} \quad c_t + k_{t+1} = F(k_t) + (1 - \delta)k_t, \quad k_0 \text{ given.}$$

6.5.1 Solving the Planner's Problem

For transparency take the case of full depreciation, $\delta = 1$, so the constraint reads $c_t + k_{t+1} = F(k_t)$. The Lagrangian is

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t U(c_t) + \sum_{t=0}^{\infty} \lambda_t (F(k_t) - k_{t+1} - c_t),$$

with first-order conditions

$$\begin{aligned} c_t : \quad & \beta^t U'(c_t) = \lambda_t, \\ c_{t+1} : \quad & \beta^{t+1} U'(c_{t+1}) = \lambda_{t+1}, \\ k_{t+1} : \quad & \lambda_{t+1} F'(k_{t+1}) = \lambda_t. \end{aligned}$$

Eliminating the multipliers gives the planner's Euler equation,

$$\frac{U'(c_t)}{\beta U'(c_{t+1})} = F'(k_{t+1}).$$

This is the cake-eating Euler equation with $1 - \delta$ replaced by the marginal product of capital $F'(k_{t+1})$: the rate at which the resource “grows” is now the technology's return rather than a fixed survival rate.

Example (The Cobb–Douglas social planner).

With $F(k_t) = A k_t^\theta$ and log utility $U(c_t) = \log c_t$, solve for the capital law of motion and the steady state.

Solution.

With log utility the Euler equation becomes

$$\frac{c_{t+1}}{\beta c_t} = F'(k_{t+1}) = A \theta k_{t+1}^{\theta-1}.$$

Using $c_t = A k_t^\theta - k_{t+1}$ from the constraint, substitute to get an equation in capital alone,

$$\frac{A k_{t+1}^\theta - k_{t+2}}{\beta (A k_t^\theta - k_{t+1})} = A \theta k_{t+1}^{\theta-1}.$$

The form of this equation suggests guessing $k_{t+1} = B k_t^\theta$ for a constant B . Then $c_t = A k_t^\theta - B k_t^\theta = (A - B) k_t^\theta$ and $c_{t+1} = (A - B) k_{t+1}^\theta$. Substituting into the Euler equation,

$$\frac{(A - B) k_{t+1}^\theta}{\beta (A - B) k_t^\theta} = \frac{B^\theta}{\beta} k_t^{\theta^2 - \theta} \stackrel{!}{=} A \theta (B k_t^\theta)^{\theta-1} = A \theta B^{\theta-1} k_t^{\theta^2 - \theta}.$$

The powers of k_t match, confirming the guess; equating coefficients,

$$\frac{B^\theta}{\beta} = A \theta B^{\theta-1} \implies B = A \beta \theta,$$

so the law of motion is

$$k_{t+1} = A \beta \theta k_t^\theta.$$

Setting $k_{t+1} = k_t = k_{ss}$ and dividing by k_{ss}^θ ,

$$k_{ss}^{1-\theta} = A \beta \theta \implies k_{ss} = (A \beta \theta)^{\frac{1}{1-\theta}}.$$

As with the OLG model, growth in the productivity level A shifts this target upward, so the steady state moves along a growth path rather than standing still.

This planner's allocation has exactly the structure of the Solow steady state, but it

is now derived from intertemporal *optimization* rather than from an assumed saving rate. That is the whole point of the neoclassical program: the conclusions survive, but they rest on choice. And because the model is built from explicit preferences and technology, its parameters can be reinterpreted to capture more—for instance, a business cycle can be introduced by making productivity stochastic, $y_t = z A k_t^\theta$ with z random (drawn from a distribution, or switching between high and low states). The planner’s solution for a single representative agent then extends to the whole economy, since every agent is identical.

6.5.2 The Two Welfare Theorems

We can now state the relationship between the planner’s allocation and the competitive equilibrium. Under suitable conditions the two coincide, and the bridge between them is exactly the set of prices that the planner’s problem does *not* contain. The planner’s resource constraint carries shadow prices—the multipliers λ_t —and these shadow prices are precisely the competitive prices that would decentralize the planner’s allocation.

Theorem 6.6: The two welfare theorems

1. **First welfare theorem.** A competitive equilibrium allocation is Pareto optimal: given the right prices, the decentralized outcome in which each household maximizes utility and each firm maximizes profit cannot be improved upon for one agent without harming another.
2. **Second welfare theorem.** Any Pareto-optimal allocation—in particular the one a benevolent social planner would choose—can be supported as a competitive equilibrium, provided prices are set appropriately (and, if necessary, lump-sum transfers redistribute initial wealth).

The economic content is that markets and planning, despite their opposite organizing principles, reach the same destination in this frictionless world. The planner allocates by resource constraint and ignores prices; the market allocates by prices and ignores any central command. Yet the shadow prices implicit in the planner’s optimization are exactly the market prices that make households and firms, acting only in their own interest, choose the planner’s allocation on their own. This equivalence is what licenses us, throughout the rest of the course, to solve a planner’s problem—which is often far easier—and read off the competitive equilibrium from it.

Remark (Where this is heading).

The neoclassical model has given the saving rate a microeconomic foundation and, with it, a way to ask whether market outcomes are efficient. The welfare theorems answer “yes” in this benchmark economy. Much of the rest of macroeconomics is the study of what happens when the theorems *fail*—when money, sticky prices, externalities, or missing markets drive a wedge between the competitive outcome and the planner’s ideal, opening room for policy to improve on the market. The Malthusian economy of Chapter 7 is the first such departure; monetary and Keynesian frictions follow.

Chapter 7

Population, Land, and the Malthusian Economy

The Solow model of Chapter 5 and its microfounded successor in Chapter 6 share two silent commitments. They build output from capital and labor alone, and they treat the size of the population as something handed to the economy from outside — a number that grows at a rate n the modeler simply writes down. For the world those models were meant to explain — a modern economy in which the binding scarce factor is reproducible capital and people respond to prosperity by having *fewer* children — both commitments are defensible. But they fail badly for the world that came before. In a pre-industrial agricultural society the scarce factor was not capital; it was *land*, which is fixed and cannot be accumulated. And the population was not exogenous: how many people a society contained was itself the outcome of an economic decision, the decision of families about how many children to raise.

This chapter studies the model that puts those two things back at the center. It is built on a stylized fact that, once seen, is hard to unsee: across the long agricultural era, technological progress did not make the average person better off. Better seeds, better ploughs, better irrigation showed up not as higher living standards but as *more people* living at the same standard as before. The mechanism is the one Thomas Malthus described in 1798. When a technical improvement raises the yield of the land, families are briefly richer, so they raise more children; the population grows; the fixed stock of land is divided among more mouths; and the average yield per person is pushed back down to where it started. Land is the binding constraint, and population is the variable that the constraint silences. The whole point of the model is to make this trap precise — and then to ask what eventually let humanity escape it.

We proceed in three steps. First we restore land to the production function and explain why a fixed factor turns rising effort into diminishing returns — the phenomenon of agricultural *involution*. Then we write down and solve the household's problem, derive the law of motion for population, and locate the steady state where the Malthusian trap bites: living standards are pinned to subsistence and any improvement in technology is spent entirely on additional people. Finally we modify the cost of children to include a *time* cost, which flattens the relationship between income and fertility and disarms the trap; combined with a quantity–quality tradeoff that turns that relationship on its head, this is the demographic

transition.

7.1 Land as the Binding Factor

The growth models we have seen so far use a production function in capital K and labor L only. To describe an agricultural economy we add a third factor, land T , and write a Cobb–Douglas technology

$$F(K, L, T) = A K^\alpha L^\beta T^{1-\alpha-\beta}, \quad (7.1)$$

where the exponents are positive and sum to one, so the technology has constant returns to scale across the three factors jointly. The term A is total factor productivity, the level of technique.

In a *modern* economy land barely matters. There is so much of it relative to what production needs that the land constraint never binds, output is governed by capital and labor, and the land share $1 - \alpha - \beta$ can be ignored with little loss — which is exactly why Chapters 5 and 6 dropped it. The situation reverses in an agricultural society. There the supply of arable land is effectively fixed, and once a growing population has put all of it under cultivation, land becomes a *hard constraint*. Adding more workers to a fixed plot then runs into the law of diminishing returns with a vengeance: each extra pair of hands raises total output by less and less, and eventually the marginal product of labor falls far below the average product.

Definition 7.1: Agricultural Involution

Involution (the “high-level equilibrium trap”) is the situation in which a fixed factor — here, land — forces ever-greater inputs of labor to deliver ever-smaller marginal gains in output. Effort and ingenuity are poured into wringing a little more from the same plot of ground; total output may even keep rising, but the marginal product of the added labor collapses, and output *per worker* stagnates. The extreme of meticulous, labor-saturated cultivation is the visible symptom of a society pressed hard against its land.

Remark (A name for a familiar pattern).

The word “involution” (and its Chinese rendering *neijuan*) has lately escaped the agricultural-history seminar to describe office workers competing harder and harder for the same fixed prizes. The economic skeleton is identical: a fixed factor, intensifying effort, and a marginal return on that effort that has been driven nearly to zero. The Malthusian model is the original, agrarian instance of the pattern.

We can now state the central empirical regularity the model exists to explain.

The Malthusian stylized fact

In an agricultural society one rarely observes any sustained improvement in living standards per person. What technological progress produces instead is a larger *population* at an unchanged standard of living. An economy with this property — where the gains from better technique are absorbed by more people rather than richer people — is called a *Malthusian economy*.

The task of the rest of the chapter is to derive this fact from optimizing behavior, so that we understand not just that it happens but exactly why.

7.2 The Household's Problem

Consider a representative family in the agricultural economy. It cares about its own consumption C_t and about the number of children n_t it raises, and it discounts the future at rate $\beta \in (0, 1)$. Lifetime utility is

$$\max_{\{C_t, n_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t (\log C_t + \log n_t). \quad (7.2)$$

The log specification carries two pieces of economics worth pausing on. That children enter utility directly captures the agrarian reality that children were both a joy and an asset — extra hands in the field and security in old age. That utility is logarithmic means consumption and children are imperfect substitutes with a unit elasticity, so the family always wants some of each and will split its resources between them in fixed proportions, as we are about to see.

Each period the family faces a budget. Raising a child is costly: it takes b units of the consumption good to bring up each one. Writing c_t for per-family consumption and y_t for per-family income, the resources spent on consumption and on children cannot exceed income,

$$c_t + b n_t = y_t, \quad (7.3)$$

where $b > 0$ is the goods cost of raising one child.

What ties this family problem to the land constraint of the previous section is the way income y_t is generated. Let P_t be the size of the population — this will be our *state variable*, the slowly-moving stock that summarizes the economy's history. The total stock of land is fixed; normalize it to one. Then land per person is simply the reciprocal of the population,

$$L_t = \frac{1}{P_t}. \quad (7.4)$$

Output per person depends on how much land each person has to work. We now specialize the production function (7.1) to the agricultural case in which labor and land are the operative factors. To keep the bookkeeping clear, write the labor input as the number of workers N_t (the symbol previously called L) and the fixed land stock as $T = 1$, so that land per head is $L_t = T/P_t = 1/P_t$ as in (7.4) — from here on the letter L denotes land per person, not labor. Dividing the two-factor output $F(N_t, T)$ by the number of workers gives

a per-person yield that is increasing and concave in land per head,

$$y_t = \frac{F(N_t, T)}{N_t} = f(L_t) = z_t L_t^\theta = \frac{z_t}{P_t^\theta}, \quad 0 < \theta < 1, \quad (7.5)$$

where z_t is the level of agricultural technology (playing the role of A above) and θ is the elasticity of yield with respect to land per head. The final equality uses $L_t = 1/P_t$ from (7.4). Equation (7.5) is the engine of the whole story: **a larger population means less land per person, hence lower income per person.** This is the channel through which the fixed factor reaches into every family's budget.

Assumption 7.2: The Malthusian Economy

The model rests on the following maintained assumptions:

- A representative family with preferences (7.2) over consumption C_t and number of children n_t , discounting at $\beta \in (0, 1)$.
- Each child costs b units of the good to raise, giving the budget $c_t + b n_t = y_t$.
- A fixed total stock of land, normalized to one, so land per person is $L_t = 1/P_t$.
- A concave per-person yield $y_t = z_t L_t^\theta = z_t/P_t^\theta$ with $0 < \theta < 1$, where z_t is the level of technology.
- Population evolves through fertility: next period's population is this period's population times the number of children per family.

7.2.1 The Static Choice: How to Split Income

The family's full problem is dynamic, because the children it raises today swell tomorrow's population and so depress tomorrow's income through (7.5). But the choice of *how to split a given income* y_t between consumption and children is a purely within-period, static problem: holding y_t fixed, the within-period payoff $\log c_t + \log n_t$ depends only on the period- t choices. We can therefore solve the allocation period by period.

Example (The within-period allocation).

Given income y_t , choose c_t and n_t to maximize $\log c_t + \log n_t$ subject to the budget $c_t + b n_t = y_t$.

Solution.

Form the Lagrangian

$$\mathcal{L} = \log c_t + \log n_t + \lambda (y_t - c_t - b n_t).$$

The first-order conditions are

$$\frac{\partial \mathcal{L}}{\partial c_t} : \frac{1}{c_t} = \lambda, \quad \frac{\partial \mathcal{L}}{\partial n_t} : \frac{1}{n_t} = \lambda b.$$

Dividing the first by the second eliminates the multiplier and gives $n_t b = c_t$: the family spends exactly as much on raising children as it does on its own consumption. This is the signature of log-log preferences — each good receives a constant budget share, here one half. Substituting $c_t = b n_t$ into the budget $c_t + b n_t = y_t$ yields $2b n_t = y_t$, so

$$n_t^* = \frac{y_t}{2b}, \quad c_t^* = \frac{y_t}{2}.$$

The two policy rules are clean and intuitive. Half of income is consumed; the other half is spent raising children, which at a unit cost b buys $y_t/(2b)$ of them. Crucially, **fertility rises with income**: a richer family raises more children. This single behavioral fact, combined with the land constraint, is everything we need to generate the Malthusian trap.

7.3 Population Dynamics and the Steady State

We now let the static rule feed the dynamics. Each family raises n_t children, so the population in the next period is

$$P_{t+1} = n_t P_t. \quad (7.6)$$

(Read n_t as children per family; with one adult per family, it is also the gross reproduction factor of the whole population.) Substituting the optimal fertility $n_t^* = y_t/(2b)$ and then the yield relation $y_t = z_t/P_t^\theta$ gives a single difference equation in the state P_t ,

$$P_{t+1} = \frac{y_t}{2b} P_t = \frac{z_t}{2b} P_t^{1-\theta}. \quad (7.7)$$

The exponent $1 - \theta$ lies strictly between zero and one, so the right-hand side is concave and increasing in P_t : the map has a unique, stable positive fixed point. Population that is too small grows (land is abundant, income is high, families have many children); population that is too large shrinks (land is scarce, income is low, families have few children). Either way the economy is driven toward a single resting point.

7.3.1 Solving for the Steady State

At a steady state the population is unchanging, $P_{t+1} = P_t = P_{ss}$, and technology is fixed at $z_t = z$. Imposing this on (7.7),

$$P_{ss} = \frac{z}{2b} P_{ss}^{1-\theta} \implies P_{ss}^\theta = \frac{z}{2b} \implies P_{ss} = \left(\frac{z}{2b}\right)^{1/\theta}.$$

Everything else follows by substitution. Steady-state income per person is, from (7.5),

$$y_{ss} = \frac{z}{P_{ss}^\theta} = \frac{z}{z/(2b)} = 2b,$$

and the optimal rules then give steady-state fertility and consumption,

$$n_{ss} = \frac{y_{ss}}{2b} = 1, \quad c_{ss} = \frac{y_{ss}}{2} = b.$$

Theorem 7.3: The Malthusian Steady State

With a fixed stock of land, technology z , and the household problem above, the economy converges to a unique steady state with population

$$P_{ss} = \left(\frac{z}{2b} \right)^{1/\theta},$$

at which income per person, fertility, and consumption per person are

$$y_{ss} = 2b, \quad n_{ss} = 1, \quad c_{ss} = b.$$

Two features of this steady state deserve emphasis. First, $n_{ss} = 1$: each family raises exactly one surviving child, so the population just replaces itself and stops growing. This is the natural reading of equilibrium — a stationary population is not assumed, it is *produced* by the economics. When land is so divided that income has fallen to $y_{ss} = 2b$, families can afford precisely one child each, and the population stabilizes. Second, and more striking, the steady-state living standard $c_{ss} = b$ is pinned to the cost of children alone. Living standards in the agricultural economy hover at a level set by biology and the cost of reproduction, not by the cleverness of the society's technology.

7.3.2 Why Technology Buys People, Not Prosperity

The payoff of the model is what it says about technological progress. Suppose the level of technique z rises permanently — better seeds, a heavier plough, a new crop rotation. Look at the steady-state expressions and ask what moves.

The steady-state population rises:

$$P_{ss} = \left(\frac{z}{2b} \right)^{1/\theta} \uparrow \quad \text{as } z \uparrow.$$

But income per person, fertility, and consumption per person are

$$y_{ss} = 2b, \quad n_{ss} = 1, \quad c_{ss} = b,$$

none of which contains z . Technology has dropped out of every welfare measure. **A better technique raises the number of people the land can support, but not the standard at which any of them lives.** This is precisely the stylized fact we set out to explain, now derived rather than asserted: in the Malthusian world, progress is paid out in population.

The mechanism, traced through the model, is a chain we can now state cleanly. A higher z raises income y_t for given population; higher income raises fertility $n_t^* = y_t/(2b)$; more children swell the population $P_{t+1} = n_t P_t$; a larger population means less land per head, $L_t = 1/P_t$; and less land per head pushes income $y_t = z/P_t^\theta$ back down. The economy slides along this chain until income has returned to $2b$ and fertility to one — at a larger population. The temporary prosperity from better technology is, in the end, entirely converted into extra mouths.

Remark (Control variables jump, state variables crawl).

It is worth noting the different speeds of adjustment. Fertility and consumption are *control* variables: the family resets them instantly each period, so when z jumps, income and fertility can jump with it. Population is a *state* variable, a stock that can only change gradually through births and deaths. So a technological improvement produces an immediate burst of prosperity and a fertility boom, followed by a long, slow climb of the population to its new, higher steady state — during which the early prosperity is gradually eroded away. The good times are real but transitory; the extra people are permanent.

7.4 Escaping the Trap: The Time Cost of Children

The model as built is relentless: every advance is swallowed by population, and living standards never escape subsistence. Yet they manifestly did escape, beginning with industrialization. Something in the mechanism must break. The model's most important shortcoming points to what.

We assumed the only cost of a child is the goods cost b . But raising a child also takes *time* — time that, in an economy where labor earns a wage, could have been spent working. The true cost of a child is therefore the goods spent on it *plus* the wages forgone while caring for it. Let w_t be the wage and let each child absorb a fraction m of a unit of time, so the forgone earnings per child are $m w_t$. The budget, now written against the family's time endowment valued at the wage, becomes

$$c_t + (b + m w_t) n_t = w_t. \quad (7.8)$$

The right-hand side is the value of the family's time if it worked the whole endowment; the term $(b + m w_t)$ is the *full* cost of a child — goods plus opportunity cost. The decisive new feature is that this full cost *rises with the wage*: in a richer economy, the time a child consumes is more valuable, so children become more expensive.

Solving the within-period problem exactly as before — log-log preferences again split spending in half between consumption and children — gives

$$c_t^* = \frac{w_t}{2}, \quad n_t^* = \frac{w_t}{2(b + m w_t)}.$$

Now trace what happens as the wage rises. Consumption rises without limit, but fertility does not. As $w_t \rightarrow \infty$, the goods cost b becomes negligible next to the time cost, and

$$n_t^* = \frac{w_t}{2(b + m w_t)} \rightarrow \frac{1}{2m}.$$

Fertility no longer explodes with income; the rising opportunity cost of children caps it. The runaway feedback that drove the Malthusian trap — richer means more children means poorer — has been disarmed at the source. As wages keep rising, the share of resources a family is willing to devote to the sheer *number* of children stops growing and the family begins to substitute toward other uses of its income, including investing more in each child. This bending and capping of fertility as incomes rise is the demographic transition.

The demographic transition

Once children carry a time cost as well as a goods cost, the full price of a child ($b + mw$) rises with the wage. Fertility then ceases to rise without bound as the economy grows rich — it flattens and asymptotes to $1/(2m)$ instead of exploding with income. The explosive Malthusian feedback is broken, and population converges rather than tracking technology forever upward. The outright *decline* in fertility seen in rich countries comes from a further mechanism, the quantity–quality tradeoff developed in the remark below. Together these are the *demographic transition*.

Remark (Why the time cost only bends fertility, and what finishes the job).

The plain log–log model with a time cost yields fertility $n^* = w/[2(b + mw)]$, which is in fact *increasing* in the wage but bounded above by $1/(2m)$. So the time cost alone removes the explosive Malthusian feedback and caps fertility; it does not by itself produce the falling fertility we observe in rich countries. The piece that delivers an outright *decline* is the quantity–quality tradeoff. Once families can also choose how much to invest in *each* child — schooling, health, human capital — a higher wage makes quality more attractive at the margin, and families substitute away from many low-investment children toward few high-investment ones. The demographic transition is thus best read as a tradeoff between the *quantity* and the *quality* of children: human capital enters the production function, child quality becomes worth buying, and chosen fertility falls.

This same logic explains a well-documented pattern within modern economies: richer families and, in particular, higher-earning women tend to have fewer children. As mental rather than manual labor becomes the key input to production and as women’s wages rise, the opportunity cost of the time children require climbs, and — once the quantity–quality tradeoff lets families substitute toward investing in each child — the quantity of children chosen falls. The rising time cost flattens the income–fertility relationship; the quantity–quality tradeoff inverts it. The Malthusian world, in which children were cheap and prosperity bred only more of them, has been turned exactly on its head.

Remark (A note on population policy).

It is tempting to read the Malthusian model as a warrant for suppressing population growth, and twentieth-century population-control programs were argued partly in these terms. But the model that justifies such a view is the *agricultural* one, in which output is bounded by a fixed factor and an extra person is one more mouth to feed from the same land. Once an economy has left agriculture — once it runs on capital, ideas, and human capital, as in the Solow and neoclassical models — that logic no longer applies. People are not merely consumers of a fixed pie; they generate large positive externalities, inventing, specializing, and expanding the very technology z that the Malthusian model held fixed. Treating population as a pure burden imports Malthusian intuitions into a world that has outgrown them. The model is a faithful description of the agrarian past, not a prescription for an industrial present.

With land and endogenous population now in hand, we have completed the long-run,

real side of the course: the Solow model, its microfounded version, and the Malthusian world that preceded modern growth. The next chapters turn from the real economy to the nominal one — money, banking, and the price level — beginning in Chapter 8.

Chapter 8

Money, Banking, and Monetary Policy

Almost everything in macroeconomics that we have studied so far — output, capital, labor, growth — could in principle be described without ever mentioning money. The Solow and neoclassical models of Chapters 5 and 6 are written entirely in terms of real goods. Yet no modern economy operates without money, and a great deal of what a government actually *does* to manage the macroeconomy in the short run is done through the central bank. This chapter is about the plumbing: what money is, how the banking system manufactures most of it, and the instruments a central bank uses to expand or contract the supply of credit.

The subject is unavoidably more institutional than the models that came before. Where there is a clean structure — the liquidity ordering of assets, the money multiplier, the central-bank balance sheet — we will state it precisely. But much of monetary policy is a matter of who does what to whom through which legal instrument, and these arrangements differ sharply across countries. The two reference systems throughout are the United States, where the central bank operates mostly through the Treasury bond market, and China, where the central bank operates mostly through the commercial banks directly. We treat the China-specific machinery in some detail, because the lecture does, and because it is the cleaner illustration of how a central bank steers an economy through the banking channel. Unless stated otherwise, “the central bank” below means the People’s Bank of China (PBoC).

Two topics that might seem to belong here have been placed elsewhere. The consumer and producer price indices (CPI and PPI), our measures of the price level, were developed in Chapter 3. The quantity theory of money, the inflation tax, and seigniorage — the consequences of *printing* money rather than the mechanics of *managing* it — are the subject of Chapter 9. Here we stay with the institutions.

8.1 Money and Liquidity

Money is best understood as an *asset* — a thing of value that one can hold — distinguished from other assets by one property above all: liquidity. An asset is liquid to the extent that it can be exchanged for goods quickly, cheaply, and at a predictable value. Cash in hand is the limiting case: it is accepted everywhere, instantly, at face value. A government bond, a

mutual-fund share, or a publicly traded stock is also a store of value, but converting it into spendable purchasing power takes time and may force a sale at an unfavorable price. The defining feature of money is that it sits at the high-liquidity end of this spectrum.

What makes an asset acceptable as money is ultimately *credibility*: people take it in exchange only because they are confident others will take it from them in turn. The wider that acceptance, the more liquid the asset. Figure 8.1 arranges the major financial assets along this dimension.

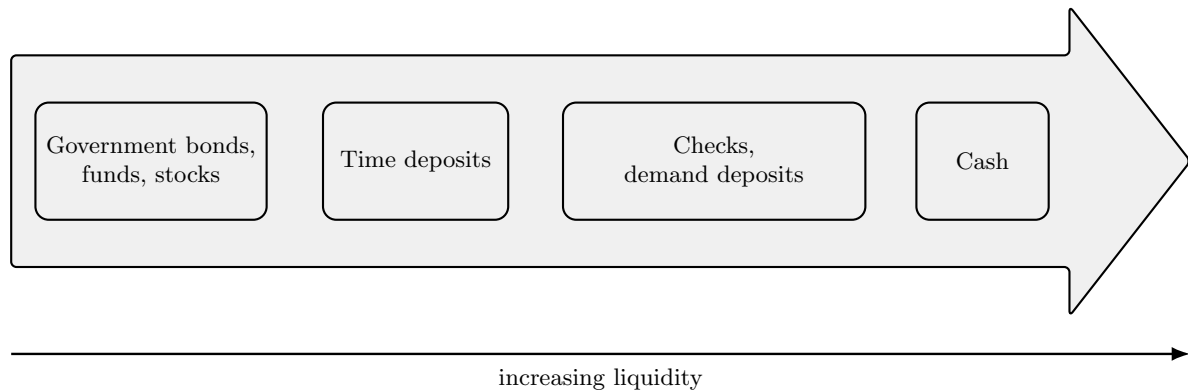


Figure 8.1: The liquidity spectrum, from the least-liquid assets (bonds, fund shares, stocks) on the left to the most-liquid asset, cash, on the right.

Definition 8.1: Liquidity

The *liquidity* of an asset is the ease with which it can be converted into a generally accepted means of payment without loss of value. Money is, by definition, the most liquid class of assets; cash is the most liquid form of money.

A useful corollary is that liquidity is priced. A less liquid asset must offer a *premium* — a higher expected return or a discount in price — to be held alongside more liquid ones; otherwise no one would accept the inconvenience of holding it. This single idea, that low-liquidity assets trade at a discount to high-liquidity ones, organizes much of what the banking system and the central bank do.

Remark ([]).

Banking as a liquidity transformation] A bank loan is collateralized: the borrower pledges a low-liquidity asset (a house, a bond, a factory) and receives high-liquidity funds in return. The borrower's illiquid wealth is, in effect, swapped for liquid purchasing power, and the *total* liquidity available to the real economy rises even though no new real asset has been created. The central bank does exactly the same thing one level up: it lends to commercial banks against collateral such as government bonds, swapping the banks' less-liquid holdings for the most-liquid asset of all. Managing the economy's liquidity is the central bank's day-to-day business. When the Treasury collects taxes, for instance, it drains liquidity out of the private sector; the central bank typically offsets this by injecting base money, so that the tax round does not cause an unintended monetary

contraction.

8.2 Monetary Aggregates

Because “money” covers assets of differing liquidity, statistical agencies do not report a single number for it. They report a nested family of *monetary aggregates*, each adding a layer of less-liquid, more deposit-like instruments to the one inside it. Figure 8.2 shows the standard nesting, $M_0 \subset M_1 \subset M_2$.

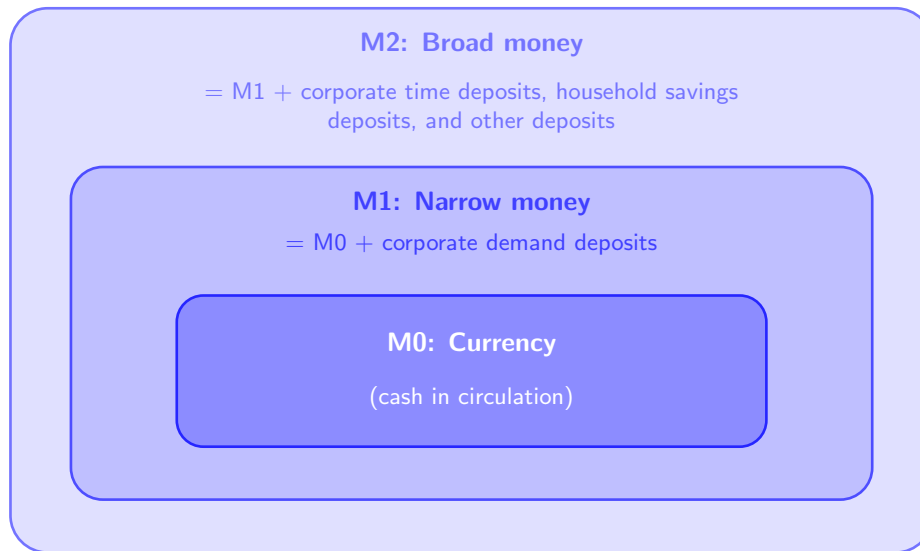


Figure 8.2: The nested monetary aggregates: currency M_0 sits inside narrow money M_1 , which sits inside broad money M_2 .

Definition 8.2: Monetary Aggregates

The monetary aggregates order money by liquidity:

- M_0 — currency in circulation: physical cash held outside the banking system, the most liquid component.
- M_1 — “narrow money”: M_0 plus demand (checkable) deposits, which can be spent on short notice.
- M_2 — “broad money”: M_1 plus time and savings deposits, which are slightly less liquid.

Each aggregate strictly contains the ones inside it.

The cash component M_0 is small and is driven mainly by short-run, seasonal factors (holidays, payroll cycles). The aggregate that matters for policy is M_2 . Cash is only a tiny fraction of M_2 ; the overwhelming bulk of “money” in a modern economy is not paper at all but *deposit balances* — entries on banks’ books. In a slogan: money is the figure on the ledger.

Remark ([].

Money held versus money in society] M_0 , M_1 , and M_2 all measure money that the public actually holds. This is not the same as the total quantity of money the system could create, nor the same as “wealth.” The aggregates are stocks of liquid claims in circulation at a point in time, no more and no less; do not read them as a census of all purchasing power in the economy.

8.2.1 Why M_2 is the policy target

M_2 is the lens through which monetary growth is viewed. The *growth rate* of M_2 is the headline measure of monetary policy; in China a target of roughly ten percent has often been set at the annual “Two Sessions.” As a rule of thumb, the growth of M_2 is expected to roughly track the growth of nominal GDP — enough new money to finance the new transactions, no more. Strictly, though, this rule cannot hold mechanically, because the whole point of monetary policy is *countercyclical*: the central bank deliberately lets money grow faster than output in a downturn and slower in a boom. A money supply that simply matched nominal GDP every quarter would be doing no stabilizing work at all.

Remark ([].

The M_2 /GDP ratio is easy to misread] China’s ratio of M_2 to GDP exceeds 200%, while the United States’ is under 100%. It is tempting to conclude that China is awash in excess liquidity and the U.S. is not. The comparison does not support that conclusion, for several reasons.

First, the two central banks behave very differently: the PBoC has historically been restrained in expanding the money supply, whereas the Fed swings aggressively (sharp cuts in crises, later reversals). Second, and more fundamentally, the *channel* through which M_2 is created differs. In common-law economies such as the U.S. and U.K., *direct finance* dominates: firms raise money chiefly by issuing bonds and stocks, which do not pass through and inflate bank-deposit aggregates. China relies on *indirect finance*: firms borrow from commercial banks, and every such loan creates a matching deposit, so a credit-driven economy mechanically carries a larger M_2 relative to GDP. The commercial banks are the central hub of the Chinese financial system. Third, M_2 is a *stock* and GDP is a *flow*; their ratio mixes units, and only their *growth rates* can be meaningfully compared. Finally, China’s rapid marketization since the reform era has steadily monetized activity that was previously non-market, pushing M_2 up for reasons unrelated to “excess.”

8.2.2 Loans and social financing

M_2 and bank lending move together, because in an indirect-finance system new loans are the principal source of new broad money. Beyond the two textbook drivers of M_2 — the supply of base money and the money multiplier, developed in the next section — M_2 growth can stall for reasons on the *demand* side: the real economy may not want to borrow (weak credit demand), or banks may be unwilling to lend (a “credit crunch,” banks hoarding liquidity). It is therefore useful to separate two related ideas.

Easy money versus easy credit

Easy money (a “loose money” stance) means an increase in the supply of base money by the central bank. *Easy credit* (a “loose credit” stance) means that the real economy can in fact obtain financing more readily — credit availability has widened. The two are not the same: if easy money fails to translate into easy credit, the *monetary transmission mechanism* is blocked, and the extra base money piles up in the banking system without reaching firms and households.

Several published series let analysts watch the credit channel directly. *New RMB loans* is the key indicator of indirect finance, split into household lending and corporate lending. Household lending is dominated by consumer credit; in recent years default rates on consumer loans and credit-card balances have risen sharply. A broader measure is *aggregate social financing*.

Definition 8.3: Aggregate Social Financing

Aggregate social financing (*shehui rongzi guimo*) measures the total financing that financial institutions provide to the real economy — to firms and individuals — and *excludes* lending among financial institutions themselves. It spans both indirect finance through the banking system (loans, trust loans, undiscounted bankers’ acceptances — the latter two are “off-balance-sheet”) and direct finance through capital markets (bonds and equities). Ordinary bank loans are recorded “on balance sheet”; trust and entrusted lending are “off balance sheet.”

The market watches three headline numbers in particular — M_2 , new RMB loans, and aggregate social financing — typically released between the tenth and fifteenth of each month.

8.3 Types of Money

Money, at bottom, is a story a society agrees to tell. What gives a particular object monetary value has changed over history, and the differences are instructive.

Commodity money. Commodity money has intrinsic value: the object is worth something in its own right (gold, silver, copper), so its monetary value is anchored to its material value. The supply of commodity money is therefore set by the technology of production — how much silver can be mined, for example — and cannot be expanded at will.

Remark ([]).

Paper money is older than you think] The year 2022 marked the thousandth anniversary of the *jiaozi*, an early Chinese paper currency. China was long constrained by a scarcity of precious metals and relied mostly on copper coinage, with little silver; the Song dynasty even experimented with iron coins, whose great defect was that they rusted. The early emergence of paper money is itself a signal: it appears precisely when commercial activity has grown too active for the available stock of precious metal to settle all the trade. Paper

is a response to a liquidity shortage.

Fiat money. Fiat money has no intrinsic value — a banknote is just printed paper — and is accepted for one of two reasons. The first is *issuer credibility*: the issuing institution is trusted enough that its notes circulate (historically, notes of highly reputable private banks). The second, and the modern basis, is *the credit of the state*: the sovereign power compels acceptance, and notes are issued by the central bank. The central bank may even delegate issuance to designated, reliable commercial banks — Hong Kong has no central bank, and its currency is issued by its note-issuing banks.

Currency in circulation is a *liability* of the central bank and an *asset* of whoever holds it — a fact we will use to read the central-bank balance sheet. Since the collapse of the Bretton Woods system, money has been pure fiat (credit) money, no longer convertible into gold.

Credit money can be created at will

The supply of commodity money is fixed by productive capacity (the mining rate of silver). The supply of fiat (credit) money is a human decision: it can be expanded essentially without limit, and a credit-money system holds together as long as the state's credit holds. If that credit collapses — if the public stops trusting the currency and abandons it for foreign money — the result is hyperinflation. So the latitude to “overdraw” on the currency is real but bounded by confidence.

Digital and crypto money. A cryptocurrency is a privately recognized money whose supply is fixed by an algorithm. Because the supply is fixed but the object is intrinsically worthless, its price is highly volatile. Its defining features are a *fixed supply* and *decentralization* — there is no central bank — not the mere fact of being “digital” (electronic payment and online transfers are digital too, but are claims on ordinary bank deposits). In this respect Bitcoin resembles gold: gold yields no return of its own, and its only payoff is the change in its price, which is why both are objects of speculation rather than investments that throw off income like stocks or bonds. Combining a store-of-value function with a transactional one, such assets carry some value, and in a world of heightened geopolitical risk that value may grow.

8.4 The Monetary Transmission Mechanism

How does a central-bank action reach an ordinary borrower? The answer — the *transmission mechanism* — runs through different intermediate markets in different countries.

Transmission chain	
China	central bank → commercial banks → economy
United States	central bank → Treasury bond market → economy

The contrast is exactly the direct-versus-indirect finance distinction from before. In the U.S., the central bank acts on the Treasury bond market — a direct-finance channel — and the rest of the economy reprices off Treasury yields. In China, the central bank acts on the commercial banks — an indirect-finance channel — and the banks pass the impulse along to firms and households.

The U.S. channel rests on a balance-sheet identity. *Government bonds are an asset of the central bank, and currency is a liability.* When the Fed *buys* bonds, its balance sheet *expands* and it pays for the bonds with newly created money, raising the money supply. When it *sells* bonds, its balance sheet *shrinks* and money is withdrawn from circulation. Buying bonds therefore loosens, selling tightens. The main counterparties in the U.S. Treasury market are themselves commercial banks; the distinctive feature is that the Fed participates as one more trading party, buying and selling on equal terms, rather than lending to the banks directly. Because cash is such a small share of M_2 , the part of the central bank that matters most is not the currency operation but the monetary-policy department that conducts these market operations.

8.5 The Reserve System and the Money Multiplier

We now reach the central mechanical fact of modern banking: *commercial banks, not the central bank, create most of the money.* They do so through the reserve system. A bank that receives a deposit is required to hold only a fraction of it as *reserves* and may lend out the rest; the loan becomes a deposit somewhere else, which is again largely lent out, and so on. A single unit of base money supports a multiple of deposits. The multiplier quantifies the multiple.

Definition 8.4: The Money Multiplier

Let C denote currency held by the public, D deposits, and R bank reserves. Broad money is $M = C + D$ and the *monetary base* (also called *high-powered money*) is $B = C + R$. The *money multiplier* is the ratio

$$\text{multiplier} = \frac{M}{B} = \frac{C + D}{C + R} = \frac{c + 1}{c + r},$$

where $c = C/D$ is the *currency-deposit ratio* and $r = R/D$ is the *reserve ratio*. The second equality follows by dividing the numerator and denominator by D .

The derivation is worth doing once, because the algebra is the whole content. Start from $M = C + D$ and $B = C + R$. Divide top and bottom of M/B by deposits D :

$$\frac{M}{B} = \frac{C/D + D/D}{C/D + R/D} = \frac{c + 1}{c + r}.$$

Two features of this expression carry the economics. First, currency C is money that has *leaked out* of the banking system; it sits in wallets and cannot be re-lent, so a higher currency-deposit ratio c shrinks the multiplier. In practice c is small, since the public holds little cash relative to deposits. Second, and the policy lever, is the reserve ratio.

Theorem 8.5: A lower reserve ratio raises the multiplier

Holding the currency–deposit ratio c fixed, the money multiplier $\frac{c+1}{c+r}$ is strictly decreasing in the reserve ratio r . A lower r lets banks lend a larger fraction of each deposit, supporting more rounds of deposit creation, and so generates a larger M_2 from the same base.

Proof. Differentiate $m(r) = (c+1)/(c+r)$ with respect to r :

$$\frac{dm}{dr} = -\frac{c+1}{(c+r)^2} < 0,$$

since $c+1 > 0$. Thus m falls as r rises and rises as r falls. $\square$

This is why the required reserve ratio is a monetary-policy instrument: cutting it, with the base unchanged, mechanically increases M_2 . The base itself is supplied by the central bank.

Definition 8.6: Base Money

Base money (high-powered money) is the money the central bank creates directly — “the original money.” In the indirect-finance system it enters the economy when the central bank lends to commercial banks. All funding the central bank releases to the banks is base money. Because it is then magnified by the money multiplier, base money generates an M_2 far larger than itself. Consequently the growth of M_2 comes chiefly from two sources: the supply of base money and the reserve ratio.

8.5.1 Why banks would set reserves too low: moral hazard and the reserve floor

Reserves trade off safety against profit. A bank that holds more reserves is safer but earns less, because idle reserves earn little; a bank that holds fewer reserves earns more but is more exposed to a run. Each bank cares about its own profit, which is private and rival. Safety, however, has an *externality*: one bank’s failure can cascade through the system. Knowing the financial system will mobilize to rescue a troubled member, each bank has an incentive to free-ride — to push its own reserve ratio lower and capture the profit while the systemic safety is provided by others. This is a classic *moral hazard*, and it breeds adverse selection: absent regulation, the banks that survive are precisely the riskiest ones, since they out-earned the prudent.

Remark ([].

The reserve floor as a mutual-aid rule] The institutional response is a coordinated reserve *floor*. Banks form, in effect, a mutual-aid association with a red line: if a member runs into trouble while its reserves are *above* the floor, the association is obliged to rescue it; if its reserves are *below* the floor, no rescue is forthcoming. The floor restores the incentive to hold adequate reserves. Historically the central bank performed exactly this lender-of-

last-resort, coordinating-rescue role, owned by the member banks like a club or industry association. The Federal Reserve, for example, has shareholders — the member banks — and remits most of its profit to the Treasury with a portion to those shareholders; yet neither the shareholders nor the President may direct its operations. A modern central bank stands on the credit of the state, not on its ownership.

8.6 Central Banks

A central bank is itself a bank, but one that does not aim at profit — though it is highly profitable in practice. It transacts only with commercial banks (and a few designated institutions), and its profit comes from the *interest spread*, chiefly between the rate it pays on reserves and the higher rate it earns on its lending facilities (in China, the spread between the reserve interest rate and the MLF rate).

Remark ([]).

Who may hold an account at the central bank] Only banks and a few non-bank institutions may open accounts at the central bank. “Banks” here includes the commercial banks and the *policy banks*, the most important of which is the China Development Bank (CDB). Policy banks fund large, low-margin, strategically important projects such as infrastructure; the CDB issues its own bonds (CDB bonds), funded ultimately by the central bank. The central Ministry of Finance may also hold an account — in effect the national treasury. The central bank designates a list of *primary dealers*, typically large commercial banks holding accounts at its head office; it transacts mainly with these primary dealers, who then on-lend funds to the rest of the banking system.

Because the central bank cannot continuously audit every commercial bank’s cash, banks keep their reserves in reserve accounts *at* the central bank, and a bank whose reserves fall below the legal requirement is penalized. Since a bank’s deposits fluctuate constantly, prudent banks hold *excess reserves* above the legal minimum as a buffer. The central bank pays interest on excess reserves, but — to stop banks profiting merely by parking funds at the central bank — the excess-reserve rate cannot exceed the required-reserve rate, and excess reserves are in fact capped.

The excess-reserve rate is the floor of the rate corridor

The interest rate on excess reserves sets the *lower bound* of the policy rate corridor: no bank will lend overnight at less than it can earn risk-free at the central bank. When the excess-reserve rate falls, market rates fall with it. A low excess-reserve ratio across the system signals that liquidity is tight — banks are holding little spare cash.

Remark ([]).

Whose interests the central bank serves] A central bank represents the *holders of monetary assets* — everyone who holds money or money-like claims — and emphatically *not* the government as such. When it lends, swapping low-liquidity assets for high-liquidity ones, it serves those claimants. Its insistence on independence is precisely a refusal to

serve the government's financing needs directly: a central bank captured by the fiscal authority would simply monetize government deficits. In China the central bank may not *purchase* government bonds outright, though it may take them as *collateral*; and the money it issues need not be backed one-for-one by collateral — it can create credit money essentially at discretion. Some countries lean toward fiscal monetization, an idea rooted in *Modern Monetary Theory* (MMT), which holds that a sovereign with ample credit faces a far higher ceiling on money issuance than is usually imagined and that the central bank should accommodate fiscal policy. The PBoC guards its independence strongly and firmly opposes fiscal monetization.

8.6.1 Mandates and structure

In compact form, monetary policy works by setting three instruments — the money supply, the policy interest rate, and the reserve ratio — to steer three targets: M_2 , aggregate social financing, and new RMB loans. A central bank's mandate fixes what it is trying to achieve, and the ordering of objectives matters.

The PBoC mandate: currency stability first

The PBoC is a government body that formulates and implements monetary policy under the guidance of the State Council. Its objective is to *maintain the stability of the currency, and thereby promote economic growth*. Currency stability — that is, controlling inflation — is the *first* objective; growth comes after. A practical corollary is that tightening tends to originate with the central bank itself, while loosening tends to be urged on it by the government.

The central bank is not directly responsible for growth; in China the lead agency for growth is the National Development and Reform Commission, followed by the Ministry of Finance. The central bank also adjusts interest rates cautiously: if the risk-free rate is set too high, risky venture investment is starved of room to develop.

The U.S. Federal Reserve performs the functions of a state agency despite having private shareholders. Its Board of Governors, headquartered in Washington, has seven members, and the term of the Fed Chair is deliberately offset from the presidential term. The Chair is not a cabinet member, and the President has no direct administrative authority over the Fed.

Definition 8.7: The FOMC and the federal funds rate

The *Federal Open Market Committee* (FOMC) sets U.S. monetary policy. It has twelve voting members: the seven Governors, plus the President of the Federal Reserve Bank of New York, plus four of the remaining eleven regional Reserve Bank presidents on a rotating basis ($7 + 1 + 4 = 12$). It meets roughly every six weeks. When the Fed “raises” or “cuts” rates, it is acting on the *federal funds rate* — the overnight interbank lending rate among U.S. banks.

In a market system the Fed cannot literally dictate the federal funds rate; what it announces is a *target range*. To keep the actual rate inside that range it uses *open-market*

operations to adjust liquidity, and these in turn move the prices of longer instruments such as the ten-year Treasury. In practice the Fed rarely lends to commercial banks directly and cannot fix most rates by decree — it can only steer a range. China’s central bank, by contrast, transacts with the banks constantly and can set rates directly.

Remark ([].

Crisis response and the over-reliance on monetary policy] The Fed’s standard crisis play-book is to cut rates sharply, flooding the system with liquidity, and accept the later reflation: this was the pattern after the 2001 dot-com bust and 9/11, the 2008 financial crisis, and the 2020 pandemic. A recurring lecture theme is that Western economies have come to lean too heavily on monetary policy because fiscal policy is hard to deploy — crisis recovery, as after 2008, is kept alive chiefly by the central bank. The point is sharpened by a contrast in governance: U.S. monetary policy is run by an insulated technocratic elite, which is fast and undemocratic, against a Congress that is democratic but slow. The balance between grassroots democracy and technocratic decisiveness recurs in many settings; in a crisis, the U.S. can act only through the fast, technocratic monetary channel.

8.6.2 Rule-based versus discretionary policy

Central banks differ in how predictable they are.

Definition 8.8: Rule-based and discretionary policy

Under *rule-based* policy the central bank follows a transparent, pre-committed rule, so the market can anticipate its moves and the bank does not surprise expectations. The leading example is a *Taylor rule*, which sets the target rate as a function of inflation and the unemployment (or output) gap. Under *discretionary* policy the objectives and instruments are deliberately left vague, so the market cannot form a precise expectation of the next move.

The Fed’s tightening intentions are openly telegraphed — it broadly follows an optimized Taylor rule keyed to inflation and unemployment — which makes U.S. policy rule-based. China’s policy goals are comparatively opaque, pursued through a toolkit of rates, reserve ratios, and the LPR, which makes it discretionary.

Remark ([].

A rough timeline of recent Fed cycles] The dot-com bust and 9/11 led to sharp U.S. rate cuts in 2001, with recovery after 2005. The 2007 subprime crisis forced another round of deep cuts, holding the rate near 0.25% for a long stretch. After 2017, with the economy fully recovered, the Fed entered a hiking cycle; in 2019, amid prosperity, President Trump repeatedly pressed for cuts. The 2022 tightening was steeper than the post-2008 path, driven on *both* sides of the market: demand-side recovery, and supply-side pressures from a falling labor-force participation rate and the energy-price spike triggered by the Russia–Ukraine war. If inflation eased, the Fed might exit the hiking path early.

8.7 Central-Bank Balance Sheets

A central bank steers the economy by changing the size and composition of its own balance sheet. The accounting is the cleanest summary of what it does.

How to read a central-bank balance sheet

On a central bank's balance sheet, *currency in circulation is a liability* and *bonds (and other purchased assets) are assets*. Therefore: *expanding* the balance sheet — buying assets, creating reserves and currency to pay for them — adds liquidity to the system and pushes interest rates *down*; *shrinking* the balance sheet withdraws liquidity and pushes rates *up*.

The two reference balance sheets are below, as T-accounts.

Federal Reserve	
Assets	Liabilities
Bonds (mainly Treasuries and MBS)	Currency in circulation (main)
	Bank reserves

Remark ([]).

How MBS got onto the Fed's balance sheet] After the 2008 crisis — whose proximate cause was subprime debt repackaged into *mortgage-backed securities* (MBS) — the Fed bought large quantities of MBS in 2009 to rescue the market. That is why MBS, alongside Treasuries, became a major asset on the Fed's balance sheet.

People's Bank of China	
Assets	Liabilities
Foreign-exchange claims	Currency in circulation (main)
Bonds	Bank reserves

The conspicuous feature of the PBoC's balance sheet is the large stock of foreign-exchange claims on the asset side, held chiefly to maintain the stability of the financial system. We will see in the China toolkit below how the accumulation of these claims drove the PBoC's balance sheet for over a decade.

8.8 Conventional and Unconventional Tools

Beyond setting reserve requirements and conducting market operations, central banks keep a graduated set of instruments, ranging from the routine to the emergency.

The discount window. In addition to open-market operations, the Fed lends directly to financial institutions; this is *discounting*, and the lending rate is the *discount rate*. The discount rate is determined through bidding (though the Fed has great influence over it), and banks then decide how much to borrow against it.

Quantitative easing. When the economy turns down — especially once the policy rate is already near zero — the Fed turns to *quantitative easing* (QE): large-scale outright purchases of government bonds and mortgage-backed securities (MBS).

Theorem 8.9: How quantitative easing works

By buying long-term securities in bulk, the central bank bids up their *prices* and so drives down their *yields*. Lower long-term yields lower the interest rate on long-term loans, which makes long-horizon investment more attractive. QE thus eases financial conditions by acting on the long end of the yield curve when the short end is already at its floor.

The liquidity trap. Easy money does not always produce growth.

Definition 8.10: Liquidity Trap

A *liquidity trap* arises when abundant liquidity fails to support real economic growth and instead inflates asset prices. In a liquidity trap the base money that banks receive does not flow through to the real economy — the transmission mechanism is blocked — so the new liquidity bids up financial and property prices rather than financing new output.

Remark ([].

A lesson of 2008: watch asset prices, not just CPI and PPI] A central lesson of the 2008 crisis was that inflation cannot be judged from CPI and PPI alone; *asset prices* must be watched too. The Taylor rule in force before the crisis did not incorporate them, and so missed the asset-price inflation building in housing.

Negative rates. The structural problems of Europe and Japan are severe: lacking domestic engines of growth, their central banks must ease whenever conditions allow but cannot easily tighten even in a recovery. Their unconventional response is the *negative interest rate*. This can take two forms: a negative overnight rate between the central bank and commercial banks (say -0.1%), or a negative rate on excess reserves (say -0.1% , a charge designed to push liquidity out of the central bank and into the real economy). A negative bond yield is also possible, though it is an outcome rather than a policy instrument.

If an economy keeps growing even as money tightens, that is a sign it has strong, self-sustaining growth momentum — the healthy case that Europe and Japan lack.

8.9 The China Toolkit

China's monetary policy operates almost entirely through the banks, with a rich and specific set of instruments. The lecture stresses two cross-cutting distinctions before the tools themselves.

The first distinction — between *price-based* and *quantity-based* policy — turns out to be less sharp than it sounds, since any quantity operation comes with a price and the two

are bound together. The more useful distinction is between *aggregate* and *structural* policy.

Aggregate versus structural policy

By definition monetary policy is *aggregate*: it has no designated direction and acts on the economy as a whole (cutting rates, buying bonds, lending to commercial banks — without specifying where the funds end up). *Structural* policy, by contrast, directs support to a particular sector or purpose. China’s past aggregate easing tended to flow into real estate, so making policy effective meant “flood irrigation” of the whole economy; the later shift toward structural tools aims at “precision drip irrigation” of targeted sectors.

8.9.1 Required reserve ratio

The classic quantity tool is the *required reserve ratio* (RRR). With base money unchanged, cutting the RRR raises M_2 (by raising the multiplier, as proved above) and raising the RRR lowers M_2 .

Remark ([]).

The differentiated reserve ratio since 2015] The RRR was the first tool given a structural form. Beginning in 2015, China adopted a *differentiated* required reserve ratio — a structural policy emphasizing macroprudential management and systemic risk. Larger banks face higher reserve requirements; smaller banks face lower ones; and city and rural commercial banks receive preferential treatment.

Remark ([]).

Why the RRR was used the way it was around 2012] China’s monetary policy shifted markedly around 2012. *Before* 2012, the balance of payments ran large surpluses, driven by trade and inbound foreign direct investment (FDI). A balance-of-payments surplus produces *imported inflation*: China had to passively absorb the inflowing foreign funds, and the PBoC was forced to expand its balance sheet (the foreign-exchange claims on the asset side), risking overheating and inflation. The central bank’s main task was therefore to *drain* liquidity, which it did by progressively *raising* the RRR and *issuing central-bank bills* — the PBoC’s IOUs, traded in the Hong Kong market. (A digression: RMB trading is a closed system. Offshore RMB, CNH, trades chiefly in Hong Kong; onshore RMB, CNY, is managed at home; CNH typically moves faster and further, though both can be intervened in. By issuing bills offshore the PBoC could buy up large amounts of RMB and reclaim imported liquidity — but doing so persistently would dry up the offshore pool.)

After 2012, the balance of payments steadied and even saw moderate outflows: large financial-account outflows (falling FDI inflows, rising Chinese investment abroad, and portfolio outflows from Chinese equity and bond markets). RMB flowed back to the central bank, liquidity contracted, and with the economy slowing the PBoC’s task reversed — now to *inject* liquidity, chiefly by *lowering* the RRR. The broader lesson is that exports matter enormously for Chinese growth and that a trade deficit would be

damaging; China has run a trade surplus throughout, with deficits only in a few months of the trade war.

8.9.2 Open-market operations: repos

China's open-market operations work mainly through *repurchase agreements* (repos), in two forms, and the naming is the reverse of U.S. usage.

Repo and reverse repo (China naming)

In China, when the *central bank lends to commercial banks* the operation is a *reverse repo*; when the *central bank borrows from commercial banks* it is a *repo*. (This is the opposite of the U.S. convention.) Reverse repos require collateral, typically government bonds — again a swap of low-liquidity assets for high-liquidity ones, *not* an outright purchase or sale. Standard maturities are 7 days (the most common), 14 days, and 28 days; there is no overnight repo.

Repos are short-term funding. Daily reverse-repo volume is typically in the tens of billions, announced around 9 a.m., with new operations issued and old ones maturing every day; around major holidays and quarter-ends the central bank may do 14-day reverse repos to meet settlement demands. If the authorities wish to signal support for a particular class of bonds, they may *widen the range of eligible collateral*: when the market fears defaults on local-government financing-vehicle (*chengtou*) bonds, for instance, the central bank's accepting more of them as collateral sends a positive signal.

The *open-market-operation rate is the reverse-repo rate*. Although nominally set by bidding, it is in fact agreed in advance and fixed by the central bank, which simply “cuts” the rate directly — in contrast to the Fed, which only sets a target range.

8.9.3 MLF and LPR

Definition 8.11: Medium-term Lending Facility (MLF)

The *medium-term lending facility* (MLF, “spicy noodles”) is a one-year lending facility. The central bank typically conducts the MLF on the 15th of each month, announcing both the volume and the rate.

The MLF rate and the reverse-repo rate move together — usually the central bank cuts the MLF rate first — and these two are the most important policy rates: when we speak of “rate cuts” or “rate hikes,” we mean the MLF and reverse-repo rates. The MLF is the more powerful of the two. In an emergency the central bank may break the routine and cut the reverse-repo rate first, even outside the usual next-morning schedule: on 8 February 2020, the reverse-repo rate was announced at 9:15 a.m., before the stock market opened at 9:30, so the operation could move the market that day. The central bank can also conduct an extra MLF to stabilize markets — as at the end of November 2020, after a Henan coal-group bond default threatened turmoil.

Definition 8.12: Loan Prime Rate (LPR)

The *loan prime rate* (LPR) is the rate at which commercial banks lend to the real economy — specifically, the rate offered to their *prime* (best) clients; other clients pay the LPR plus a spread. The LPR is quote-based and is set as the *MLF rate plus a spread*:

$$\text{LPR} = \text{MLF rate} + \text{spread}.$$

The mechanics run on a monthly cycle: the MLF is conducted and its rate fixed on the 15th, and the LPR quote is fixed on the 20th. The central bank polls eighteen commercial banks, discards the extreme quotes, and takes the arithmetic mean of the spread. After the LPR reform, the transmission of policy to the real economy runs

$$\text{MLF rate} \rightarrow \text{LPR} \rightarrow \text{lending rate}.$$

Remark ().

The LPR replaced the old benchmark rate] The old *benchmark lending rate* was simply “parachuted in” by decree; the LPR is grounded in actual market quotes, and China no longer uses the benchmark rate. Within the LPR, the MLF rate plays the role the benchmark rate once did. The reform raised the degree of marketization, but the central bank still strongly influences the LPR: banks have little incentive to quote it *down* on their own, so around the 20th the central bank communicates frequently with them to transmit its intentions. The one-year LPR reflects short-term lending; the five-year LPR mainly reflects mortgages. The two usually move together but may be cut asymmetrically — a cut in the five-year LPR alone signals that policy is turning to support real estate.

8.9.4 SLF, PSL, and re-lending**Definition 8.13: Standing Lending Facility (SLF)**

The *standing lending facility* (SLF, “hot-and-sour noodles”) is a short-term facility, under one month, through which the central bank lends to financial institutions. It is open to *all* financial institutions and provides *emergency liquidity*, but at a very high rate. The 7-day SLF rate is regarded as the *upper bound* of the rate corridor. In practice it is used little.

All financial institutions may open accounts at the central bank, but at different tiers: a small bank may have an account only at a regional branch, while a primary dealer holds one at the head office. Together with the excess-reserve rate as the floor, the SLF rate as the ceiling defines the *interest-rate corridor* within which short-term rates are meant to trade.

Definition 8.14: Pledged Supplementary Lending (PSL)

Pledged supplementary lending (PSL) is a *directed* (structural) facility, extended to *policy banks* — chiefly the China Development Bank. It funds targeted government priorities through the policy-bank channel.

Remark (I).

PSL and the property cycle] PSL was central to the housing “destocking” campaign — clearing the inventory of unsold property, in part by resettling residents of urban villages into it. The campaign was so large that it drove a broad rise in house prices, especially in second- and third-tier cities. After the 2018 Central Economic Work Conference tightened the property stance with the slogan “houses are for living in, not for speculation,” PSL volume shrank. It rose again in 2022, chiefly to fund the CDB’s infrastructure push; after July 2022, amid large-scale property-debt defaults and the social-stability risks they posed, the central bank issued PSL to the CDB to finance the “ensure delivery of homes” effort.

Why not simply fund these priorities through ordinary government bonds, given that CDB-bond yields are close to Treasury yields? Because fiscal scale is set in advance: in early 2022 the budget was set small, and a looser fiscal stance cannot be enacted mid-year, so the fiscal impulse could only be delivered through the CDB. The CDB thus partly substitutes for the fiscal authority, bypassing the National People’s Congress process — which is, in essence, the central bank channeling funds to fiscal purposes. The underlying logic is again MMT: government debt is treated as essentially unconstrained, backed by an unlimited power to issue currency, so long as the state’s credit holds.

Definition 8.15: Re-lending

Re-lending is a structural facility in which the central bank channels funds to specific firms *through* commercial banks: a commercial bank lends to a designated firm, and the central bank then lends the same amount to that commercial bank. The scale of re-lending exceeds five trillion yuan; examples include re-lending to support agriculture, small and micro enterprises, and pandemic response.

8.9.5 Interbank overnight lending

Once funds reach a commercial bank, the bank manages its liquidity in the *interbank market*, where banks borrow and lend among themselves through various products, including repo transactions. It is important to keep two things separate.

Interbank lending versus central-bank repos

Interbank *overnight lending* exists so that *banks* can fund one another; central-bank *repos* exist so that the *central bank* can inject or withdraw liquidity. Central-bank repos have no overnight tenor — their shortest maturity is 7 days — whereas interbank

lending is by nature same-day overnight.

Definition 8.16: DR001 and DR007

DR001 is the overnight interbank lending rate — the Chinese analogue of the U.S. federal funds rate. Formally it is the weighted-average rate on overnight repos collateralized by interest-rate bonds, transacted between banks. *DR007* is the corresponding 7-day weighted-average repo rate. Both are *actual transaction* rates. (By contrast, Shibor is only a *quoted* rate and does not represent realized transaction prices.)

These benchmark rates close the loop. The central bank acts on base money and the policy rates (reverse repo, MLF); the banks transmit the impulse through the interbank market (DR001, DR007) and on to the real economy through the LPR and the loans built on top of it. That chain — from the central bank, through the banks, to firms and households — is the monetary transmission mechanism with which the chapter began, now filled in instrument by instrument.

Chapter 9

The Quantity Theory of Money and Inflation

Chapter 8 described how money is created — how the central bank and the commercial banks together expand the stock of currency and deposits that circulates in the economy. This chapter asks the natural sequel: what happens to the price level when that stock grows? The short answer, and the organizing idea of classical monetary economics, is that the price level is ultimately a monetary phenomenon. When the quantity of money rises faster than the quantity of goods, each unit of money chases more goods than before, and prices climb. A sustained, broad rise in the price level is what we call *inflation*, and over long horizons it is driven by the growth rate of money.

Before building the model that makes this precise, we pause on a prior question that students of macroeconomics almost always get backwards: *why does inflation matter at all?* The popular complaint is that inflation eats into our purchasing power, leaving us poorer. We will see that this intuition, taken literally, is largely a confusion — in a frictionless world incomes rise alongside prices, and the real costs of inflation lie elsewhere. Identifying where those costs really are, and where there are even some offsetting benefits, sharpens our understanding of what monetary policy can and cannot do. The chapter then develops the *quantity theory of money*, summarized by the equation $MV = PY$, derives a simple expression for the inflation rate, recasts the same equation as a theory of money demand, and closes with the *classical dichotomy* and the *neutrality of money* — the proposition that money does not affect real quantities in the long run. That proposition holds only when prices have fully adjusted; the breakdown of neutrality in the short run, when prices are sticky, is exactly the doorway into the Keynesian models of the following chapters.

9.1 Inflation as a Tax

A useful first lens on inflation is fiscal. When a government cannot or will not finance its spending through ordinary taxes or borrowing, it can simply print money to pay its bills. Each new unit of money it issues dilutes the purchasing power of every unit already in circulation. The holders of money are thereby made poorer in real terms, and the resources that money commanded are transferred to the government. The effect is indistinguishable

from a tax levied on money balances.

Definition 9.1: Seigniorage (the Inflation Tax)

Seigniorage is the real revenue a government obtains by issuing money. Equivalently, it is the *inflation tax*: when the money stock grows and the price level rises, the real value of the money the public holds falls, and that lost purchasing power accrues to the issuer. The “tax base” is the real money balances M/P the public chooses to hold, and the “tax rate” is the inflation rate π that erodes them.

This is the traditional view that inflation amounts to levying an inflation tax. It is worth holding onto because it explains why governments under fiscal stress so often resort to the printing press, and why episodes of hyperinflation are almost always episodes of fiscal collapse: the inflation tax is the revenue source of last resort. It also frames the central question of the quantity theory in fiscal language — if the government keeps expanding M to finance spending, what does that do to P ?

9.2 The Costs of Inflation

If inflation transfers resources and distorts decisions, where exactly are the costs borne? It is helpful to separate the genuine, first-order costs that economic theory identifies from the cost the public most fears, which turns out to be more apparent than real.

9.2.1 The Real Costs

1. **Shoe-leather costs (rising transaction costs).** When prices are expected to keep rising, holding money becomes expensive, and people economize on their money balances — making more frequent trips to the bank, holding wealth in interest-bearing or real assets, and shortening the horizon of contracts. Long-term contracts become harder to write because both parties must constantly account for the changing price level. The name evokes the wear on one’s shoes from extra trips to the bank; more broadly it captures all the real resources spent managing money holdings that would be unnecessary under stable prices.
2. **Menu costs (the cost of adjusting prices).** Firms must physically change their posted prices — reprinting menus and catalogues, relabeling shelves, reprogramming systems — and each such change consumes real resources. The higher the inflation rate, the more often prices must be revised, and the larger these costs grow.
3. **Distorted resource allocation.** Prices are the signals that guide resources to their most valued uses. When the general price level is rising, and rising at different rates for different goods, it becomes hard to tell a change in a good’s *relative* price — which should redirect resources — from a change in the overall price level, which should not. The *signaling role of prices fails*, and resources are misallocated as a result.
4. **Redistribution between debtors and creditors.** Inflation that is higher than was anticipated when a loan was signed lowers the *real* interest rate actually paid. A

debtor repays in money that is worth less than expected, while a creditor receives less real value than the contract promised. Inflation therefore redistributes wealth from creditors (lenders) to debtors (borrowers): *borrowers gain and lenders lose*.

9.2.2 Why “Inflation Shrinks My Income” is Not a First-Order Cost

Notice what is *absent* from the list above: the reason most people give for fearing inflation, namely that rising prices shrink the purchasing power of their incomes and so make them poorer. In theory this concern does not hold up. If all prices in the economy rise in proportion, then wages, rents, and profits — which are themselves prices — should rise in the same proportion. A worker whose wage and whose grocery bill both double is no worse off. Real income, the ratio of nominal income to the price level, is unchanged. Inflation that is uniform and fully anticipated does not by itself reduce anyone’s real income.

Remark (Where the popular worry becomes real).

The theoretical argument assumes all prices and incomes move together and instantly. Reality departs from this in two ways. First, inflation is often *structural*: different prices rise at different speeds, so the basket of goods a particular household buys may rise faster than its particular source of income. Second, wages and many prices are *sticky* — they do not adjust immediately. When the price of goods rises but a worker’s nominal wage is fixed by contract for a year, that worker’s real income genuinely falls for the duration. These frictions are exactly why the popular worry has bite in the short run. They are also the foundation of the Keynesian claim that fiscal and monetary policy can move real output in the short run: if prices adjusted instantly, there would be no short-run real effects to exploit. We return to this when we take up money neutrality at the end of the chapter, and develop it fully in Chapters 10–12.

9.3 The Benefits of Inflation

Inflation is not all cost. A moderate rate of inflation carries two benefits that are worth stating explicitly, both of which work through the same channel that produced the costs above.

1. **It stimulates the demand for credit.** By lowering the real interest rate on existing debt, inflation makes borrowing cheaper in real terms and encourages firms and households to take on loans and invest. A higher demand for credit can support spending and economic activity.
2. **It adds flexibility to the labor market.** Nominal wages are notoriously difficult to cut: workers resist outright pay reductions, and contracts and norms make them *downward-rigid*. Yet sometimes the labor market requires a fall in the real wage — for instance when a firm faces weaker demand and would otherwise have to lay workers off. Inflation provides a way out. If the price level rises while nominal wages hold steady, the *real* wage falls without anyone’s pay being formally cut. Lower real wages make it cheaper for firms to employ workers, raising labor demand and supporting employment. In this way moderate inflation greases the wheels of the labor market.

9.4 The Quantity Theory of Money

We now turn from why inflation matters to what determines it. The classical answer is the *quantity theory of money*, which begins from an accounting identity linking the quantity of money to the value of transactions it carries out.

Definition 9.2: The Quantity Equation

The *quantity equation* (or equation of exchange) states that the quantity of money times the speed at which it circulates equals the money value of transactions:

$$M \times V = P \times T,$$

where M is the nominal money stock, V is the *velocity of money* (the average number of times a unit of money changes hands per period), P is the price level, and T is the volume of transactions.

The volume of transactions T is hard to measure directly, so in practice we replace it with real output (real GDP), denoted Y . With this substitution the right-hand side $P \times Y$ is just *nominal* GDP, and the quantity equation becomes the form we will use throughout:

$$M \times V = P \times Y. \tag{9.1}$$

What turns this identity into a *theory* is a set of statements about which terms are free to move and which are fixed in the short run.

Assumption 9.3: The Quantity-Theory Assumptions

Output is set by the supply side. Real output Y is determined by the economy's potential — its capital, labor, and technology — and so is taken as given in the short run, independent of the money stock.

Velocity is set by habit. The velocity V is governed by the transaction technology and payment habits of the economy (how people are paid, how often, and how they hold money). These habits change only slowly, so V is roughly constant in the short run.

Remark (Velocity and the interest rate).

Velocity is not literally a constant. It rises when the opportunity cost of holding money rises: when the interest rate climbs — that is, when the price of liquidity goes up — people hold less money relative to their spending, so each unit of money turns over more often and V increases. The assumption is only that, over short horizons, V moves little compared with the swings in M and P that interest us. This dependence of V on the interest rate is precisely what the liquidity-preference reformulation in Section 9.5 will make explicit.

With V and Y fixed in the background, equation (9.1) ties the price level directly to the

money stock. To see this cleanly, take logarithms of both sides,

$$\log M + \log V = \log P + \log Y,$$

and differentiate with respect to time. Since the time derivative of $\log X$ is the growth rate $\Delta X/X$, this gives

$$\frac{\Delta M}{M} + \frac{\Delta V}{V} = \frac{\Delta P}{P} + \frac{\Delta Y}{Y}, \quad (9.2)$$

or in the language of percentage changes,

$$\% \Delta M + \% \Delta V = \% \Delta P + \% \Delta Y.$$

Equation (9.2) expresses the quantity equation as a relationship among *growth rates*: the sum of money growth and velocity growth equals the sum of inflation and real output growth. Two conclusions follow.

Two conclusions of the quantity theory

1. **Inflation is driven by money growth.** Holding the growth rates of velocity and real output fixed, any increase in the money growth rate $\Delta M/M$ passes through one-for-one to the inflation rate $\Delta P/P$. Sustained inflation is, at bottom, a consequence of sustained money creation.
2. **Money growth tracks nominal GDP growth.** Because $\Delta V/V \approx 0$, money growth $\Delta M/M$ moves roughly together with the growth of nominal GDP, $\% \Delta P + \% \Delta Y$. In practice the rate at which a central bank expands the money supply broadly matches the growth of nominal output.

Setting velocity growth $\Delta V/V$ to zero in (9.2) and recalling that the inflation rate π is the growth rate of the price level, $\pi = \Delta P/P$, we can solve for inflation directly:

$$\pi = \frac{\Delta M}{M} - \frac{\Delta Y}{Y}. \quad (9.3)$$

Definition 9.4: The Inflation Rate from Money Growth

Under constant velocity, the inflation rate equals the gap between money growth and real output growth,

$$\pi = \frac{\Delta M}{M} - \frac{\Delta Y}{Y}.$$

Equation (9.3) gives a rough but illuminating estimate of inflation. Read it as follows: in theory the quantity of money in circulation “ought” to grow in step with the economy’s real output, so that there is just enough money to carry out the additional real transactions at unchanged prices. The portion of money growth that *exceeds* real growth has no extra goods to chase, and so spills over into higher prices. That excess is the inflation rate.

9.5 Money Demand and Liquidity Preference

The quantity equation has so far been read as a theory of the price level. The same equation, rearranged, is also a theory of *money demand* — of how much money people wish to hold. Rearranging $MV = PY$ to isolate real balances,

$$\frac{M}{P} = \frac{Y}{V}. \quad (9.4)$$

The left-hand side M/P is the *real money supply* — the purchasing power of the money stock — and in equilibrium it equals the quantity of real balances the public demands. The quantity equation thus describes *liquidity preference*: the amount of money the economy needs to keep itself running. Two observations follow. First, the equation is fundamentally about money *demand*. Second, it characterizes an *equilibrium*: when the money market clears, the M in the equation represents both the supply of money and the demand for it.

Keynes generalized the right-hand side of (9.4) into an explicit money-demand function. Rather than pinning real balances to Y/V with a fixed velocity, he let the demand for real balances depend on income and on the interest rate through a *liquidity-preference* function L :

$$\left(\frac{M}{P}\right)^d = L(i, Y) = L(r + \pi^e, Y). \quad (9.5)$$

Here i is the nominal interest rate, which by the Fisher relation is the sum of the real interest rate r and expected inflation π^e , so $i = r + \pi^e$. The signs of the two arguments are central:

1. **Money demand is increasing in income Y .** A higher level of real income means more spending and more transactions to finance, and so a greater need for money to carry them out. Thus L rises with Y .
2. **Money demand is decreasing in the nominal interest rate i .** The interest rate is the opportunity cost of holding money: money held as cash or in non-interest-bearing accounts forgoes the return i available on bonds and other assets. When i rises, holding money becomes more costly, so people hold less of it — equivalently, velocity rises and each unit of money does more work. Thus L falls with i .

Remark (Why money demand uses *expected* inflation).

The opportunity cost that matters for a decision made *today* about how much money to hold over the coming period is the *nominal* interest rate, and the relevant nominal rate is the one that compensates for the inflation a saver *expects* to occur, not the inflation that will be realized after the fact. Money demand is an *ex ante* (forward-looking) concept, so it depends on *expected* inflation π^e , which is why the second argument of L is written $r + \pi^e$ rather than $r + \pi$.

On the other side of the market, the nominal money supply M is set by the central bank. We treat this as an exogenous, independent policy decision: monetary authorities choose M through administrative action, and that choice is taken as given by the rest of the model. Equilibrium in the money market requires that the real supply of money equal the real demand for it,

$$\frac{M}{P} = L(i, Y) = L(r + \pi^e, Y). \quad (9.6)$$

With the nominal money supply M fixed by policy, the interest rate i and the price level P adjust so that (9.6) holds. This equilibrium condition is the engine of the LM relation we will build in Chapter 11.

9.6 The Classical Dichotomy and the Neutrality of Money

We can now state the central long-run proposition of monetary theory, and the precise sense in which it can fail in the short run — the bridge to everything that follows.

Definition 9.5: The Classical Dichotomy

The *classical dichotomy* is the separation of economic variables into *real* quantities (output, employment, relative prices, the real interest rate) and *nominal* quantities (the money stock, the price level, nominal wages). The dichotomy holds that nominal variables do not affect real variables: changes in the quantity of money move the price level but leave the real allocation untouched.

If the classical dichotomy holds, then the price level cannot alter real economic behavior. A change in the money stock and the policy that produces it has no effect on output, employment, or consumption — they merely scale all nominal magnitudes up or down together. Money is then said to be *neutral*.

Theorem 9.6: Neutrality of Money

Money is *neutral* if a change in the nominal money stock changes all nominal variables in the same proportion and leaves all real variables unchanged. Neutrality holds when *every* price in the economy adjusts at the same time, in the same direction, and by the same proportion, so that no relative price is disturbed.

The condition for neutrality is demanding: it requires that all prices — goods prices, wages, asset prices — move together, instantaneously and proportionately. When that condition is met, doubling the money stock simply doubles the price level and every nominal wage and contract, leaving real wages, real output, and the real interest rate exactly where they were. This is the same neutrality result encountered in the simple monetary model of Chapter 8, now stated at the level of the whole economy.

Reality, however, is full of frictions. Prices and wages are sticky: they are set in advance, revised infrequently, and do not all move at once. When the money stock changes but prices have not yet caught up, relative prices are temporarily disturbed, and the change in money *does* move real quantities. In the short run, therefore, *money is not neutral*, and monetary policy can influence the level of output and employment.

Neutrality is a long-run, not a short-run, property

- **Short run:** prices are sticky, the classical dichotomy fails, and money is *not* neutral. Monetary and fiscal policy can move real output and employment.
- **Long run:** given enough time, every sticky price adjusts until the economy reaches a new equilibrium with no remaining price rigidity. The classical dichotomy is re-

stored and money is neutral. Policy moves only the price level, not real quantities.

This distinction is the organizing principle of the rest of the course. The economy has a *long-run* equilibrium, anchored at potential output and governed by the supply side, where money is neutral and the classical dichotomy holds. It also has *short-run* fluctuations — the business cycle — during which sticky prices let demand-side disturbances (and, less controllably, supply-side shocks such as movements in total factor productivity) push the economy away from potential. Over time, adjustment in prices, the interest rate, and related variables pulls a short-run equilibrium that has strayed from potential output back toward it.

Remark (Keynes on the short run and the long run).

Keynes built his theory on exactly this division between a short run and a long run, distinguished by whether prices are sticky. In the short run, supply is flexible and prices are rigid, so monetary and fiscal policy can shift aggregate demand and thereby change the level of output; money is not neutral. In the long run, prices are fully flexible, aggregate supply is rigid at potential output, and money is neutral. The promise of the Keynesian framework is that policy can smooth the short-run fluctuations of the business cycle even though it cannot change the economy's long-run potential. The models of the following chapters — the Keynesian cross (Chapter 10), IS-LM (Chapter 11), and aggregate demand and supply (Chapter 12) — are the machinery that makes this promise operational.

Chapter 10

The Keynesian Cross and the Multiplier

The growth models of the preceding chapters described where the economy is headed. In the Solow and neoclassical frameworks (Chapters 5 and 6), and again whenever we wrote

$$Y = F(K, L, \text{TFP}),$$

output is determined by the supply side: by the capital stock, the labor force, and the technology that turns them into goods. This is the right picture for the *long run*, when prices and wages have had time to adjust and the economy sits at its productive potential. But it says nothing about why output rises and falls from one year to the next, why recessions happen, or why a government might think a tax cut could pull the economy out of a slump. For those questions we need a model of the *short run*, in which output is determined not by what the economy is able to produce but by how much buyers want to spend.

This chapter develops the simplest such model, due to Keynes. Its organizing identity is the demand-side decomposition of output already familiar from the national accounts (Chapter 2),

$$Y = C + I + G + \text{NX},$$

read now as a statement about *planned* spending rather than realized accounting. We will specify how each of the four components — consumption, investment, government purchases, and net exports — depends on income, add them into a single planned-expenditure schedule, and impose the requirement that planned spending equal output. The result is the *Keynesian cross*, a diagram whose intersection pins down equilibrium output. Its most consequential lesson is the *multiplier*: because part of any extra spending becomes someone's income and is partly spent again, a small push to demand moves output by more than the push itself.

A word of warning about what this model is. Keynes's framework is, at bottom, *static*. It contains no clean notion of the future: consumption responds only to *current* income, investment is treated as independent of current output, and intertemporal considerations such as interest rates and lifetime wealth are either suppressed or folded into a single behavioral parameter. That is a genuine limitation, and later chapters (the IS–LM model of

Chapter 11 and the dynamics of Chapter 14) restore some of what is missing here. But the very crudeness of the model is also the source of its durability: it captures the one force — spending out of current income — that the short-run data turn out to display most strongly, and it does so in a form simple enough to reason with.

10.1 Two Horizons for Output

It is worth stating plainly the contrast that motivates everything below. In the long run, output is a supply-side quantity: $Y = F(K, L, \text{TFP})$ is fixed by the stocks of factors and the state of technology, and demand can do no more than determine prices. In the short run, with prices sticky and factors potentially idle, the causation runs the other way. The level of demand — how much households, firms, the government, and foreigners wish to buy — determines how much firms actually produce, and hence output and employment.

The accounting identity $Y = C + I + G + \text{NX}$ is true by definition in *every* period, because the national accounts record realized purchases. The Keynesian move is to reinterpret the right-hand side as *planned* or *desired* expenditure and to ask when plans are consistent with the output they presuppose. Because spending corresponds to demand, we will treat planned expenditure as the economy's aggregate demand,

$$\text{AD} = C + I + G + \text{NX},$$

identifying the two without further apology. The identification is not perfectly rigorous — aggregate demand in a fuller model also responds to the price level and the interest rate — but in the short run demand fluctuates so much, and so much faster than potential output, that for the purposes of this chapter we may treat short-run movements in output as movements in demand and need not separate a short-run from a long-run schedule.

10.2 The Components of Planned Expenditure

We now write down a behavioral equation for each of the four components, building up the planned-expenditure schedule one term at a time. Throughout, Y denotes income (equivalently output), T denotes taxes net of transfers, and $Y - T$ is therefore *disposable income*, the income households have left to spend or save.

10.2.1 Consumption

What determines how much a household consumes? In principle, a great deal: its current income, its expected future income (in the limit, its entire *lifetime* income), its accumulated wealth (a stock, positively related to consumption), and intertemporal terms such as the interest rate and the rate at which it discounts the future. Keynes's model deliberately ignores almost all of this. Because it has no real dynamic structure — no future periods in which to enjoy saved resources — it cannot represent interest rates or lifetime income as separate forces. Consumption is made to depend on *current* disposable income alone:

Definition 10.1: The Keynesian Consumption Function

Household consumption is a linear function of current disposable income,

$$C = C_0 + \text{MPC} (Y - T),$$

where $C_0 > 0$ is *autonomous consumption* — the amount consumed even at zero disposable income — and $\text{MPC} \in (0, 1)$ is the *marginal propensity to consume*, the fraction of each additional dollar of disposable income that is spent. The term $\text{MPC}(Y - T)$ is *induced consumption*, the part driven by income.

The crucial modeling convention is that *every variable is current*: there is no expected future income in the equation. Whatever role the future, wealth, or interest rates play in shaping behavior is absorbed wholesale into the single parameter MPC. A household that is patient, wealthy, or optimistic about the future simply has a different MPC; the model does not say where that number comes from.

Remark (A crude model that fits the data).

The consumption function is plainly incomplete — it omits wealth, expectations, and the interest rate, all of which a richer theory would include. Keynes did not claim otherwise. His defense was empirical: the function is built from observation rather than from first principles, and disposable income is, in fact, the single factor that explains the most variation in aggregate consumption. One should read $C = C_0 + \text{MPC}(Y - T)$ as the *base* model, to which further terms (an interest-rate effect, a wealth effect) can be appended as the application demands.

Remark (Tax cuts and the marginal propensity to consume in China).

The size of the MPC has direct policy consequences, and they are institution-specific. Chinese tax relief has historically operated mainly through cuts to *corporate* taxes — which is also where the bulk of revenue is raised — rather than through transfers to individuals; direct cash subsidies to households are uncommon. Combined with a comparatively low household MPC, this means a given tax cut tends to feed through to consumption only weakly. Investment-based stimulus, by contrast, can act on both margins: in the short run it raises employment and demand, and in the long run it can shift the supply side by adding to productive capacity.

Saving is simply the part of disposable income not consumed, so the consumption function immediately implies a *saving function*:

$$S = (Y - T) - C = -C_0 + (1 - \text{MPC})(Y - T).$$

The slope $1 - \text{MPC}$ is the *marginal propensity to save*: every dollar of disposable income is either spent or saved, so the two propensities sum to one.

10.2.2 Investment

A firm undertakes an investment project if it is profitable to do so. The natural criterion is net present value: discount the project's future income stream and its future cost stream back to the present and invest when the former exceeds the latter,

$$\text{NPV}(I) = \sum_{t=1}^{\infty} \frac{\text{Income}_t}{(1+r)^t} - \sum_{t=1}^{\infty} \frac{\text{Cost}_t}{(1+r)^t}.$$

The trouble is the income side. Future revenues are deeply uncertain and largely resist quantification; in practice the decision rests on something closer to instinct than to calculation — the famous *animal spirits* Keynes invoked, by analogy with an animal's impulse to pursue prey. The cost side is more tractable. Investment must be financed, and the cost of finance rises with the interest rate r , so higher rates make fewer projects clear the hurdle. We therefore model investment as a decreasing function of the interest rate,

$$I = I(r), \quad I'(r) < 0,$$

without committing to a particular functional form — Keynes himself proposed none.

Investment does not depend on current income

In the Keynesian model, investment is a function of the interest rate alone: $I = I(r)$, decreasing in r . It does *not* depend on current output, household income, or wealth. Within the static model this is a deliberate simplification that keeps the determinants of I and Y separate.

Remark (The real-world caveat).

The assumption that investment is independent of current income is a clean theoretical stance, not a literal description of the world. In practice investment is strongly correlated with output: it responds to expectations of future demand, to government policy, to the wealth and confidence of those making the decisions, and to the availability of internal funds. Treating I as exogenous to Y is best read as a way of isolating one channel at a time; it is relaxed once we let investment respond to the interest rate that the IS–LM model of Chapter 11 determines jointly with output.

10.2.3 Government Purchases

Government purchases of goods and services, G , are taken to be the outcome of a political and administrative process rather than a response to current economic variables. They are set by decision, not by the model, and we therefore treat G as *exogenous*: a constant, to be chosen by the policymaker. Taxes T are treated the same way, as a policy instrument set from outside the model.

10.2.4 Net Exports

Net exports are exports minus imports,

$$NX = X - M.$$

Exports X are purchases of domestic goods by foreigners and so depend on *foreign* income, with which they are positively related; from the point of view of the domestic model they are exogenous. Imports M are domestic purchases of foreign goods and rise with domestic disposable income,

$$M = \text{MPM} (Y - T),$$

where MPM is the *marginal propensity to import*, the fraction of each additional dollar of disposable income spent on foreign goods. For much of the analysis it is convenient, and common, to suppress this income dependence and treat NX as an exogenous constant as well; we will do so when deriving the multiplier and note in a remark how the income-sensitive case modifies the result.

10.3 Planned Expenditure and the Keynesian Cross

Adding the four components gives the planned-expenditure schedule as a function of income. Writing it out with the income-dependent terms displayed,

$$AD = C + I + G + NX = \underbrace{C_0 + I(r) + G + X}_{\text{autonomous}} + \underbrace{(\text{MPC} - \text{MPM})}_{\text{slope}} (Y - T).$$

The schedule is increasing in income but with a slope strictly less than one: a dollar more of income raises planned spending by $\text{MPC} - \text{MPM} < 1$, because part of the extra dollar is saved and part leaks abroad. In a diagram with planned expenditure on the vertical axis and income Y on the horizontal axis, this is an upward-sloping line that is flatter than the 45° line. (If imports are folded into the exogenous constant, the slope is simply $\text{MPC} < 1$; what matters is only that it is positive and below one.)

Equilibrium requires that plans be self-consistent: firms produce exactly what buyers wish to buy, so that planned expenditure equals output and there is no unintended accumulation or run-down of inventories.

Definition 10.2: Short-Run Equilibrium

The economy is in short-run equilibrium when planned aggregate expenditure equals output,

$$AD = Y,$$

that is, when the planned-expenditure schedule meets the 45° line along which expenditure and income are equal.

The 45° line is the locus of all points at which the quantity demanded equals the quantity produced; it is the constraint that defines equilibrium. Because the planned-expenditure schedule has slope below one while the 45° line has slope exactly one, the two lines must

cross exactly once. Their intersection is the equilibrium level of output. This diagram is the *Keynesian cross*, shown in Figure 10.1.

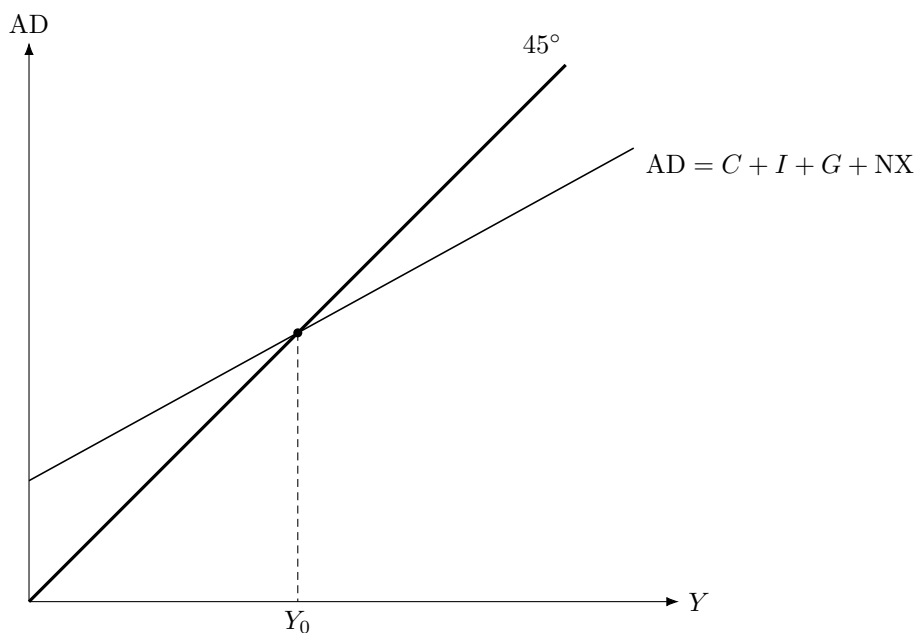


Figure 10.1: The Keynesian cross: equilibrium output occurs where the planned-expenditure schedule (slope below one) crosses the 45° line along which expenditure equals income.

10.3.1 Stability of the Equilibrium

The intersection is not merely a point where the equations happen to balance; it is a *stable* resting point, toward which the economy is pushed from either side. The adjustment mechanism runs through inventories and the production decisions of firms. Suppose output is temporarily above its equilibrium level, $Y_1 > Y_0$. At Y_1 the planned-expenditure line lies *below* the 45° line, so planned spending falls short of output: buyers want less than firms have produced. Goods pile up as unsold inventory, firms cut back production, and output drifts down toward Y_0 . Conversely, if output is too low, $Y_2 < Y_0$, planned spending exceeds output, inventories are drawn down, firms expand production, and output rises back toward Y_0 . From either side the economy converges to the crossing point, as illustrated in Figure 10.2.

Remark (Equilibrium output need not be potential output).

In the long run the Keynesian-cross equilibrium ought to coincide with potential output — the supply-determined level $Y = F(K, L, TFP)$. In the short run there is no such guarantee. The economy can settle at a stable equilibrium below potential, with idle capacity and unemployed labor, or above it, with the economy overheating. This gap between the equilibrium the model delivers and the potential output the long-run model delivers is precisely what gives short-run demand management its rationale.

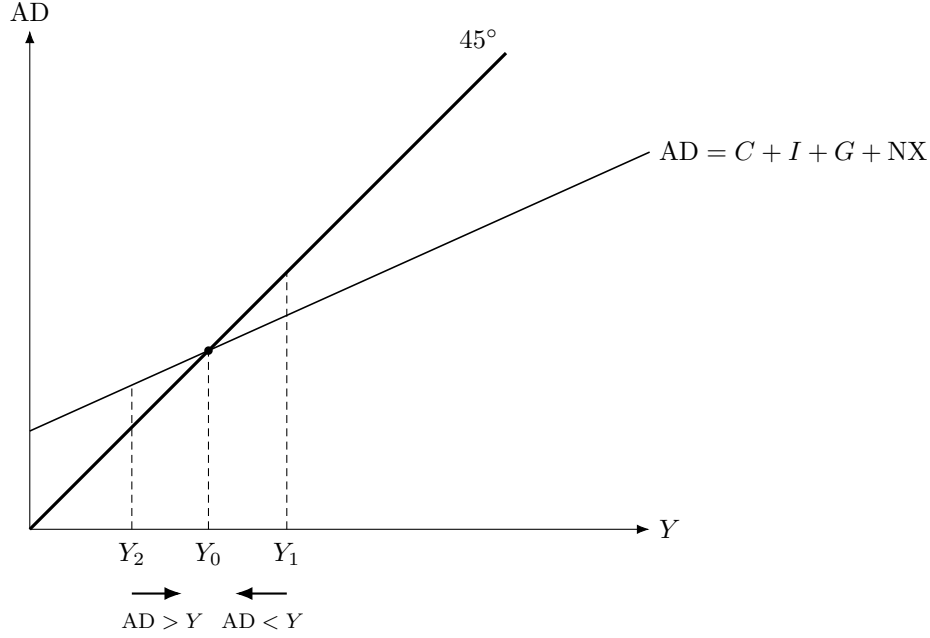


Figure 10.2: Stability of the Keynesian-cross equilibrium: above Y_0 planned spending falls short of output and firms cut back; below Y_0 spending exceeds output and firms expand; output converges to Y_0 from either side.

10.4 The Multiplier

The relative slopes of the two lines do more than guarantee that they cross. They also govern *how far* the crossing point moves when the planned-expenditure schedule shifts. Suppose some component of autonomous spending rises, shifting the whole schedule up by an amount ΔD . Because the schedule is flatter than the 45° line, the horizontal distance the intersection travels, ΔY , exceeds the vertical shift ΔD that caused it: a small increase in autonomous spending buys a larger increase in output. This amplification is the *multiplier effect*, and the factor by which the shift is magnified is the *multiplier*. The geometry is shown in Figure 10.3.

To find the multiplier algebraically, treat NX as an exogenous constant and impose the equilibrium condition $AD = Y$ on the planned-expenditure schedule

$$AD = C_0 + MPC(Y - T) + I + G + NX.$$

Setting $AD = Y$ and solving for Y :

$$\begin{aligned} Y &= C_0 + MPC(Y - T) + I + G + NX \\ Y - MPCY &= C_0 - MPC T + I + G + NX \\ (1 - MPC)Y &= C_0 - MPC T + I + G + NX, \end{aligned}$$

which gives the reduced-form solution for equilibrium output,

$$Y = \frac{1}{1 - MPC} (C_0 - MPC T + I + G + NX). \quad (10.1)$$

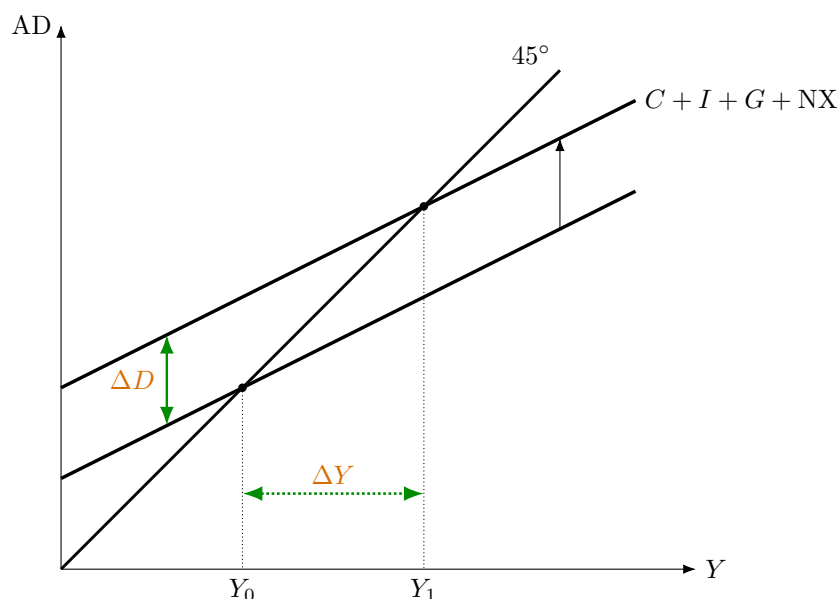


Figure 10.3: The multiplier: an upward shift ΔD of the planned-expenditure schedule raises equilibrium output by $\Delta Y > \Delta D$, because the schedule is flatter than the 45° line.

Each multiplier is now read off as a partial derivative of Y with respect to the corresponding exogenous variable.

Theorem 10.3: The Keynesian Multipliers

In the equilibrium (10.1), the responses of output to the exogenous components are:

$$\text{Spending multiplier: } \frac{\partial Y}{\partial G} = \frac{\partial Y}{\partial I} = \frac{\partial Y}{\partial C_0} = \frac{1}{1 - \text{MPC}} > 1,$$

$$\text{Tax multiplier: } \frac{\partial Y}{\partial T} = \frac{-\text{MPC}}{1 - \text{MPC}} < 0,$$

$$\text{Balanced-budget multiplier: } \left. \frac{dY}{dG} \right|_{dT=dG} = 1.$$

The spending multiplier exceeds one for any $\text{MPC} \in (0, 1)$; the tax multiplier is negative, and its *magnitude* $\frac{\text{MPC}}{1 - \text{MPC}}$ exceeds one if and only if $\text{MPC} > \frac{1}{2}$.

Proof. The spending multipliers follow by differentiating (10.1): since C_0 , I , G , and NX all enter with coefficient $1/(1 - \text{MPC})$, each has the same multiplier $1/(1 - \text{MPC})$, which exceeds one because $0 < \text{MPC} < 1$. The economic content is the round-by-round leakage story: an extra dollar of autonomous spending becomes a dollar of someone's income, of which a fraction MPC is re-spent, of which MPC is re-spent again, and so on. The induced output is the geometric series

$$1 + \text{MPC} + \text{MPC}^2 + \cdots = \frac{1}{1 - \text{MPC}}.$$

Differentiating (10.1) with respect to T gives the tax multiplier $-\text{MPC}/(1 - \text{MPC})$, negative because a tax increase cuts disposable income and hence spending. Its magnitude is smaller than the spending multiplier by exactly the factor MPC : a tax change acts on demand only after passing through the consumption function, so its first-round effect is $\text{MPC} \Delta T$ rather than the full ΔG . Solving $\text{MPC}/(1 - \text{MPC}) > 1$ gives $\text{MPC} > 1 - \text{MPC}$, i.e. $\text{MPC} > \frac{1}{2}$.

For the balanced-budget multiplier, increase spending and taxes together by the same amount, $\Delta G = \Delta T \equiv \Delta$. By the two multipliers just found,

$$\Delta Y = \frac{1}{1 - \text{MPC}} \Delta G - \frac{\text{MPC}}{1 - \text{MPC}} \Delta T = \frac{1 - \text{MPC}}{1 - \text{MPC}} \Delta = \Delta.$$

A fully tax-financed increase in government purchases raises output by exactly the amount of the increase: the balanced-budget multiplier is one. $\square$

The intuition behind the difference between the spending and tax multipliers is worth dwelling on, because it underlies the entire fiscal-policy debate. An increase in G or in I pushes the planned-expenditure schedule up by its full amount and so is magnified by the full factor $1/(1 - \text{MPC})$. A tax cut, by contrast, raises spending only indirectly: households receive the extra disposable income but spend only a fraction MPC of it, saving the rest. The schedule shifts up by only $\text{MPC} |\Delta T|$, so the tax change is the weaker instrument dollar for dollar. The balanced-budget result then says something striking: even a stimulus that is fully paid for by raising taxes is not neutral — it still raises output one-for-one, because the spending injection works in full while the tax withdrawal is partly absorbed by lower saving.

Remark (Net exports that respond to income).

If imports are kept income-sensitive rather than folded into the constant, $\text{NX} = X - \text{MPM}(Y - T)$, the equilibrium condition becomes $Y = C_0 + \text{MPC}(Y - T) + I + G + X - \text{MPM}(Y - T)$, and solving gives

$$Y = \frac{1}{1 - \text{MPC} + \text{MPM}} (C_0 - (\text{MPC} - \text{MPM})T + I + G + X).$$

The spending multiplier is now $1/(1 - \text{MPC} + \text{MPM})$, smaller than before: imports are an extra leakage, so each round of re-spending loses not only the saved fraction but also the part spent abroad. The qualitative lessons — a positive spending multiplier above one, a smaller-magnitude negative tax multiplier — are unchanged.

Why the multiplier matters for policy

The multiplier says that the effect of a change in spending, taxes, or investment is not confined to its own size: a small change in autonomous demand levers a larger change in output. When assessing any of these, one must account for the multiplied effect, not just the direct one. The asymmetry between the multipliers also has a policy edge — for a given budgetary cost, an increase in government purchases moves output more than a tax cut of the same size. This is one reason that, when the economy faces downward pressure, the typical response is a fiscal expansion weighted toward

spending. We return to fiscal policy in detail in Chapter 13.

The Keynesian cross is, however, only half the short-run story. We have taken the interest rate r as fixed when computing investment, and with it the whole autonomous block. But the interest rate is itself an equilibrium price, set in the market for money. Allowing r to vary traces out a whole schedule of equilibrium output levels — one Keynesian-cross equilibrium for each interest rate — and combining that schedule with the money market is the task of the IS–LM model in Chapter 11.

Chapter 11

The IS–LM Model

The Keynesian cross of Chapter 10 determines output for a *given* interest rate: it tells us where planned expenditure crosses the 45° line, but it takes the interest rate r as a number handed down from outside. Yet r is itself an equilibrium price — the price of liquidity — and it moves when output moves. The IS–LM model closes this loop. It holds the price level fixed (we are still in the short run) and asks for the pair (Y, r) at which two markets clear simultaneously: the market for goods, summarized by the *IS* curve, and the market for money, summarized by the *LM* curve. Fiscal policy will turn out to act on the first curve and monetary policy on the second, and the interaction of the two is what makes the simple multiplier of the previous chapter an overstatement.

11.1 The IS Curve

The IS curve is the set of all (Y, r) pairs at which the goods market is in equilibrium — that is, at which planned expenditure equals output. Every point on it is a Keynesian-cross equilibrium for a particular interest rate. To see its shape, recall that investment falls when the interest rate rises, $I = I(r)$ with $I'(r) < 0$. Solving the Keynesian cross with investment written as a function of r ,

$$Y^* = \frac{1}{1 - \text{MPC}} (C_0 - \text{MPC} \cdot T + I(r) + G + \text{NX}),$$

so a lower interest rate raises investment, shifts the planned-expenditure line upward, and — magnified by the multiplier — raises equilibrium output. Output and the interest rate therefore move in opposite directions along the goods-market equilibrium locus.

Definition 11.1: The IS Curve

The *IS curve* is the locus of (Y, r) pairs at which planned expenditure equals output (the goods market clears). Because a lower interest rate raises investment and hence equilibrium output, the IS curve slopes *downward*.

Figure 11.1 stacks the two panels that produce it. Anything other than the interest rate that shifts the Keynesian-cross equilibrium — a change in G , in taxes, in autonomous

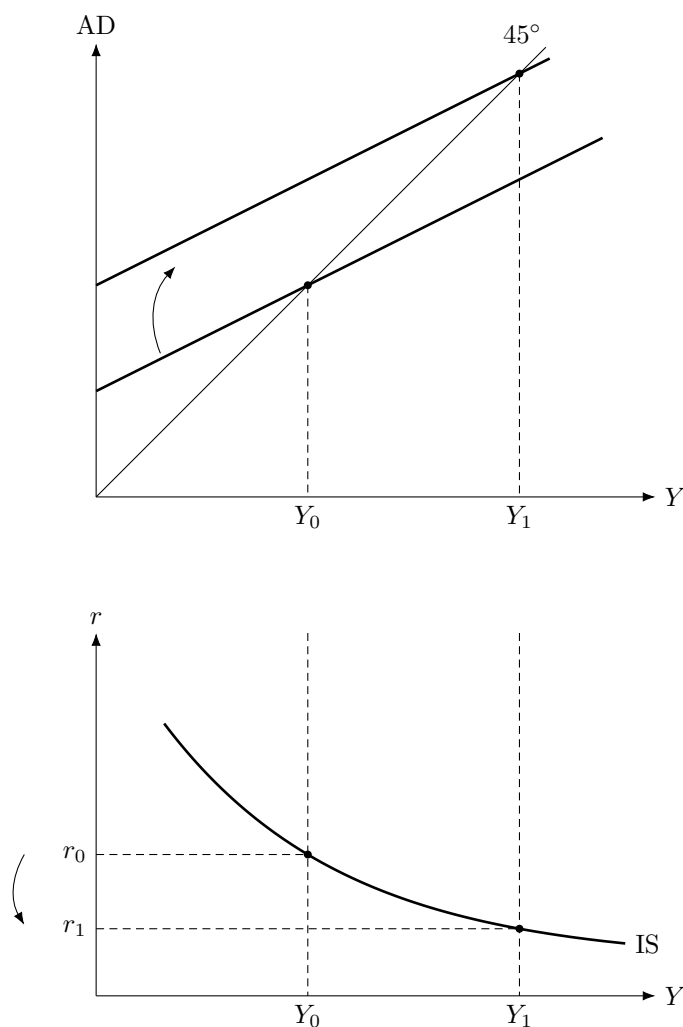


Figure 11.1: Deriving the IS curve. In the upper panel a fall in the interest rate from r_0 to r_1 raises investment, shifting planned expenditure up and moving the Keynesian-cross equilibrium from Y_0 to Y_1 . Plotting each interest rate against the output it implies (lower panel) traces out the downward-sloping IS curve.

consumption or investment, or in net exports — shifts the *whole* IS curve; a change in the interest rate is a movement *along* it.

11.2 The LM Curve

The LM curve comes from the money market. The quantity theory, $MV = PY$, can be read as a statement about money demand: to carry out a nominal volume of transactions PY at a velocity V , the public wishes to hold

$$M^d = \frac{PY}{V}.$$

Velocity is not a constant here; it rises with the interest rate, $V = V(r)$, $V'(r) > 0$, because a higher interest rate raises the opportunity cost of holding non-interest-bearing money, so each unit of money is turned over faster and less of it is demanded. The supply of money, by contrast, is set by the central bank and is independent of the interest rate: in the diagram it is a vertical line.

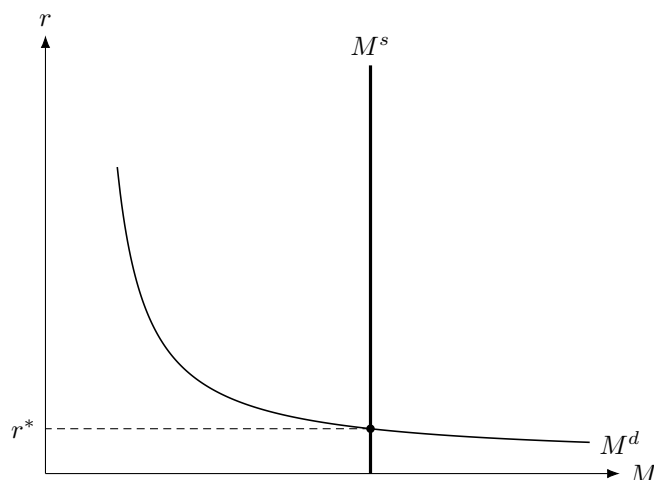


Figure 11.2: The money market. The money supply M^s is vertical (set by the central bank); money demand M^d slopes down in the interest rate. Their intersection fixes the equilibrium interest rate.

Equilibrium in Figure 11.2 pins down the interest rate that clears the money market for a given level of output and prices. Now raise output. Higher Y means more transactions and thus greater money demand at every interest rate, shifting the demand curve to the right. With supply fixed, the interest rate must rise to choke money demand back to the available stock.

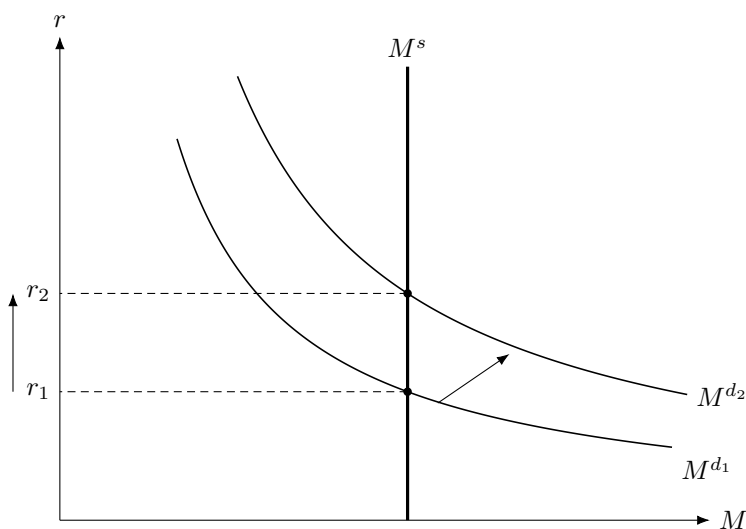


Figure 11.3: A rise in output shifts money demand to the right; with the money supply fixed, the equilibrium interest rate rises from r_0 to r_1 .

Collecting these (Y, r) pairs — higher output requiring a higher interest rate to keep the money market in balance — traces out the LM curve.

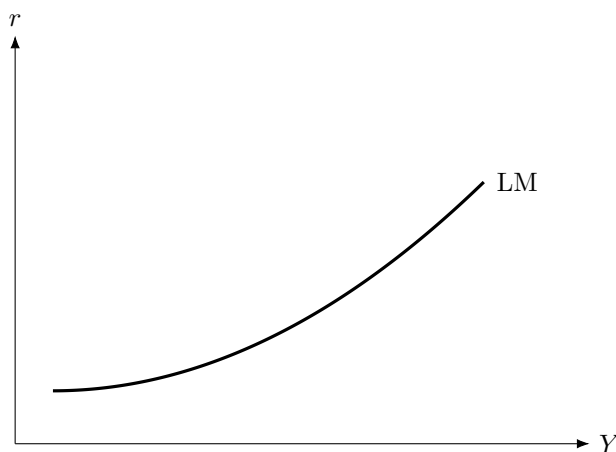


Figure 11.4: The LM curve: the locus of output and interest-rate pairs at which the money market clears. It slopes upward because higher output raises money demand and hence the market-clearing interest rate.

Definition 11.2: The LM Curve

The *LM curve* (liquidity preference–money supply) is the locus of (Y, r) pairs at which the money market clears for a given real money supply. Because higher output raises money demand and therefore the equilibrium interest rate, the LM curve slopes *upward*.

What shifts the entire LM curve? Anything that changes the real money supply M/P . An increase in the nominal money supply means that, at any level of output, the market clears at a lower interest rate, so the LM curve shifts down (equivalently, to the right).

A rise in the price level works in the opposite direction. The relevant quantity is the *real* money supply M/P : a higher P shrinks M/P exactly as a cut in the nominal money supply would, raising money demand relative to supply and pushing the LM curve up.

Reading the two curves

The IS curve summarizes the goods market and is shifted by *fiscal* variables (G , T , autonomous spending). The LM curve summarizes the money market and is shifted by *monetary* variables — the nominal money supply and the price level, both acting through the real money supply M/P .

11.3 Equilibrium

Plotting the two curves together, with output on the horizontal axis and the interest rate on the vertical axis, the unique crossing point gives the pair (Y_0, r_0) at which *both* markets

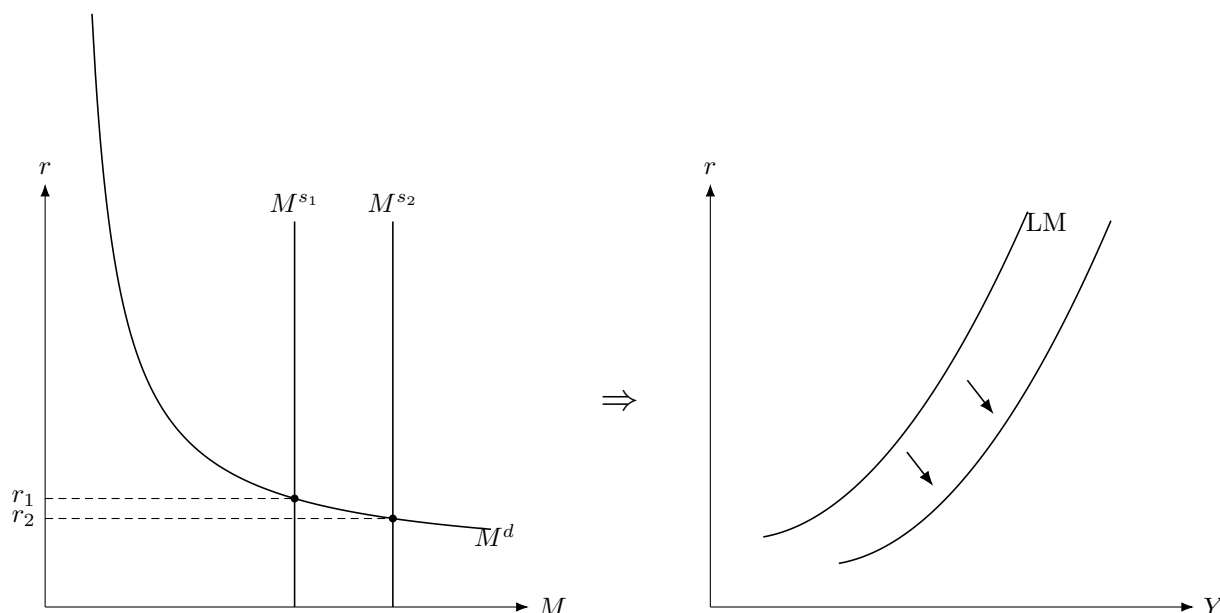


Figure 11.5: An increase in the money supply (from M^{s1} to M^{s2}) lowers the market-clearing interest rate at each level of output, shifting the LM curve down and to the right.

clear. The arrow of causation runs both ways: the interest rate helps determine output through investment, and output helps determine the interest rate through money demand.

The equilibrium is stable. Suppose some force pushed output above Y_0 — say a policy interest rate set too low. At that higher output, money demand would outstrip the fixed supply and the interest rate would be bid up, choking off investment until output returned to the equilibrium. A change in the interest rate or in output moves the economy *along* the curves; a change in any other determinant shifts a curve bodily.

11.4 Fiscal Policy in IS–LM

Consider an increase in government purchases. In the goods market this shifts the IS curve to the right. If the interest rate stayed at r_0 , output would rise all the way to Y_1 , the full Keynesian-cross multiplier acting on the larger G . But the move along the new IS curve toward its intersection with the unchanged LM curve tells a more sober story: as output rises, money demand rises, the interest rate is pushed up to r_2 , and the higher interest rate crowds out private investment, pulling output back from Y_1 to Y_2 , with $Y_0 < Y_2 < Y_1$.

Definition 11.3: The (First) Crowding-Out Effect

The *crowding-out effect* is the reduction in private investment caused by the rise in the interest rate that a fiscal expansion induces through the money market. It makes the realized output gain ($Y_2 - Y_0$) smaller than the naive multiplier prediction ($Y_1 - Y_0$).

A tax cut shifts the IS curve outward in the same way (through the tax multiplier

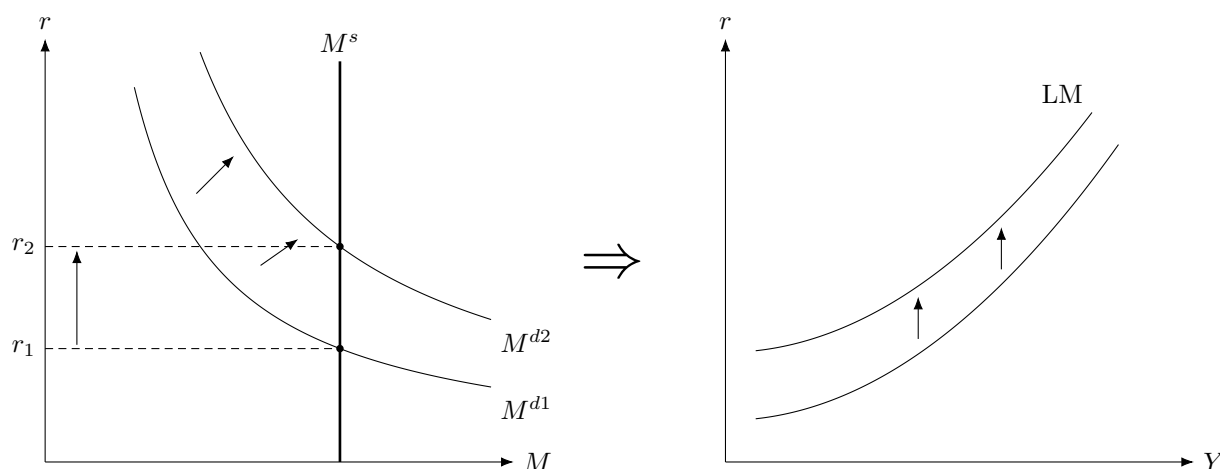


Figure 11.6: A rise in the price level raises nominal money demand (lowers the real money supply M/P); the market-clearing interest rate rises at each level of output and the LM curve shifts up.

rather than the spending multiplier), and the same crowding-out logic applies. The unifying statement is that *fiscal policy acts on the IS curve*.

11.5 Monetary Policy in IS-LM

Now let the central bank increase the money supply. The IS curve is unaffected; the LM curve shifts down, because more money clears the market at a lower interest rate for any output. At the initial output the interest rate would fall; the lower rate stimulates investment and output rises along the IS curve to Y_1 , with a new, lower equilibrium interest rate r_1 .

A rise in the price level runs the same machinery in reverse. Because it reduces the real money supply M/P , it shifts the LM curve up, raising the interest rate and lowering output.

This last experiment is worth pausing on, because it is the hinge between IS-LM and the aggregate-demand curve of the next chapter: *a higher price level is equivalent, for the money market, to a smaller money supply*. Holding the nominal money stock fixed and varying P therefore traces a relationship between the price level and equilibrium output — which is exactly aggregate demand.

Summary

Fiscal policy shifts the IS curve; monetary policy (and the price level) shift the LM curve. A fiscal expansion raises output but also the interest rate, crowding out investment; a monetary expansion raises output by lowering the interest rate. In every case the interest rate adjusts so that both markets clear at once.

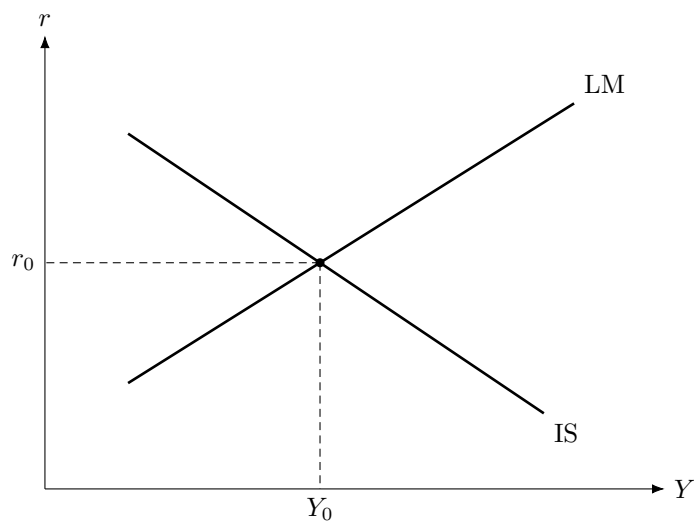


Figure 11.7: IS-LM equilibrium: the downward-sloping IS curve and the upward-sloping LM curve cross at the unique pair (Y_0, r_0) at which both the goods and money markets clear.

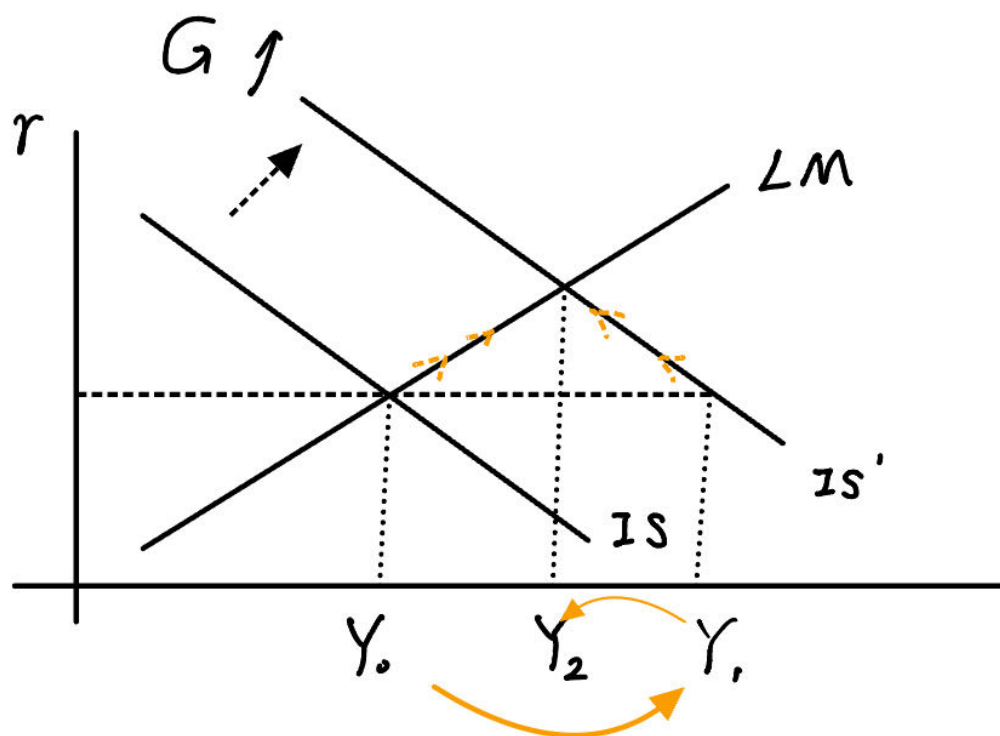


Figure 11.8: A rise in government purchases shifts the IS curve to the right. At a fixed interest rate output would reach Y_1 ; but the induced rise in the interest rate to r_2 crowds out investment, so the new equilibrium is at Y_2 , with $Y_0 < Y_2 < Y_1$. The gap $Y_1 - Y_2$ is the first crowding-out effect.

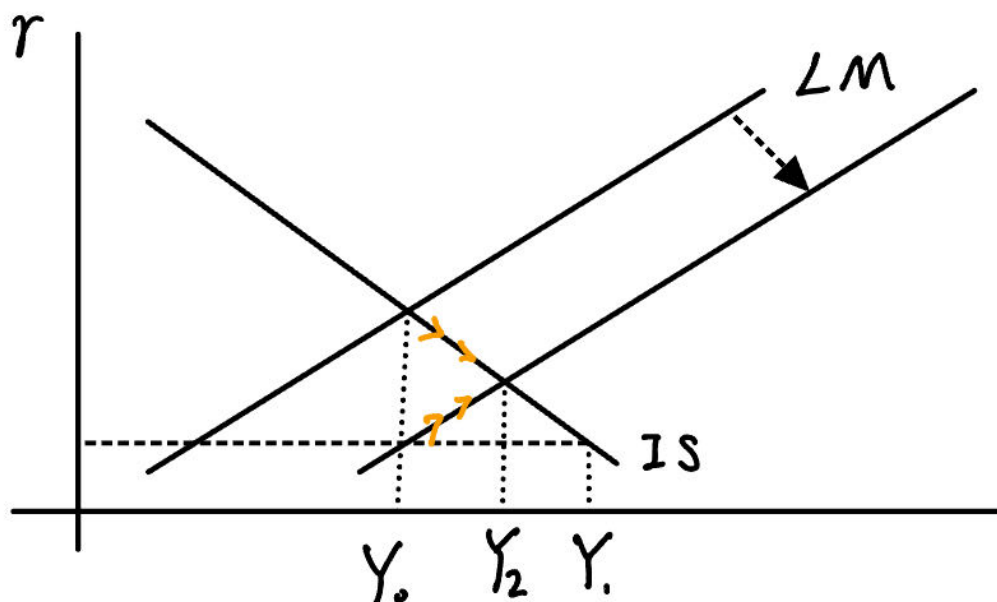


Figure 11.9: An increase in the money supply shifts the LM curve down. Output rises from Y_0 to Y_1 and the interest rate falls from r_0 to r_1 .

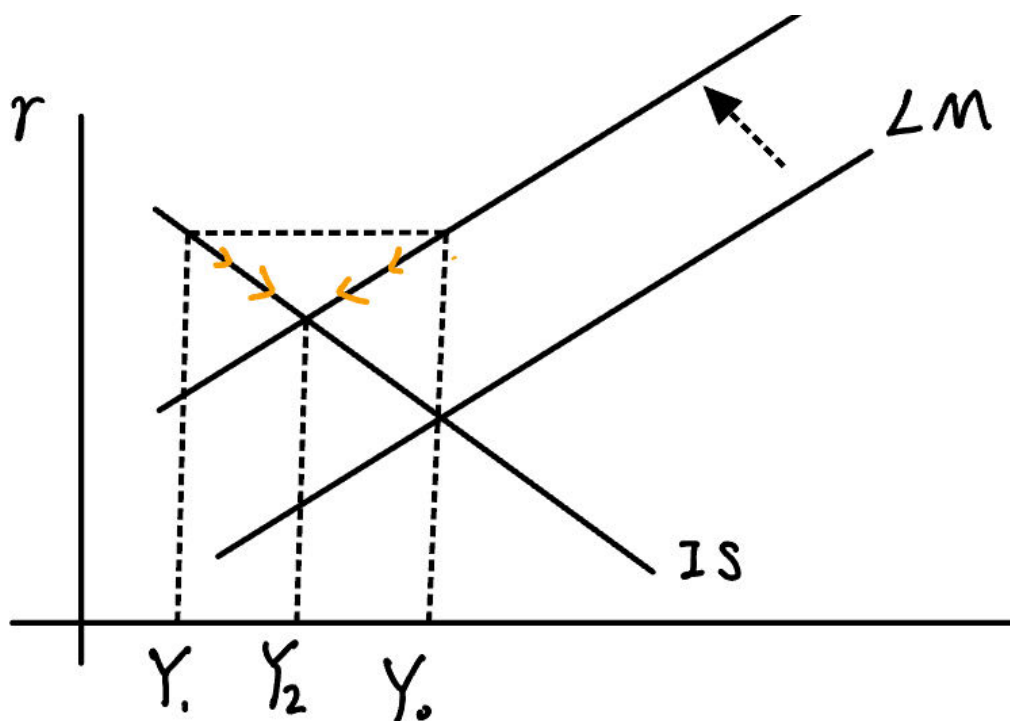


Figure 11.10: A rise in the price level (a fall in the real money supply M/P) shifts the LM curve up: the interest rate rises to r_1 and output falls to Y_1 .

Chapter 12

Aggregate Demand, Aggregate Supply, and the Supply Side

The IS–LM model held the price level fixed. To complete the short-run picture we let the price level move and ask how output and prices are jointly determined. The answer is an aggregate-demand curve, inherited from IS–LM, and an aggregate-supply curve, which encodes how firms respond to prices. Their intersection determines the short-run equilibrium; the long-run equilibrium, where prices have fully adjusted, sits on a vertical long-run supply curve at potential output. With this apparatus we can finally trace a fiscal expansion all the way through — multiplier, then two crowding-out effects — and see why, in the long run, demand stimulus changes only the price level.

12.1 The Aggregate-Demand Curve

Eliminating the interest rate between the IS and LM equations leaves a single relationship between the price level and equilibrium output. We met its mechanism at the end of the last chapter: a higher price level shrinks the real money supply M/P , shifts the LM curve up, raises the interest rate, and reduces investment and output. Equilibrium output is therefore a decreasing function of the price level, $Y^d = Y(P)$.

Remark (Why AD slopes down).

The aggregate-demand curve is *not* the microeconomic story in which buyers see a higher price and demand less. Here money may be perfectly neutral and the public perfectly rational, and the curve still slopes down: a higher price level reduces real balances, raises the interest rate, and crowds out investment. The downward slope is a statement about the money market and the interest rate, not about relative prices.

Definition 12.1: The Aggregate-Demand Curve

The *aggregate-demand (AD) curve* is the locus of (Y, P) pairs consistent with IS–LM equilibrium. It slopes downward because a higher price level reduces real money balances, raising the interest rate and lowering investment and output. Monetary and fiscal policy both shift the AD curve; a change in the price level is a movement along it.

12.2 The Aggregate-Supply Curve

In the long run, output is determined by capital, labor, and technology, not by the price level, so the long-run aggregate-supply (LRAS) curve is vertical at potential output $\bar{Y}$. In the short run, prices are sticky and supply responds to them. When the price level rises faster than costs (wages adjust with a lag), profit margins widen and firms expand production beyond the potential level, so the short-run aggregate-supply (SRAS) curve slopes upward. Its slope measures how sensitive supply is to prices: the more idle resources the economy has, the flatter SRAS is; when resources are fully employed, SRAS is nearly vertical.

Triple equilibrium

In a full long-run equilibrium the long-run supply curve, the short-run supply curve, and the aggregate-demand curve all intersect at a single point, with output at its potential level $\bar{Y}$.

12.3 A Fiscal Expansion from Start to Finish

We can now follow a rise in government purchases through all three diagrams at once — the Keynesian cross, IS–LM, and AD–AS — and watch output pass through four levels.

The economy begins in long-run equilibrium at potential output Y_0 . Government purchases then rise.

1. *The multiplier* ($Y_0 \rightarrow Y_1$). Planned expenditure rises, and through the multiplier output would reach Y_1 — if the interest rate and the price level were held fixed. (When the IS curve fixes its horizontal shift we hold the interest rate fixed; when the AD curve fixes its shift we hold the price level fixed.)
2. *IS–LM, the first crowding-out* ($Y_1 \rightarrow Y_2$). Allowing the interest rate to respond to the higher money demand, the rise in r crowds out private investment and output falls back to Y_2 .
3. *AD–AS, the second crowding-out* ($Y_2 \rightarrow Y_3$). At Y_2 aggregate demand still exceeds short-run supply, so the price level rises. The higher price level shrinks real balances, raising the interest rate further and crowding out still more investment; output settles at $Y_3 < Y_2$. This is the *second crowding-out effect*.

4. *Short run to long run* ($Y_3 \rightarrow Y_0$). Over time, costs (notably wages) rise toward the higher price level, profit margins are squeezed, and the short-run supply curve shifts left. Each leftward shift recreates an excess of demand over supply and pushes prices up again, until supply and demand meet back at potential output.

The two crowding-out effects

First crowding-out: a demand expansion raises money demand, raises the interest rate, and lowers investment. Second crowding-out: the same demand expansion raises the *price level*, which lowers real balances, raises the interest rate again, and lowers investment further. The multiplier is thus reduced twice, and the net effect on output can be small.

A few cautions are worth stating plainly. Prices change *only* when supply and demand are out of balance. The IS–LM equilibrium is necessary for full equilibrium but not sufficient — the price level must also clear the goods market against supply. And the long-run lesson is stark: with supply vertical, a pure demand stimulus raises only the price level, not output.

Remark (“In the long run we are all dead”).

Keynes’s quip is a defense, not a concession. Keynesian policy is aimed at the *short run*: it is countercyclical, meant to pull an economy out of a slump quickly rather than to manufacture a boom on top of one. Whether the two crowding-out effects leave much of the multiplier intact depends on the relative slopes of the curves — a flat LM or AD curve, or abundant idle resources, leaves the stimulus largely intact. When the LM curve is so flat that even large changes in output barely move the interest rate, the economy is in a *liquidity trap*, monetary policy is impotent, and fiscal policy is at its most effective.

12.4 The Supply Side

The experiments above all shifted demand. Shifts in supply behave very differently. Along a given short-run Phillips curve, inflation and unemployment move in opposite directions, so a demand-driven boom trades higher prices for lower unemployment. A negative *supply* shock breaks this trade-off.

When short-run supply contracts, output falls (raising unemployment) and the price level rises (raising inflation) simultaneously — the “lose–lose” outcome called *stagflation* (stagnation plus inflation). Demand policy cannot fix it: stimulus would worsen the inflation, contraction would worsen the unemployment. The cure must come from the supply side — raising firms’ willingness and ability to produce. This is the logic of the 1980s supply-side school: cut marginal tax rates to restore the incentive to work and invest, exploiting the fact (developed in Chapter 13) that beyond a point lower rates need not mean lower revenue.

An increase in long-run supply — genuine growth in productive capacity — is the mirror image: output rises and the price level falls. This is the deep meaning of economic growth: more output at a lower price level, the only “free lunch” in the model.

When analyzing a short-run supply shock we hold long-run supply fixed; the short-run diagram can still display long-run changes as a comparison of two static snapshots.

12.5 Microfoundations of the Short-Run Supply Curve

Why does short-run supply slope up at all? Three models give three reasons, but all deliver the same reduced form,

$$Y = \bar{Y} + \alpha (P - P^e), \quad \alpha > 0,$$

in which output exceeds potential only when the price level exceeds what was expected.

12.5.1 The Sticky-Wage Model

Workers and firms sign nominal wage contracts before the price level is known, fixing the nominal wage at $W = \omega P^e$, where ω is the target real wage. The realized real wage is then

$$\frac{W}{P} = \omega \frac{P^e}{P}.$$

When $P > P^e$ the real wage falls below target, firms hire more and output rises above potential; only when $P = P^e$ are employment and output at their potential levels. The model predicts a *counter-cyclical* real wage — falling in booms — which the data contradict in the long run, though over short horizons the mechanism has some bite.

12.5.2 The Imperfect-Information Model

Here all wages and prices are perfectly flexible, but each producer observes only the nominal price of its own good, not the aggregate price level (which is learned only with a lag). Supply depends on the *relative* price; lacking the price level, the firm uses its expectation P^e . If the general price level rises but the firm cannot yet see it, it mistakes the higher nominal price of its own good for a favorable *relative*-price signal and produces more, so aggregate output exceeds potential. The assumption that the price level is known only after the fact is a realistic one.

12.5.3 The Sticky-Price Model

Output and prices move together, but the causation can run from Y to P as easily as from P to Y . Each firm would like to set its price according to the price level and the state of demand,

$$p = P + \alpha (Y - \bar{Y}), \quad \alpha > 0,$$

charging more when output runs above potential. A firm free to change its price at will follows this rule. A firm that must commit to a price in advance (because of menu costs, long-term contracts, or customer relations) instead sets it on expectations,

$$p = P^e + \alpha (Y^e - \bar{Y}),$$

and for simplicity expects output at potential. Let a fraction s of firms have sticky prices and $1 - s$ be fully flexible. The equilibrium price level is the weighted average of the two groups' prices,

$$P = s P^e + (1 - s)[P + \alpha(Y - \bar{Y})],$$

which rearranges to

$$P = P^e + \left[\frac{(1-s)\alpha}{s} \right] (Y - \bar{Y}), \quad \text{equivalently} \quad Y = \bar{Y} + \frac{s}{\alpha(1-s)} (P - P^e).$$

If every firm were flexible ($s = 0$), an increase in nominal income would pass entirely into prices: money would be neutral and supply vertical. It is the presence of price-stickers that gives the supply curve its positive slope. Unlike the sticky-wage model, the sticky-price model implies a *pro-cyclical* real wage: a negative demand shock leads flexible firms to cut prices and sticky firms to cut output and labor demand, and the leftward shift in labor demand lowers the real wage — which matches the data better.

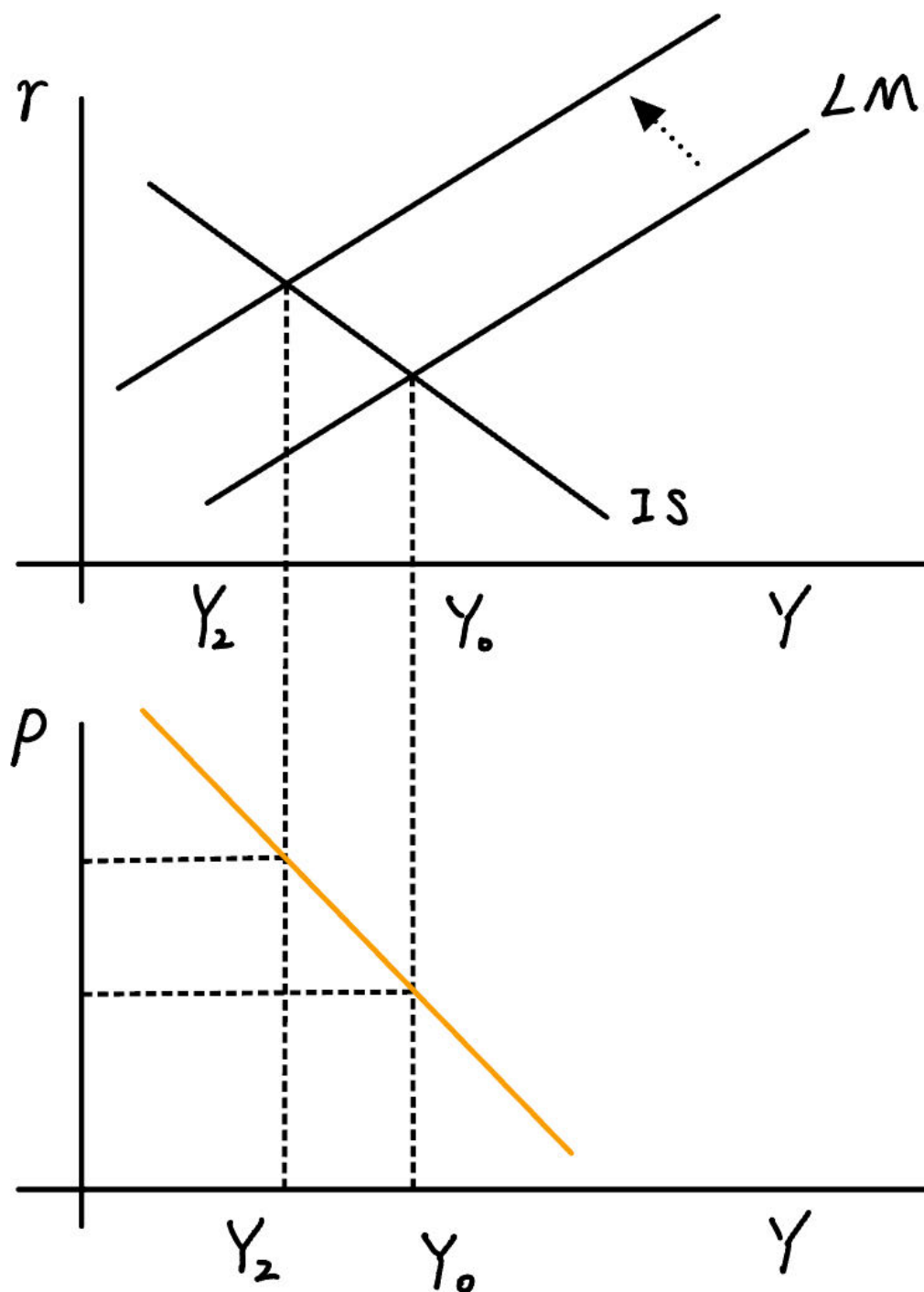


Figure 12.1: Deriving the aggregate-demand curve. In the upper panel a higher price level lowers the real money supply, shifting LM up and reducing output from Y_0 to Y_2 . Plotting price against the output it implies (lower panel) traces out the downward-sloping AD curve.

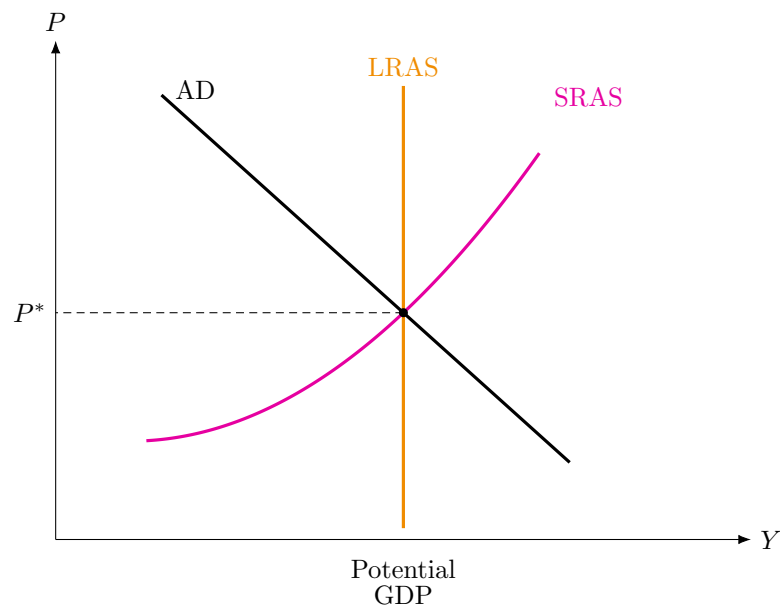


Figure 12.2: Aggregate supply and demand. Long-run aggregate supply (LRAS) is vertical at potential output; short-run aggregate supply (SRAS) slopes upward; aggregate demand (AD) slopes downward. In a full equilibrium all three pass through the same point.

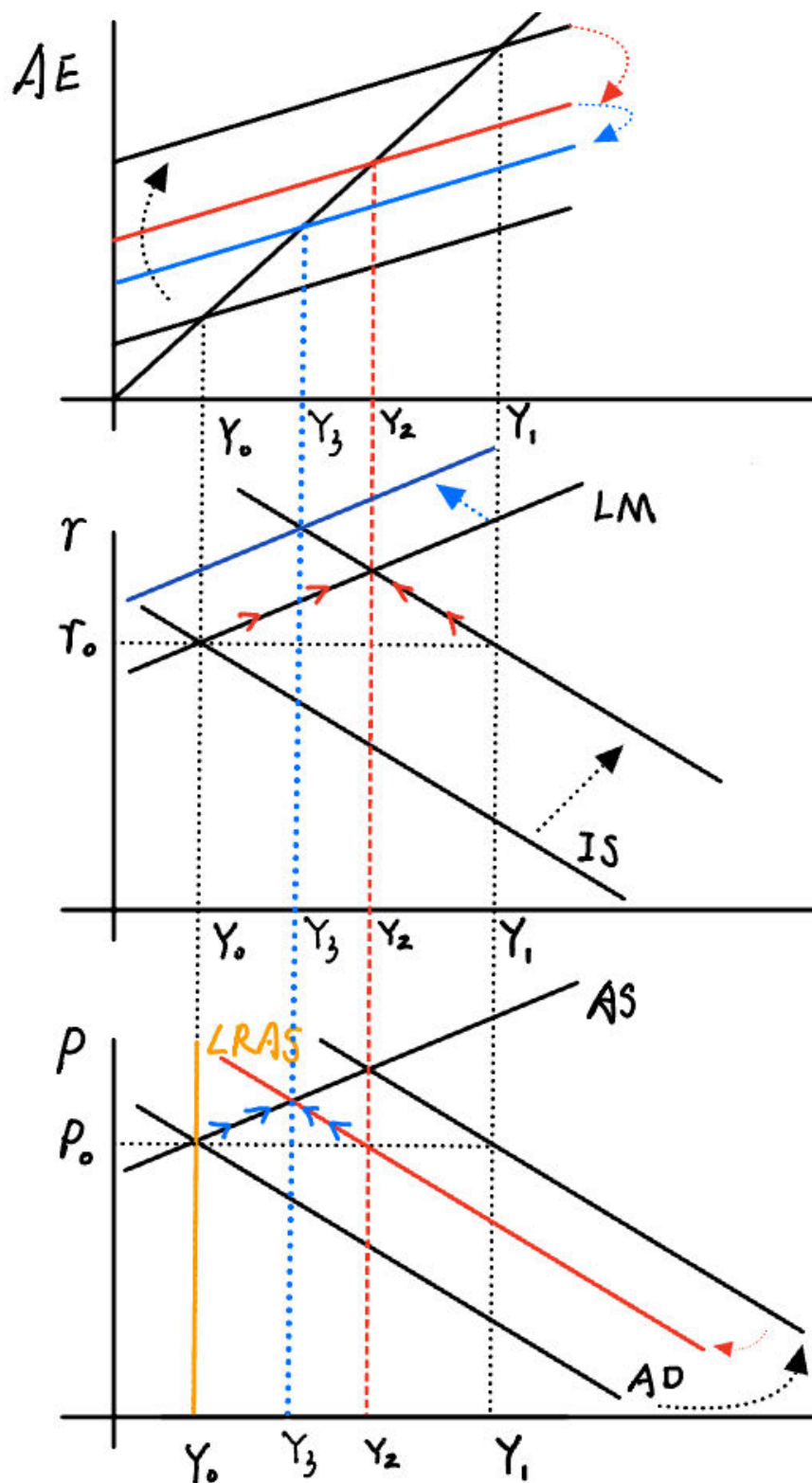


Figure 12.3: The complete adjustment to a fiscal expansion. Black: the multiplier raises output to Y_1 at a fixed interest rate and price level. Red: allowing for liquidity preference, the interest rate rises and the first crowding-out effect pulls output back to Y_2 . Blue: allowing the price level to rise, the second crowding-out effect pulls output back to Y_3 . In the long run, rising costs return output to potential, Y_0 . Throughout, $Y_0 < Y_3 < Y_2 < Y_1$.

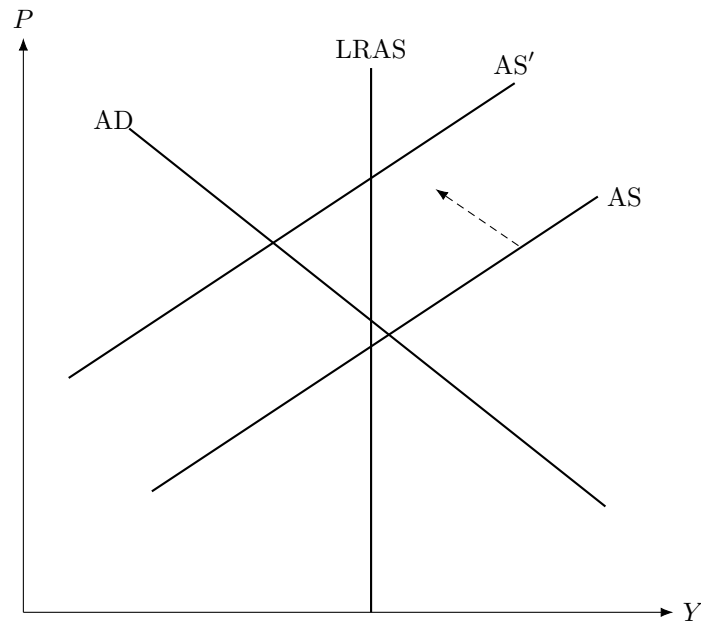


Figure 12.4: A negative supply shock. The short-run supply curve shifts left: output falls and the price level rises at the same time — stagflation.

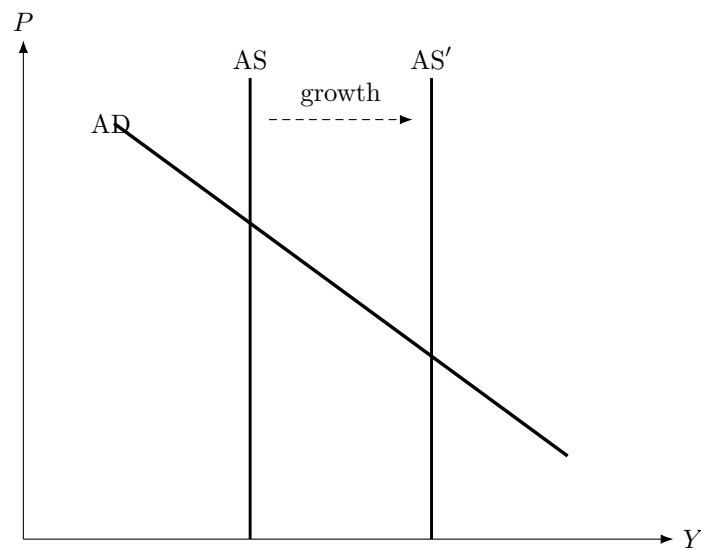


Figure 12.5: Long-run supply growth. The supply curve shifts right: output rises and the price level falls — the “win-win” of genuine economic growth.

Chapter 13

Fiscal Policy: Ricardian Equivalence, the Laffer Curve, and Government Debt

The Keynesian apparatus treated a fiscal expansion as a shift of the IS curve, financed by some unspecified mix of taxes and borrowing. The classical tradition objects that the financing is not a detail at all. A forward-looking household understands that a deficit today is a tax tomorrow, and plans accordingly — so much so that, under sharp enough assumptions, the choice between taxes and debt has no effect on anything real. This chapter develops that claim, called Ricardian equivalence, in a clean two-period model; identifies the assumptions it leans on, and hence the realistic reasons it fails; and then turns to two questions the equivalence result reframes rather than settles: how high taxes can go before they become self-defeating (the Laffer curve), and whether government debt is a burden.

13.1 Ricardian Equivalence

Classical economists — Smith, Ricardo — held that markets solve most problems and that government is largely unnecessary. They could not, however, explain the Great Depression, which is where Keynes entered with a case for using government spending to prop up demand. The classical reply, later given rigorous microfoundations by the new classical school, is that government spending crowds out private consumption whether it is financed by taxes or by debt, because rational consumers see through the deficit: a bond sold today must be redeemed by a tax tomorrow.

Consider an economy with a single representative consumer and a government, lasting two periods. The government spends only in the first period, $G_1 = G$ and $G_2 = 0$, financing it with some combination of a first-period tax t_1 and bonds b , so that $G = t_1 + b$. In the second period it raises a tax $t_2 = (1 + r)b$ to repay the bonds with interest. Its budget can be written in present value as

$$G = t_1 + \frac{t_2}{1 + r},$$

the present value of taxes equals the present value of spending.

Definition 13.1: Ricardian Equivalence

Ricardian equivalence is the proposition that, for a given path of government spending, the choice between tax finance and bond finance leaves the consumer's budget set — and hence consumption and welfare — unchanged. Only the present value of spending matters, not how it is financed.

The consumer has preferences $U(C_1, C_2) = u(C_1) + \beta u(C_2)$ and faces the period budgets

$$C_1 = y_1 - t_1 - s, \quad C_2 = y_2 - t_2 + (1 + r)s,$$

where s is private saving. Since the only asset is the government bond, in equilibrium private saving equals the bonds issued, $s = b$. Combining the two period budgets to eliminate s , and then substituting the government budget, gives the consumer's lifetime budget constraint

$$C_1 + \frac{C_2}{1 + r} = y_1 + \frac{y_2}{1 + r} - \left(t_1 + \frac{t_2}{1 + r} \right) = y_1 + \frac{y_2}{1 + r} - G.$$

The present value of consumption equals the present value of income less the present value of spending. The split of G between t_1 and b has dropped out entirely.

The key point

The consumer's lifetime budget depends on G alone, not on whether G is raised by taxes or by bonds. Two financing schemes with the same G yield the same (C_1, C_2) . That is Ricardian equivalence.

Example (Log utility).

Take $u(C) = \log C$, so the consumer solves

$$\max_s \log(y_1 - t_1 - s) + \beta \log(y_2 - t_2 + (1 + r)s).$$

The first-order condition is

$$\frac{1}{y_1 - t_1 - s} = \frac{\beta(1 + r)}{y_2 - t_2 + (1 + r)s},$$

with solution

$$b = s^* = \frac{\beta(1 + r)(y_1 - t_1) - (y_2 - t_2)}{(1 + \beta)(1 + r)}.$$

Compare the two polar financing schemes. Under *pure taxation*, $G = t_1$, $b = 0$, $t_2 = 0$, and the optimum requires $G = y_1 - \frac{y_2}{\beta(1 + r)}$. Under *pure bond finance*, $b = G$, $t_1 = 0$, $t_2 = (1 + r)G$, and substituting gives exactly the same condition $G = y_1 - \frac{y_2}{\beta(1 + r)}$. The government spends the same amount in both cases, so C_1 and C_2 are identical: the equivalence holds, and by the same argument it holds for any tax–bond mix in between.

13.1.1 The Assumptions, and Why They Fail

Equivalence is a theorem, and like any theorem it has premises.

Assumption 13.2: Premises of Ricardian Equivalence

1. **Perfect foresight / rationality:** the consumer anticipates $t_2 = (1 + r)b$ and optimizes over both periods.
2. **The consumer lives both periods:** the same agent who benefits from the deficit also bears the future tax.
3. **The taxpayer and the bondholder are the same consumer.**
4. **Taxes are lump-sum.**

Each assumption, when it fails, breaks equivalence in an instructive way.

- *Finite lives.* If consumers do not live to pay the future tax, a bond-financed deficit lets the current generation consume more and forces the next generation to consume less; consumption is no longer smoothed across the two. (Altruistic bequests can partially restore equivalence by linking the generations.)
- *Taxpayer $\neq$ bondholder.* Two effects appear. A *distributional* effect: those who buy fewer bonds now consume more, while those who must buy more later consume less. And an *effect on saving and capital*: when a deficit is not fully matched by extra private saving, government borrowing absorbs funds that would have financed investment, reducing the capital stock.
- *Distortionary taxes.* If the tax is not lump-sum but an income or *ad valorem* tax (e.g. a progressive tax), it changes relative prices and distorts the allocation away from the efficient one. Taxing now creates a distortion now; borrowing and taxing later creates a distortion later. Because the timing of distortions now matters, the financing choice is no longer neutral — which is the basis for *tax smoothing*: spread distortionary taxes thinly over time, and lean on debt to do it.

13.2 The Laffer Curve

Once taxes are distortionary, their level — not just their timing — matters, and raising the rate need not raise revenue. A static labor-supply model makes the point. A consumer with utility $U(C, l) = C + \log l$ over consumption and leisure l faces a proportional tax τ on labor income; with the wage w and total time normalized to one, the budget is

$$C = (1 - \tau)(1 - l)w.$$

Maximizing $C + \log l$ over l gives the first-order condition $-(1 - \tau)w + 1/l = 0$, so the optimal leisure and labor supply are

$$l^* = \frac{1}{(1 - \tau)w}, \quad 1 - l^* = 1 - \frac{1}{(1 - \tau)w}.$$

Rewriting the budget as $C + (1 - \tau)wl = (1 - \tau)w$ shows that a higher τ *lowers* the effective price of leisure $(1 - \tau)w$, so by the substitution effect leisure rises and labor supply falls. Tax revenue is the rate times the (taxed) labor income,

$$R = \tau w(1 - l^*) = \tau w - \frac{\tau}{1 - \tau}.$$

The two terms have a clean reading: τw is the *potential-revenue effect*, the revenue if the consumer worked all the time; and $-\frac{\tau}{1 - \tau}$ is the *distortion effect*, with τ the rate and $1 - \tau$ the distortion it induces.

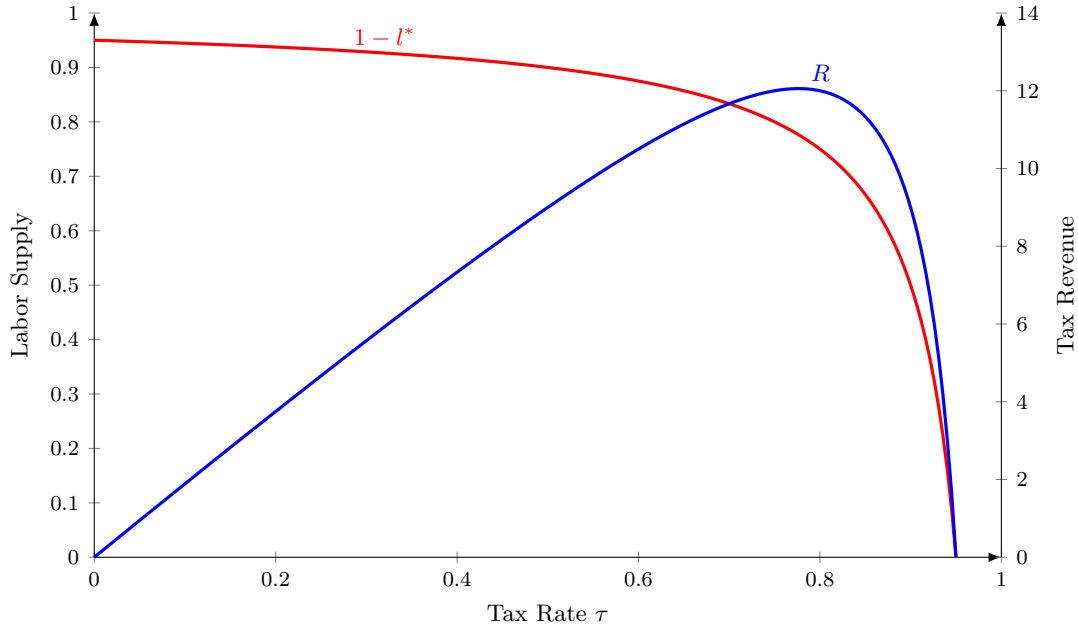


Figure 13.1: The Laffer curve. As the tax rate rises, labor supply (left axis) falls, slowly at first and then steeply; tax revenue (right axis) traces an inverted U, rising while the rate effect dominates and falling once the shrinking tax base dominates.

Differentiating, $dR/d\tau = w - 1/(1 - \tau)^2$, which is zero at the revenue-maximizing rate; beyond it, revenue falls. Figure 13.1 shows the inverted-U shape: past the peak the tax base shrinks faster than the rate rises, so revenue declines. A *prohibitive* tax rate is one beyond the peak, where labor supply has collapsed so far that raising the rate lowers total revenue; there the distortion of the optimal allocation is severe.

Implication

High tax rates are very distortionary, so governments should *smooth* distortions over time and rely more on debt. In wartime, when spending must spike, the textbook prescription is to raise rates only modestly and finance the bulk with bonds — precisely tax smoothing in action.

13.3 Government Debt

Three questions about public debt recur.

Is government debt sustainable? A sovereign's debt is unlike a household's. As long as the state exists, its debt can in principle be rolled over indefinitely — new bonds issued to redeem old ones — and never repaid in full. Comparisons of debt to GDP are common but imperfect: GDP is a flow while debt is a stock; governments hold large assets behind the debt; and as long as the regime and its credit endure, the debt is backed. Broadly, the debt is manageable, though a downgrade by ratings agencies can strike even when there is fiscal room.

Is the debt a burden on the economy? When the government borrows to spend, some people consume less today, and the burden is shared across generations through the future taxes. The usual occasion is a slump, in which the government spends to support demand. For the public, holding the bonds is itself a positive-return asset, secure so long as the debt is rolled over.

Is there an optimal debt-to-tax ratio? Under Ricardian equivalence, taxes and bonds are equivalent, so no optimal mix exists. But the equivalence relies on assumptions the real world violates — chiefly lump-sum taxation — so once taxes are distortionary the tax-smoothing argument returns and there is a genuine trade-off. Historically, the United States financed itself mainly through taxes around the Second World War; since the 1970s the balance has shifted toward debt, with steady tax revenue but expanding outlays and persistent deficits.

13.4 Social Security

Social-security spending is a large part of the government budget, and its design echoes the financing question.

Definition 13.3: Two Social-Security Systems

A *defined-benefit* (pay-as-you-go) system funds today's retirees out of today's workers' contributions — the young support the old. A *defined-contribution* (fully funded) system invests each worker's contributions and pays them back, with investment returns, in retirement — one supports one's own future self.

A crucial fact about pay-as-you-go systems is that the money in the “account” is not actually held in reserve; it has already been spent on current retirees. Such a system is robust to investment risk but acutely vulnerable to demographic change. As a population ages — fewer workers supporting more retirees, the old-age dependency ratio rising — a pay-as-you-go system tends toward deficit. Remedies are all unpalatable: higher contribution rates burden workers and firms; cutting benefits is politically impossible; raising the retirement age may be unavoidable. A fully funded system avoids the demographic squeeze but bears inflation and investment risk instead.

Chapter 14

The Phillips Curve, Expectations, and the Conduct of Policy

The aggregate-supply curve says output rises above potential only when prices surprise on the upside. Translated into the language of inflation and unemployment, that same surprise becomes the Phillips curve — the short-run menu between inflation and unemployment that policy appears to exploit. Whether it can be exploited turns entirely on expectations, and that question — how the public forms its expectations, and whether policy can shape them — organizes the modern debate over how monetary and fiscal policy should be run: actively or passively, by rule or by discretion. This final chapter follows that thread from the Phillips curve to the Taylor rule.

14.1 The Phillips Curve

Empirically, inflation and unemployment move in opposite directions over the cycle. The modern, expectations-augmented Phillips curve states this as

$$\pi - \pi^e = -\beta(u - u^n) + v, \quad \beta > 0,$$

where $u - u^n$ is cyclical unemployment (the deviation of unemployment from its natural rate), v is a supply shock, and π^e is expected inflation. Inflation runs above expectations when unemployment is below its natural rate.

Definition 14.1: The Phillips Curve

The *expectations-augmented Phillips curve* relates surprise inflation to cyclical unemployment, $\pi - \pi^e = -\beta(u - u^n) + v$. The trade-off is between inflation and *cyclical* unemployment, and it is stated relative to expected inflation.

14.1.1 Derivation

The Phillips curve is the supply curve in disguise. Start from short-run aggregate supply, $Y = \bar{Y} + \alpha(P - P^e)$, and solve for the price level,

$$P = P^e + \frac{1}{\alpha} (Y - \bar{Y}).$$

Because the relationship was read off the data, append the supply shock v as the residual term and difference against last period's price level P_{-1} ,

$$P - P_{-1} = (P^e - P_{-1}) + \frac{1}{\alpha} (Y - \bar{Y}) + v,$$

which, written in inflation rates, is $\pi - \pi^e = \frac{1}{k\alpha} (Y - \bar{Y}) + v$ for a constant of proportionality k . Finally invoke *Okun's law*, the empirical regularity that output and unemployment move together — roughly, each extra percentage point of unemployment corresponds to about two percentage points of lost GDP. Okun's law lets us replace the output gap with cyclical unemployment,

$$\frac{1}{k\alpha} (Y - \bar{Y}) = -\beta (u - u^n),$$

and substituting gives the Phillips curve. The relationship is correlational, not causal, but it quantifies a link that policy has repeatedly tried to use.

14.2 Expectations

Everything hinges on π^e . In a world of imperfect information or sticky prices, agents act on what they expect, and the entire position of the Phillips curve moves with that expectation. Two theories of expectation formation give very different policy conclusions.

14.2.1 Adaptive Expectations

Adaptive expectations are formed by extrapolating the past. The simplest version sets $\pi^e = \pi_{-1}$, which turns the Phillips curve into a law of motion for inflation,

$$\pi = \pi_{-1} - \beta (u - u^n) + v.$$

Inflation has inertia: absent shocks, past inflation feeds into present and future inflation. This is essentially what a regression on historical data delivers, and therein lies its weakness — it presumes the past relationships will continue to hold.

Remark (The Lucas Critique).

Forecasting policy effects from historically estimated relationships is unreliable precisely because policy changes expectations. In Lucas's words: "Forecasting the effects of policy changes has often been done using models estimated with historical data. Such predictions would not be valid if the policy change alters expectations in a way that changes the fundamental relationships between variables." When a regime changes, the historical correlations exhibit a *structural break*, and adaptive forecasts fail.

14.2.2 Rational Expectations

Rational expectations use the model and all available information, not just the past — expectations are, in effect, a forecast of one's own decisions and everyone else's. Analysis then cannot rest on regression alone; it requires a general-equilibrium model that, while it may not pin down individual causes, embeds every mechanism and produces a coherent shock-response. Under rational expectations a *painless disinflation* is possible in principle: if the central bank can directly shift expectations with a credible signal, the supply curve shifts and inflation falls without a recession — the sacrifice ratio need not bind. In the United States, rational expectations amounts to “trust the Fed.” Where the central bank's signals are too opaque to support quantitative expectations — the policy logic hard to read — rational expectations degenerate into rumor, and markets gyrate accordingly.

14.3 The Natural-Rate Hypothesis and Hysteresis

Underlying all of this is the *natural-rate hypothesis*: there exists a potential level of output (and a natural rate of unemployment) at which the economy is in equilibrium, reached in the long run and always eventually returned to, with the short run a deviation from it. It is one of the organizing assumptions of macroeconomics.

The competing view is *hysteresis*: temporary shocks can move the long-run level itself, so potential output exists but is not a fixed number — it depends on the economy's history. The pandemic is a natural example. A long spell of unemployment erodes workers' skills, so that when demand recovers they cannot find matching work and lose bargaining power; a prolonged downturn shifts consumption habits toward precautionary saving; supply chains relocate, capital depreciates, investment turns cautious. Each channel lets a transitory shock leave a permanent mark.

14.4 The Conduct of Policy

14.4.1 Active versus Passive

An *active* stance holds that in a slump the government has a duty to act — to ease suffering and stimulate the economy. Two problems complicate activism.

First, policy operates with *lags*. The *inside lag* is the time to recognize the need and enact a response — confirming that the economy is contracting takes several indicators, and execution (especially of fiscal policy) requires coordination. The *outside lag* is the time for the policy to affect the economy, which may overshoot. Second, the economy contains *automatic stabilizers* — institutions that smooth the cycle without deliberate action and without lags. A progressive income tax restrains the multiplier in a boom by taking a rising share of extra income; unemployment insurance supports income, and hence consumption, in a downturn; under a floating exchange rate, a strong export boom appreciates the currency and tempers demand.

To use active policy well, a government must identify the phase of the cycle early — watching leading indicators (composite indices, investment, PMI, real-estate financing) that turn before the cycle does, and, when the environment changes enough that historical relationships break, using a structural model rather than extrapolation. It must also *lead* the

public's expectations: if a rational public already anticipates a policy, the policy's effect is blunted. Suppose past data show that money growth lowers unemployment, so the public expects it; if the central bank then expands the money supply, expected inflation rises in step and unemployment does not fall — the anticipated policy is neutral, and a different or surprising policy is needed.

14.4.2 Rules versus Discretion

The deeper choice is whether policy should follow a published *rule* or be set at *discretion*.

Definition 14.2: Rules and Discretion

A policy conducted *by rule* is set and adjusted according to an explicit, public formula, so it can be anticipated. A policy conducted *by discretion* responds case by case to circumstances and cannot be anticipated.

The canonical rule is the *Taylor rule* for the federal funds rate, which sets the target rate from inflation and the output gap,

$$r_{ff} = 2 + 0.5(\pi - 2) - 0.5 \cdot \text{GDP gap}, \quad \text{GDP gap} = 100 \cdot \frac{\bar{Y} - Y}{\bar{Y}},$$

where 2% is taken as the natural rate of inflation. Note the sign convention: the gap is defined as the *shortfall* of output below potential, so subtracting it means that when output runs *above* potential (a boom, $Y > \bar{Y}$, gap negative) the rule *raises* the rate, and when output runs below potential it lowers the rate — exactly the stabilizing response, equivalent to adding the conventional output gap $(Y - \bar{Y})/\bar{Y}$. This is a transparent, credible rule: the rate is set by formula, with independence and predictability, without second-guessing the source of an inflation. China, by contrast, aims to “match” M_2 growth to nominal GDP growth — a target that sits awkwardly with the countercyclical role that the quantity theory assigns to money, and whose “matching,” and the forecast of nominal GDP it rests on, is itself discretionary.

The case for rules

Several arguments favor rules over discretion. Policymakers may lack the requisite expertise; their judgments may rest on vague or shifting principles, possibly out of step with the public's; and they may be time-inconsistent, undermining the *credibility* on which expectations — and hence the policy's own effectiveness — depend. Where the competence and consistency of discretionary policymakers are in doubt, a well-designed rule can do better.